MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL TTL, COUNTERS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, binary and decode counters. Three product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit
01	Divide-by-12 Counter
02	4-Bit Binary Counter
03	Synchronous 4-Bit Decade Counter (Asynchronous Clear)
04	Synchronous 4-Bit Binary Counter (Synchronous Clear)
05	Synchronous 4-Bit Decade Counter (Synchronous Clear)
06	Synchronous 4-Bit Binary Counter (Asynchronous Clear)
07	High-speed Decade Counter
08	Synchronous 4-Bit Up/Down Decade Counter
09	Synchronous 4-Bit Up/Down Binary Counter

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline, (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4" flat package)
B	F-3 (14-lead, 1/8" x 1/4" flat package)
C	D-1 (14-lead, 1/4" x 3/4" dual-in-line package)
D	F-2 (14-lead, 1/4" x 3/8" flat package)
E	D-2 (16-lead, 1/4" x 7/8" dual-in-line package)
F	F-5 (16-lead, 1/4" x 3/8" flat package)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage range	-0.5 Vdc to 7.0 Vdc
Input voltage range	-1.5 Vdc at -12 mA to 5.5 Vdc
Storage temperature range	-65° to +150°C
Maximum power dissipation (P_D) 1/	
Device types 01, 02, 07	268 mW
Device types 03 through 06	500 mW
Device types 08 and 09	490 mW
Lead temperature (soldering 10 seconds)	300°C
Thermal resistance, junction to case (θ_{JC})	0.09°C/mW for flat package; 0.08°C/mW for dual-in-line package
Junction temperature (T_J)	175°C

1.4 Recommended operating conditions.

Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.8 Vdc
Normalized fanout (each output)	
Device types 01, 02, 07, 08, 09	10 maximum
Device types 03, 04, 05, 06	
Low-level	10 maximum
High level	20 maximum
Width of input count pulse, $t_{p(in)}$	
Device types 01, 02, 07	50 ns minimum
Device types 08, 09	25 ns minimum
Width of reset pulse, $t_{p(reset)}$	
Device types 01, 02, 07	50 ns minimum
Input clock frequency, f_{CLOCK}	
Device types 01, 02, 07	0 to 10 MHz
Device types 03, 04, 05, 06, 08, 09	0 to 20 MHz
Width of clock pulse, $t_{w(CLOCK)}$	
Device types 03, 04, 05, 06	30 ns minimum
Width of clear pulse, $t_{w(CLEAR)}$	
Device types 03, 04, 05, 06, 08, 09	25 ns minimum
Setup time, $t_{(SETUP)}$,	
Device types 03, 04, 05, 06	
Data inputs	20 ns minimum
Enable P	25 ns minimum
Load	30 ns minimum
Clear (04 and 05 only)	25 ns minimum
Setup time, $t_{(SETUP)}$,	
Device types 08, 09	
Data inputs	25 ns minimum
Hold time at any input, t_{HOLD}	
Device types 03, 04, 05, 06, 08, 09	0 ns minimum
Case operating temperature range	-55° to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

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STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagram. The terminal connections and logic diagrams shall be as specified on figures 1 and 2 respectively.

3.2.2 Truth tables and timing diagrams. The truth tables and timing diagrams shall be as specified on figures 3 and 4 respectively.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III. (Subgroups 7 and 8 testing requires only a summary of attributes data.)

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 5 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 of MIL-STD-883, as applicable, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions	Device type	Limits		Unit
				Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IN} = 0.8 V I_{OH} = -400 µA	01,02,07, 08,09	2.4		V
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IN} = 0.8 V I_{OH} = -800 µA	03,04, 05,06	2.4		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 16 mA, V_{IN} = 2.0 V	01,02,03, 04,05,06, 07		0.4	V
		V_{IN} = 0.8 V	08,09		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -12 mA T_C = 25°C	01,02,03, 04,05,06, 07,08,09		-1.5	V
High-level input current at reset inputs	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	01,02,07		40	µA
High-level input current at reset inputs	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01,02,07		100	µA
High-level input current at input A	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	01,07		80	µA
High-level input current at input A	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01,07		200	µA
High-level input current at input BC or BD	I_{IH5}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	01,07		160	µA
High-level input current at input BC or BD	I_{IH6}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01,07		400	µA
High-level input current at input A or B	I_{IH7}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	02		80	µA
High-level input current at input A or B	I_{IH8}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	02		200	µA
High-level input current at clock or enable T	I_{IH9}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	03,04, 05,06		80	µA
High-level input current at clock or enable T	I_{IH10}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	03,04, 05,06		200	µA
High-level input current at other inputs	I_{IH11}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	03,04,05, 06,08,09		40	µA
High-level input current at other inputs	I_{IH12}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	03,04,05, 06,08,09		100	µA
Low-level input current at reset inputs	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01,02,07	-0.4	-1.6	mA
Low-level input current at input A	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01	-0.4	-3.2	mA
Low-level input current at input BC or BD	I_{IL3}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	08,09	-0.7	-1.6	mA
			01	-0.4	-6.4	mA
Low-level input current at input A or E	I_{IL4}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	07	-1.4	-6.4	mA
			02	-0.4	-3.2	mA
Low-level input current at clock, count up or count down	I_{IL5}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	03,04 05,06	-1.0	-2.3	mA
			08,09	-0.7	-1.6	mA
Low-level input current at data inputs	I_{IL6}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	03,04,05, 06	-0.4	-1.3	mA
			08,09	-0.7	-1.6	mA
Low-level input current at clear, enable, and load	I_{IL7}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	03,04,05, 06,08,09	-0.5	-1.6	mA
Low-level input current at count enable T	I_{IL8}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	03,04,05, 06	-1.0	-3.2	mA
Short-circuit supply current	I_{OS}	V_{CC} = 5.5 V 1/	01,02,03, 04,05,06, 07	-20	-57	mA
			08,09	-20	-65	mA
Supply current	I_{CC}	V_{CC} = 5.5 V, V_{IN} = 4.5 V	01,02,07		44	mA
			08,09		89	mA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions	Device type	Limits		Unit
				Min	Max	
High-level supply current	I_{CCH}	V_{CC} = 5.5 V 2/	03,04, 05,06		85	mA
Low-level supply current	I_{CCL}	V_{CC} = 5.5 V 3/	03,04, 05,06		91	mA
Maximum frequency of input count pulses	F_{MAX}	V_{CC} = 5.0 V C_L = 50 pF ±10% R_L = 390Ω ±5%	01,02,07	10		MHz
Maximum input clock frequency	F_{MAX}		03,04,05, 06,08,09	20		MHz
Propagation delay time to logical H level from input count pulse to output D	t_{PLH1}		01,07	20	115	ns
Propagation delay time to logical L level from input count pulse to output D	t_{PHL1}		01,07	20	115	ns
Propagation delay time to logical H level from input count pulse to output D	t_{PLH2}		02	20	135	ns
Propagation delay time to logical L level from input count pulse to output D	t_{PHL2}		02	20	135	ns
Propagation delay time, low-to-high-level carry output from clock	t_{PLH3}		03,04, 05,06	3	58	ns
Propagation delay time, high-to-low-level carry output from clock	t_{PHL3}		03,04 05,06	3	58	ns
Propagation delay time, low-to-high-level Q output from clock	t_{PLH4}		03,04, 05,06	3	36	ns
Propagation delay time, high-to-low-level Q output from clock	t_{PHL4}		03,04, 05,06	3	38	ns
Propagation delay time, low-to-high-level carry output from enable T	t_{PLH5}		03,04, 05,06	3	27	ns
Propagation delay time, high-to-low-level carry output from enable T	t_{PHL5}		03,04, 05,06	3	27	ns
Propagation delay time, high-to-low-level Q output from clear	t_{PHL6}		03,04, 05,06	3	47	ns
Propagation delay time, high-to-low-level Q output from clear	t_{PHL7}		08,09	3	57	ns
Propagation delay time, low-to-high-level Q output from load input	t_{PLH8}		08,09	3	66	ns
Propagation delay time, high-to-low-level Q output from load input	t_{PHL8}		08,09	3	62	ns
Propagation delay time, high-to-low-level carry output from count up input	t_{PHL9}		08,09	3	39	ns
Propagation delay time, low-to-high-level carry output from count up input	t_{PLH9}		08,09	3	45	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions	Device type	Limits		Unit
				Min	Max	
Propagation delay time, high-to-low-level borrow output from count down input	t_{PHL10}	V_{CC} = 5.0 V C_L = 50 pF ±10% R_L = 390Ω ±5%	08,09	3	39	ns
Propagation delay time, low-to-high-level borrow output from count down input	t_{PLH10}		08,09	3	42	ns
Propagation delay time, low-to-high-level, Q output from count up or count down input	t_{PLH11}		08,09	3	63	ns
Propagation delay time, high-to-low-level, Q output from count up or count down input	t_{PHL11}		08,09	3	71	ns

1/ Not more than one output should be shorted at a time.
2/ I_{CCH} is measured: (a) With the load input high; and (b) Then again with the load input low with all other inputs high and all outputs open.
3/ I_{CCL} is measured: (a) With the clock input high; and (b) Then again with the clock input low with all other inputs low and all outputs open.

TABLE II. Electrical test requirements.

MIL-STD-883 Test requirements	Subgroup (see table III)		
	Class S devices	Class B devices	Class C devices
Interim electrical parameters (Pre Burn-In) (Method 5004)	1	1	None
Final electrical test parameters (Method 5004)	1*,2,3,7, 9,10,11	1*,2,3,7,9	1,7
Group A test requirements (Method 5005)	1,2,3,7,8, 9,10,11	1,2,3,7, 9,10,11	1,2,3,7,9
Group B test requirements (Method 5005)	1,2,3 9,10,11	N/A	N/A
Group C end point electrical parameters (Method 5005)	N/A	1,2,3	1
Additional electrical subgroups for Group C periodic inspections	N/A	None	10,11
Group D end point electrical parameters (Method 5005)	1,2,3	1,2,3	1

*PDA applies to subgroup 1 (see 4.2c).

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a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 5, or equivalent.

(2) T_A = 125°C, minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Percent defective allowable (PDA) - The PDA for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualifiction inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class C devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

c. Steady state life test (method 1005 of MIL-STD-883) conditions, or equivalent.

(1) Test condition D or E, using the circuit shown on figure 5, or equivalent..

(2) T_A = 125°C, minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End point electrical parameters shall be as specified in table II herein.

4.4.5 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of MIL-M-38510.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for the packaging of microcircuits shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logistic support of existing equipment.

6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

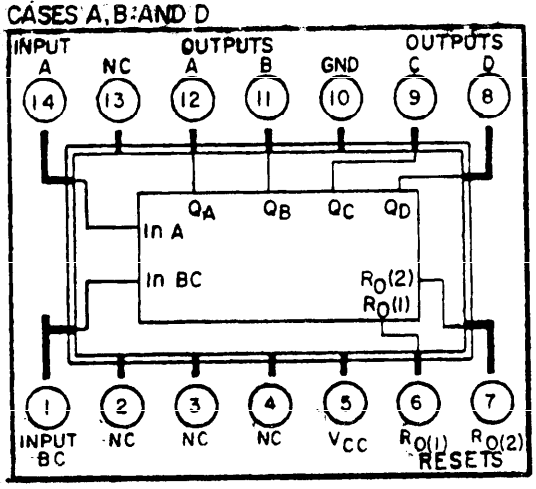
6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - Electrical ground (common terminal)
V_{IN} - - - - - - - - - - - Voltage level at an input terminal

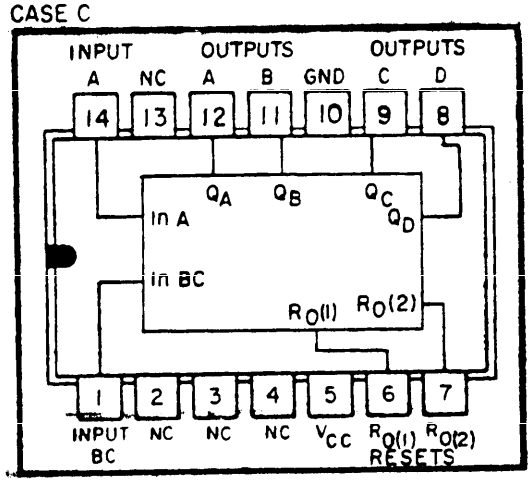
6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be procured to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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CASES A, B AND D

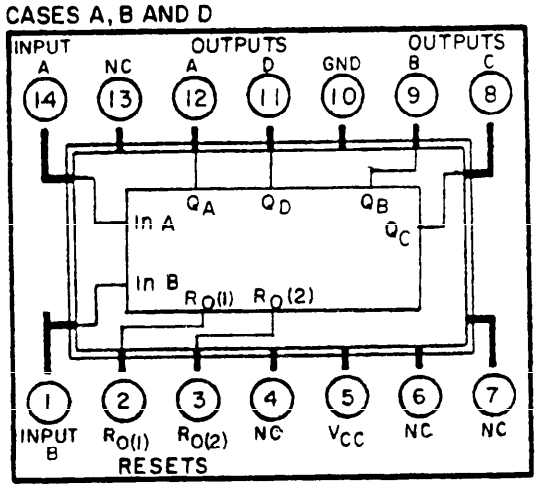


CASE C

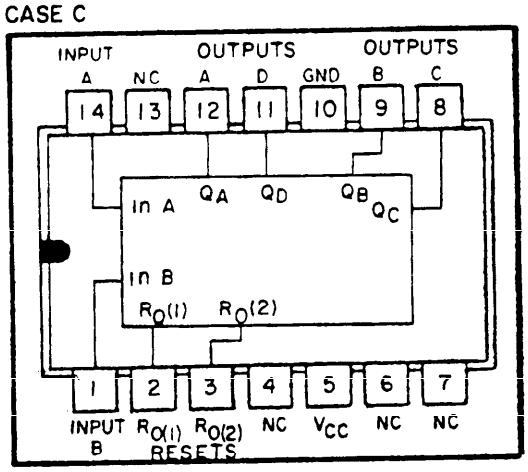


DEVICE TYPE 01

CASES A, B AND D

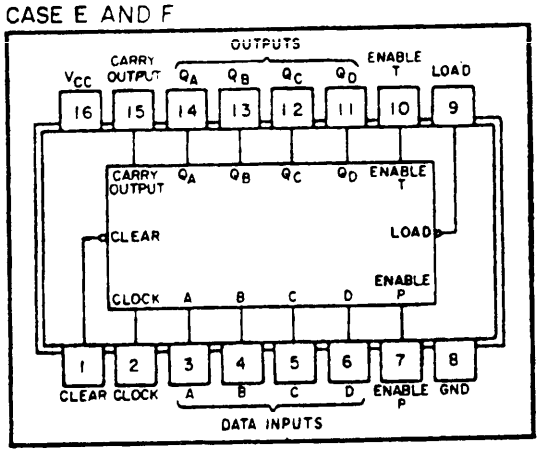


CASE C



DEVICE TYPE 02

CASE E AND F

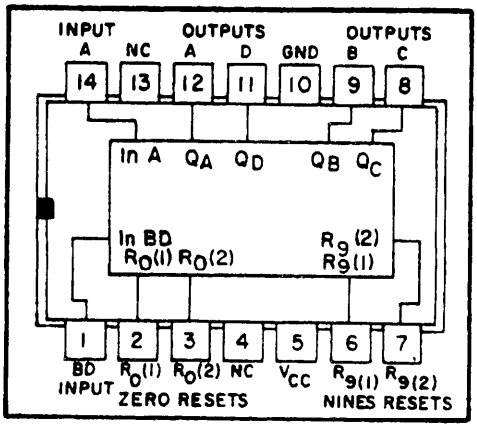


DEVICE TYPES 03, 04, 05, 06

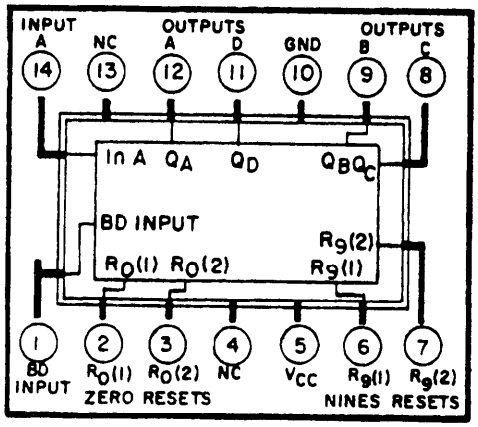
FIGURE 1. Terminal connections (top view).

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CASE C

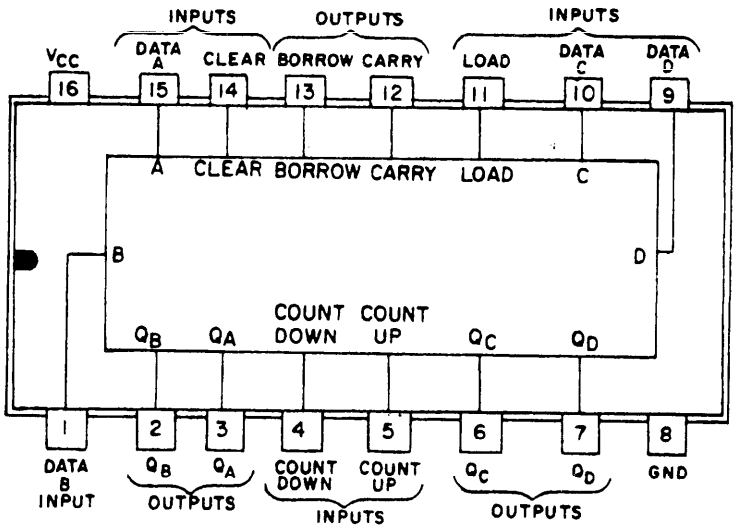


CASES A AND D



DEVICE TYPE 07

CASES E AND F

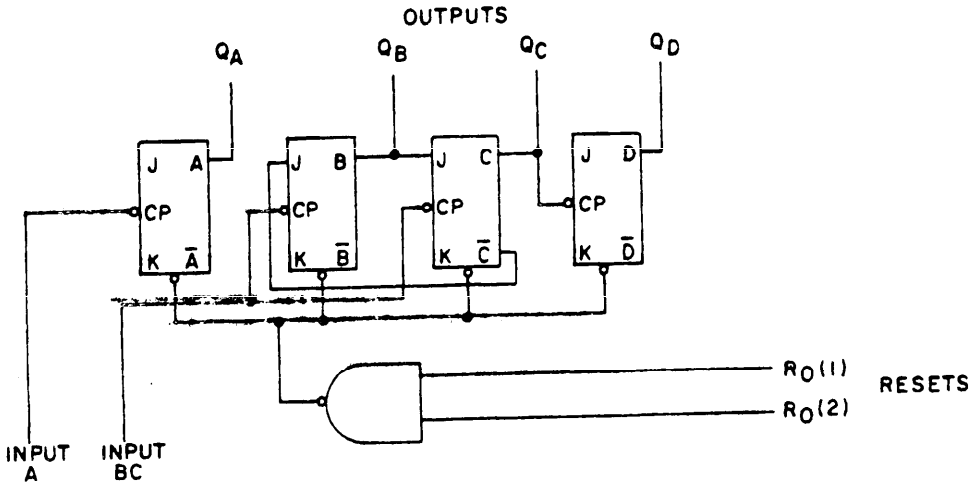


LOGIC: Low input to load sets Q_A = A, Q_B = B, Q_C = C, and Q_D = D

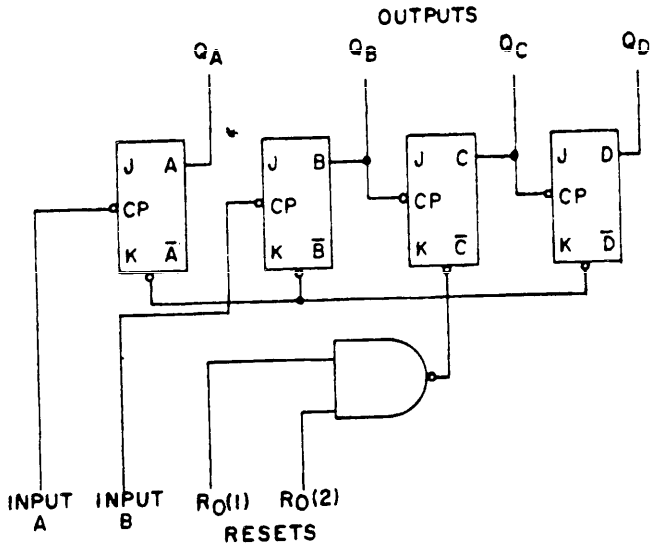
DEVICE TYPES 08 and 09

FIGURE 1. Terminal connections - Continued.

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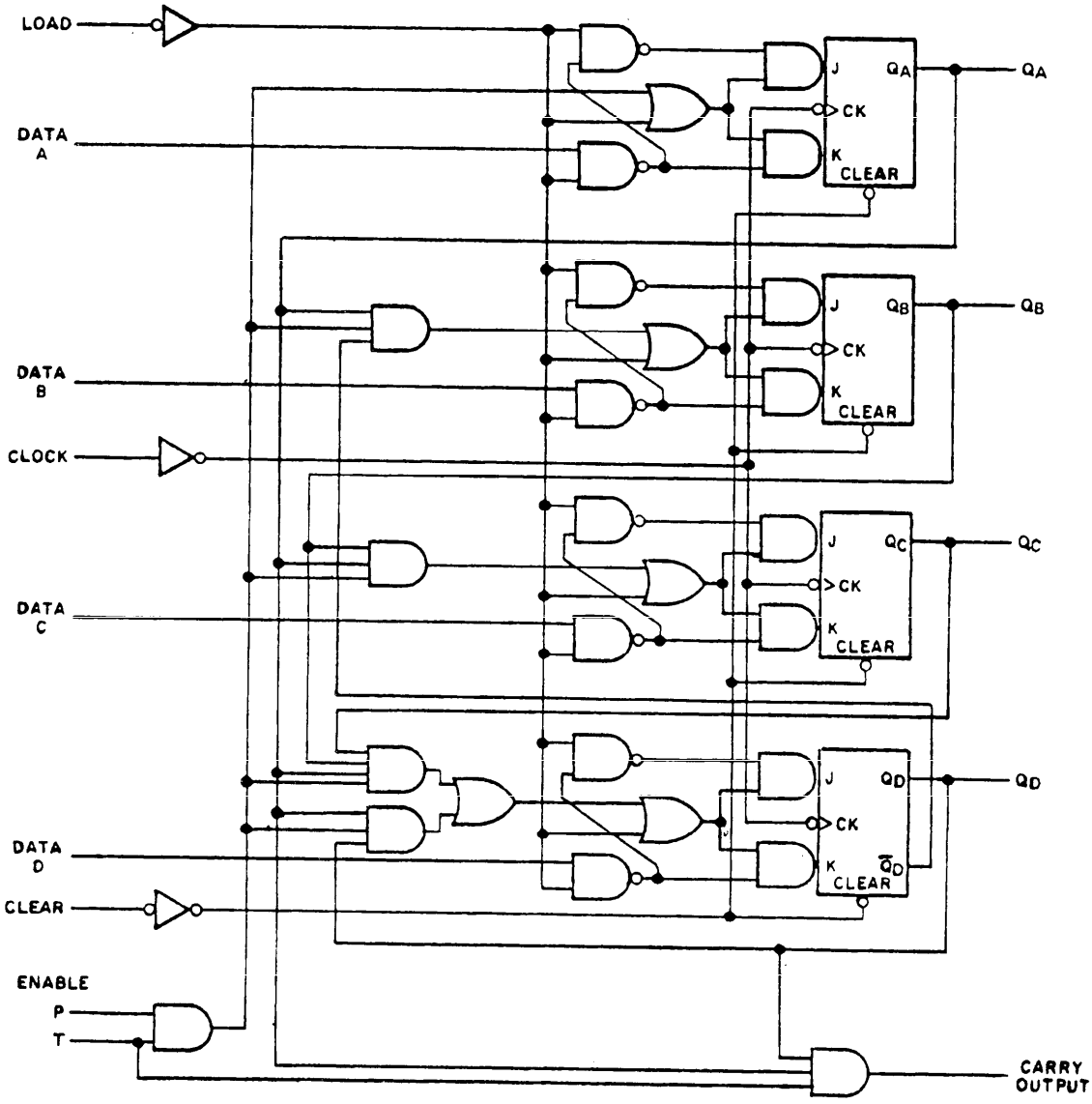
DEVICE TYPE 01



DEVICE TYPE 02

FIGURE 2. Logic diagrams.

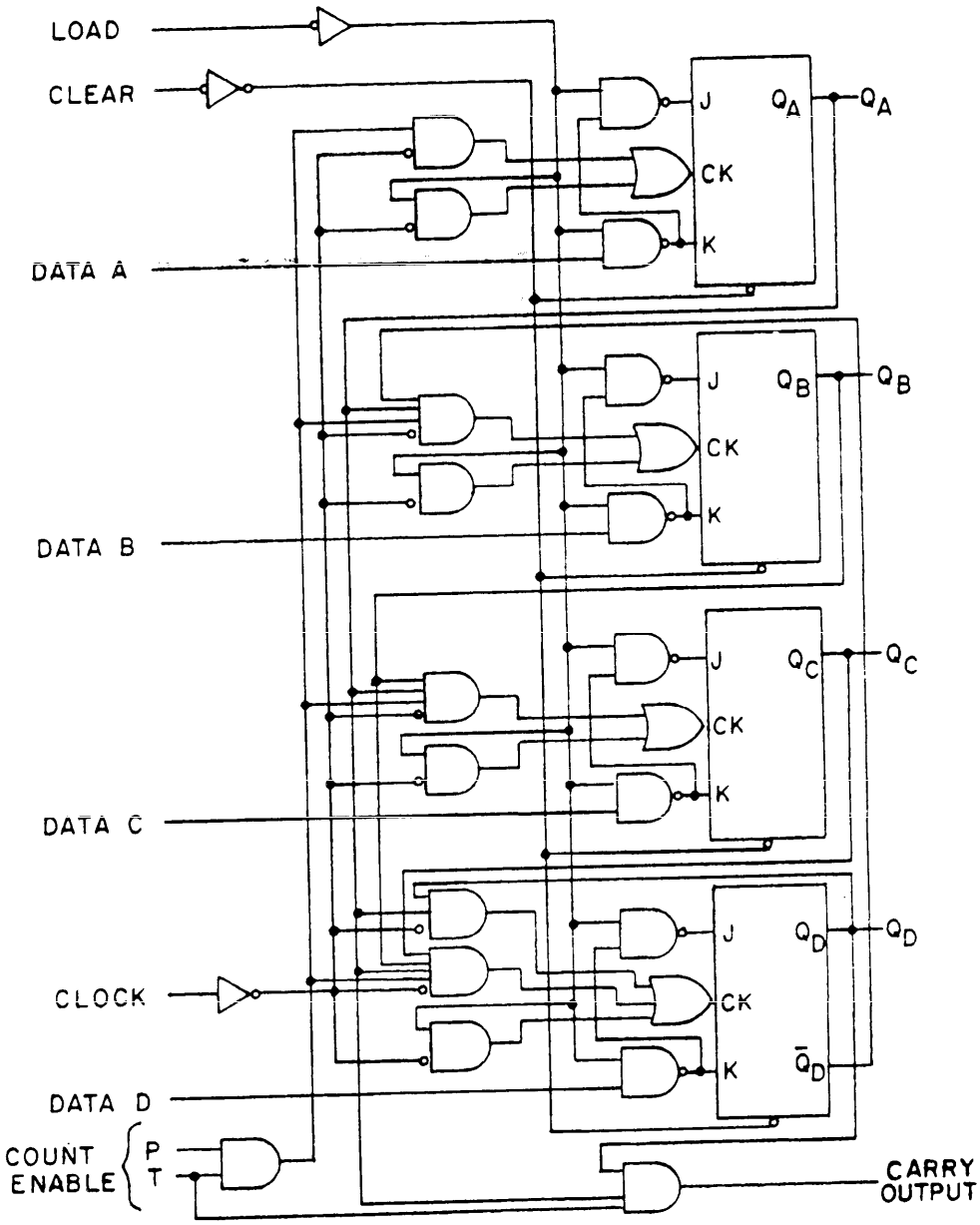
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DEVICE TYPE 03

FIGURE 2. Logic diagrams - Continued.

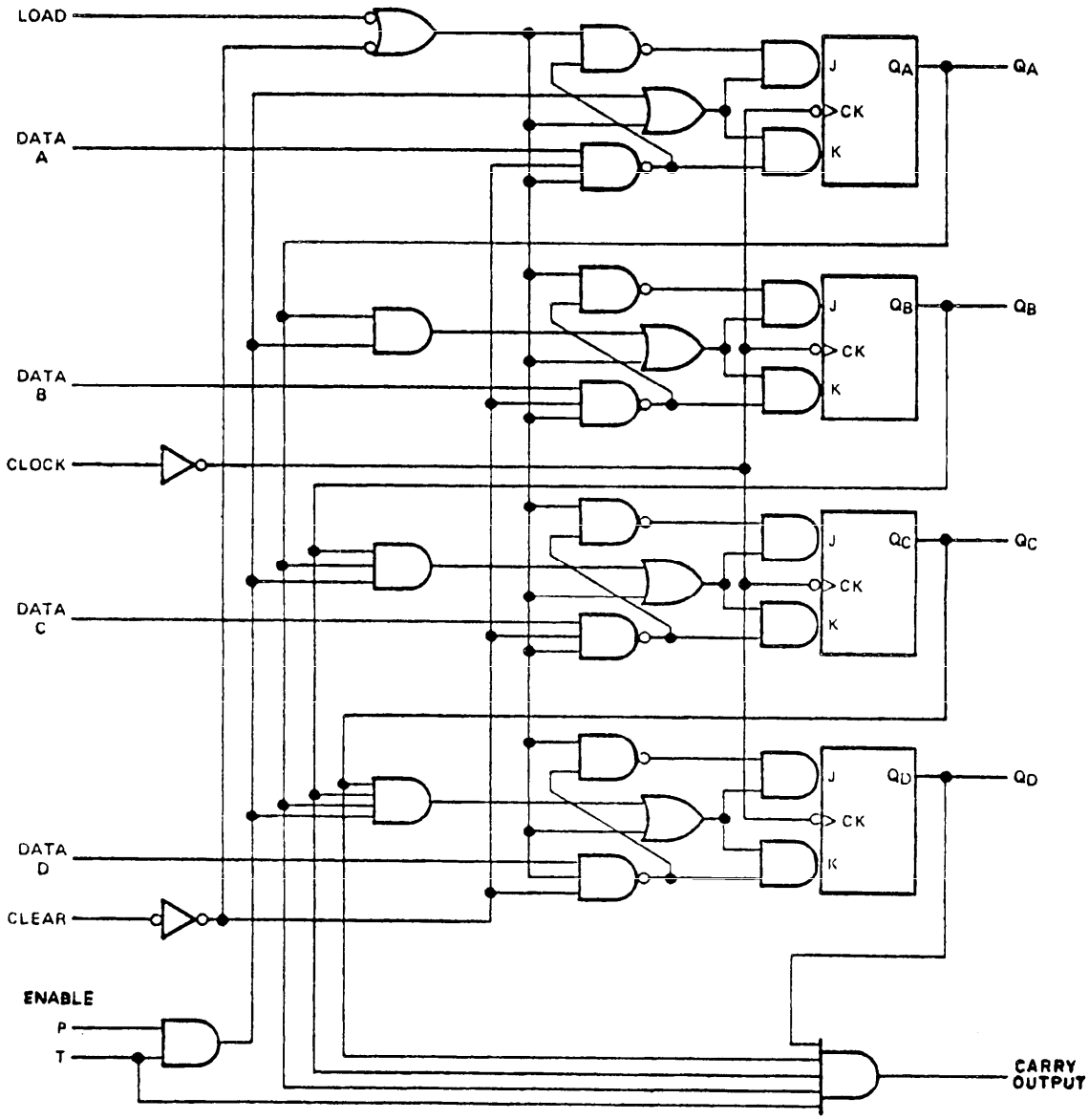
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DEVICE TYPE 03

FIGURE 2. Logic diagrams - Continued.

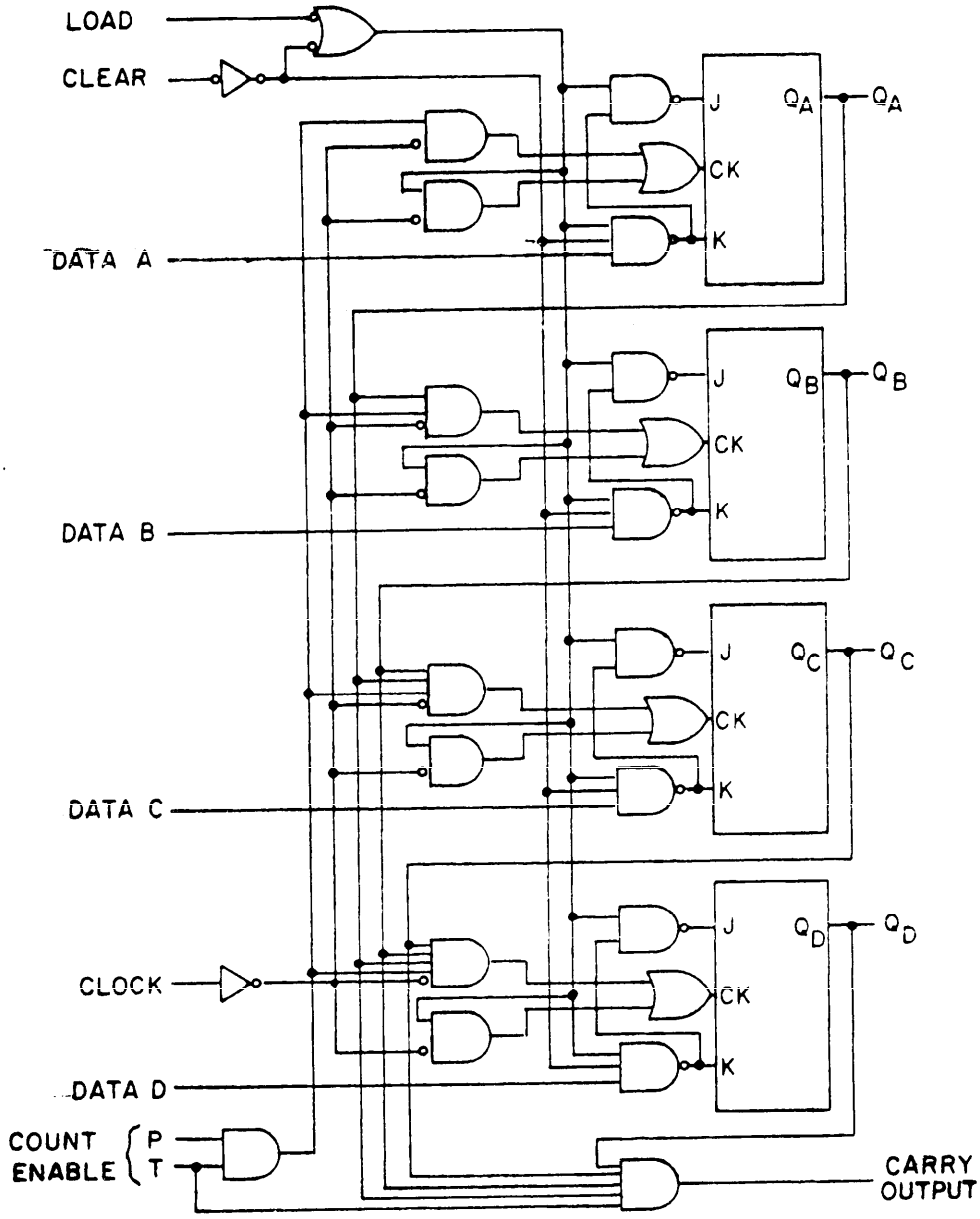
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DEVICE TYPE 04

FIGURE 2. Logic diagrams - Continued.

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DEVICE TYPE 04

FIGURE 2. Logic diagram - Continued.

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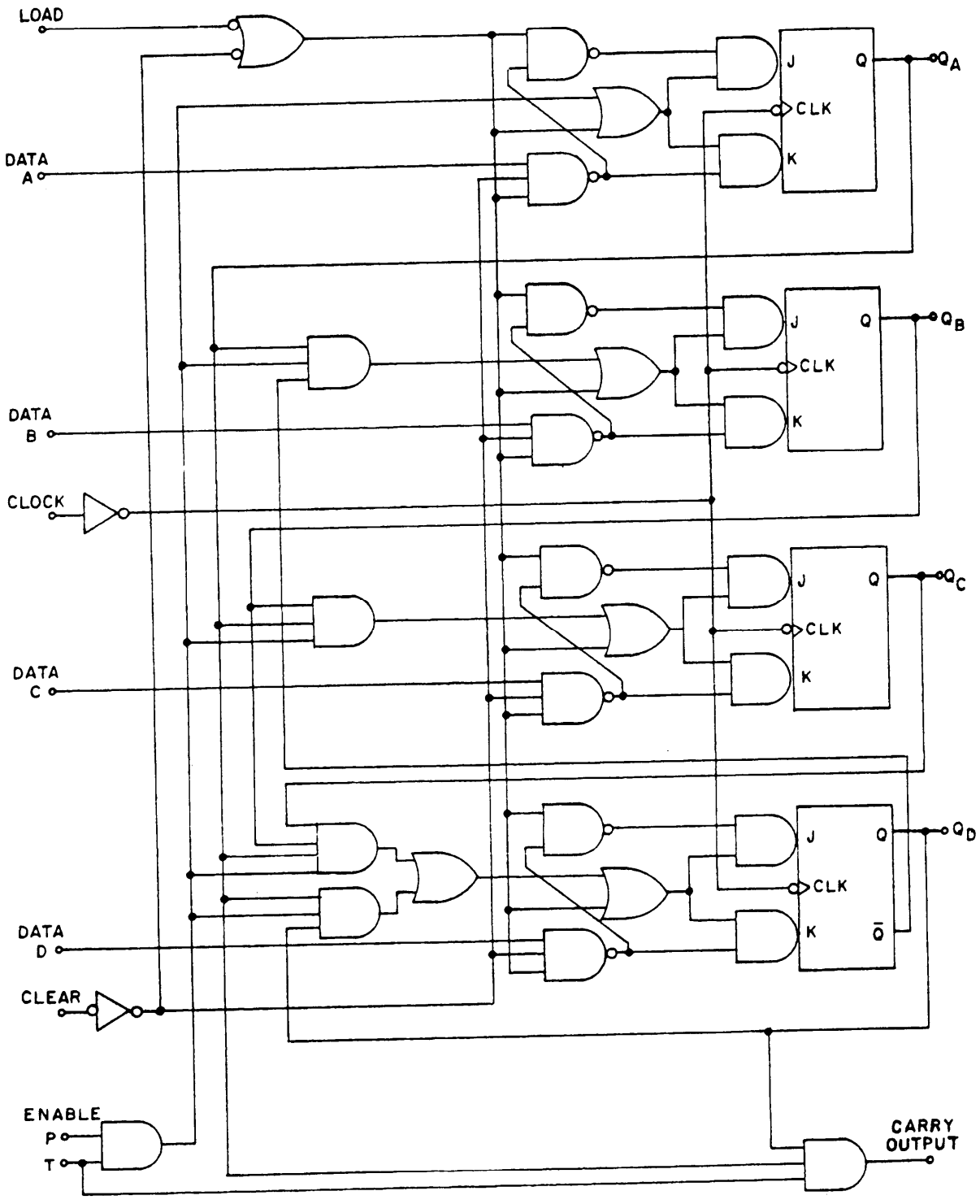
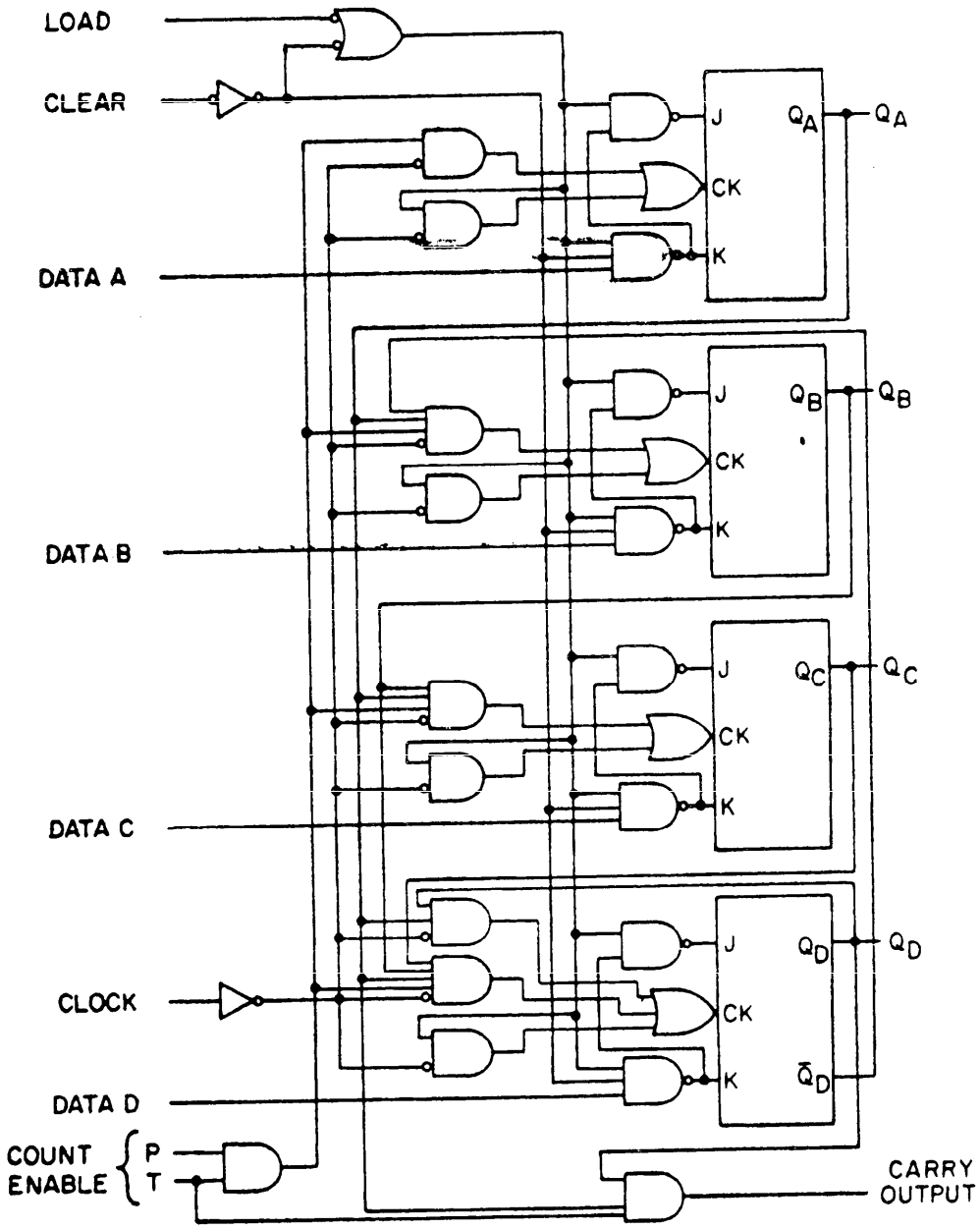


FIGURE 2. Logic diagram - Continued.

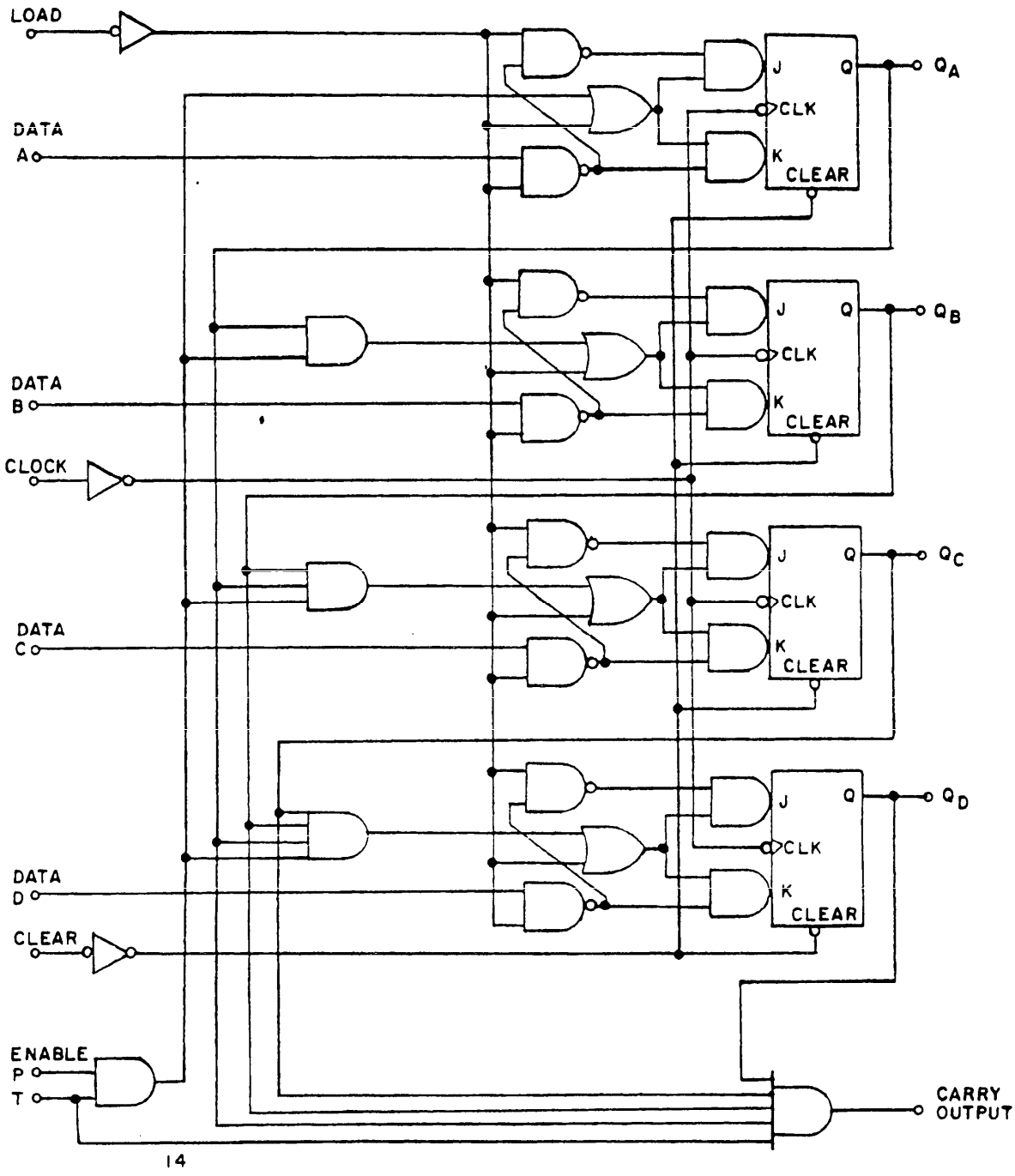
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DEVICE TYPE 05

FIGURE 2. Logic diagrams - Continued.

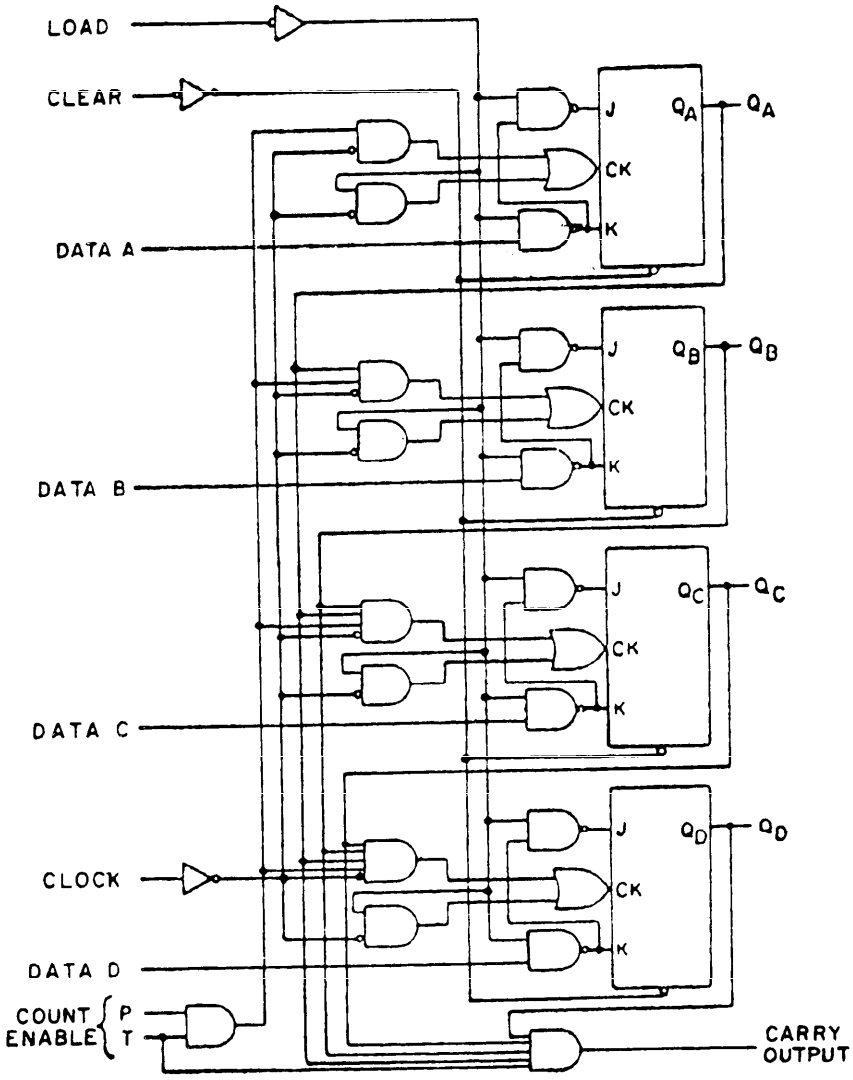
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DEVICE TYPE 06

FIGURE 2. Logic diagrams - Continued.

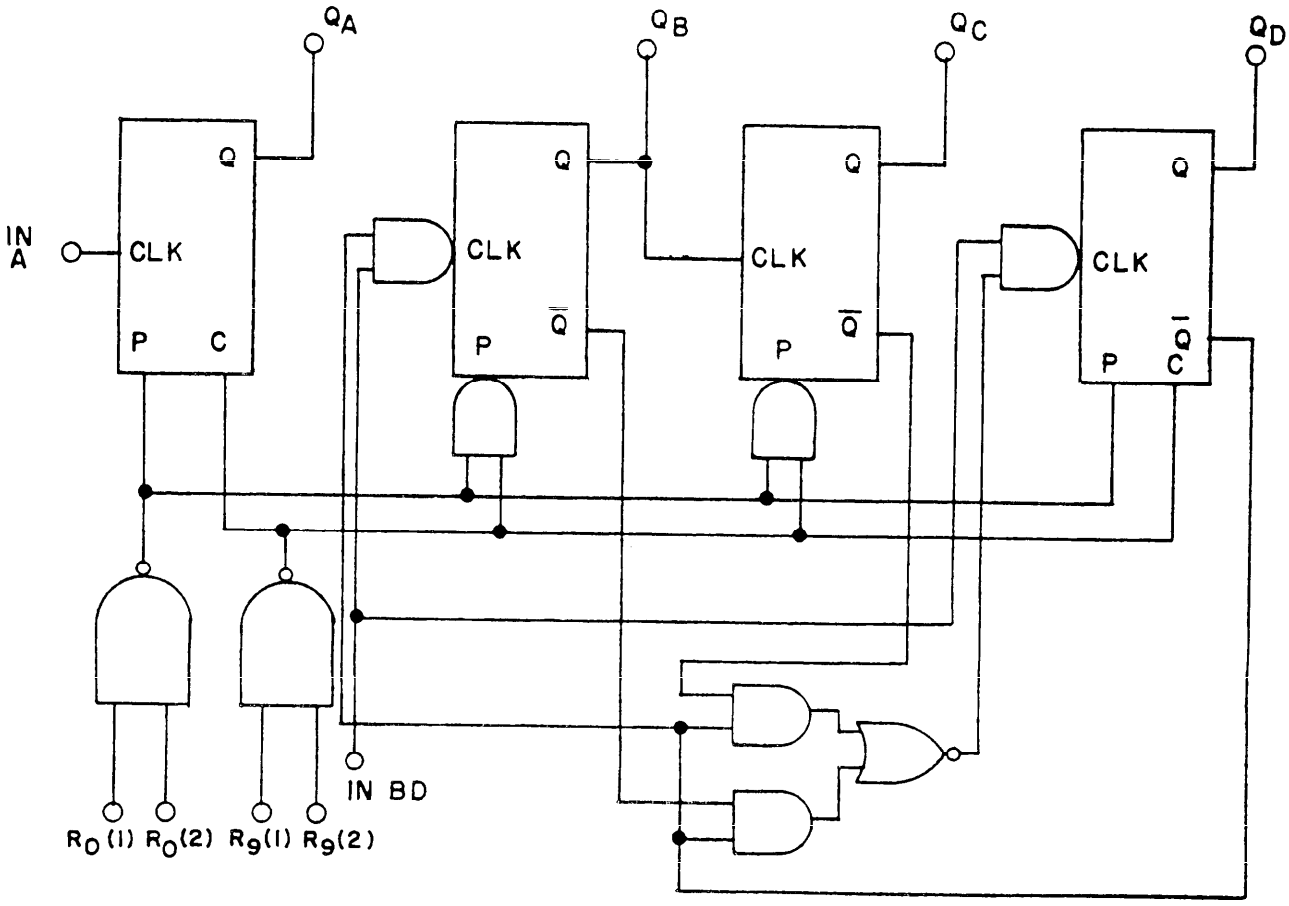
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DEVICE TYPE 06

FIGURE 2. Logic diagrams - Continued.

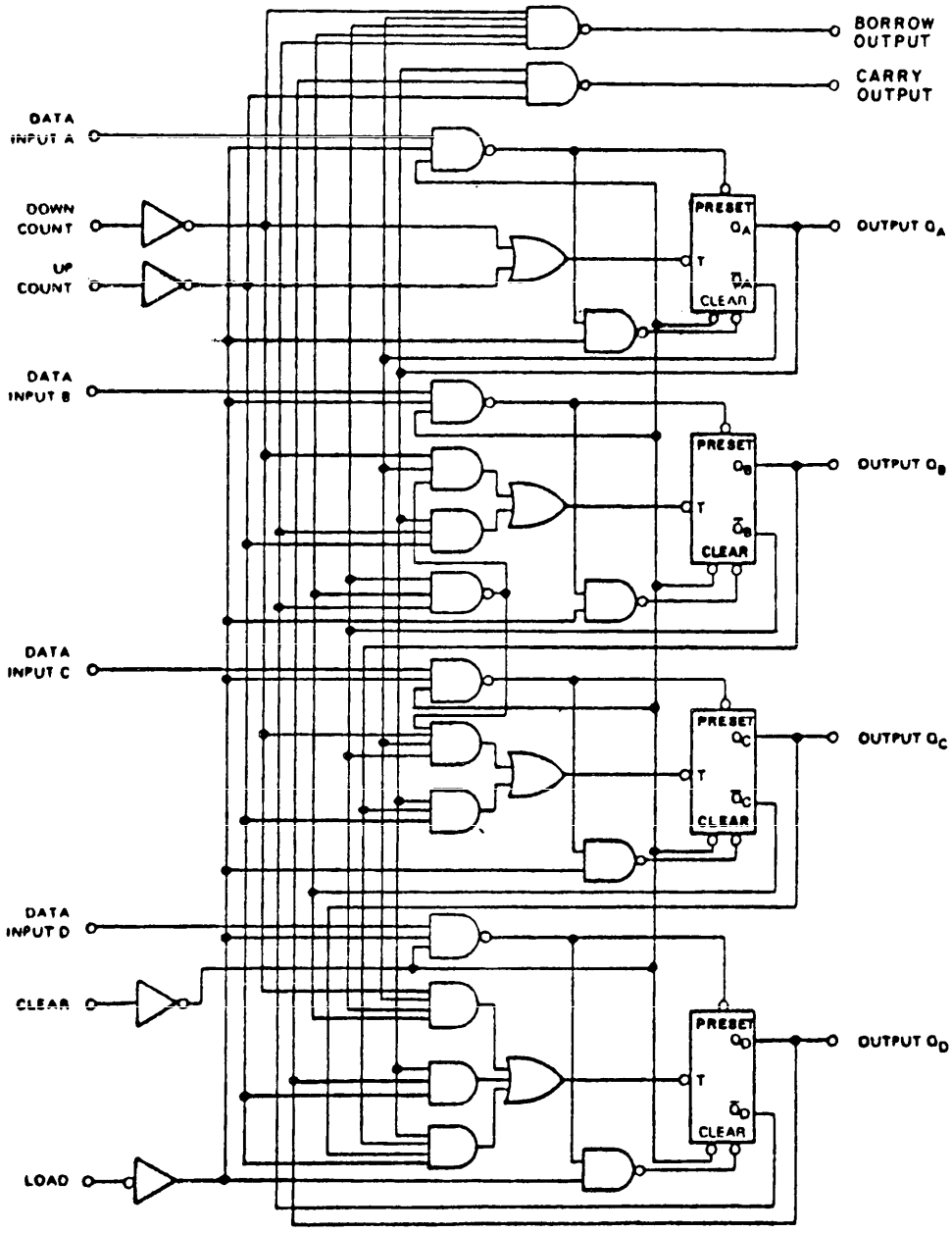
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DEVICE TYPE 07

FIGURE 2. Logic diagram - Continued.

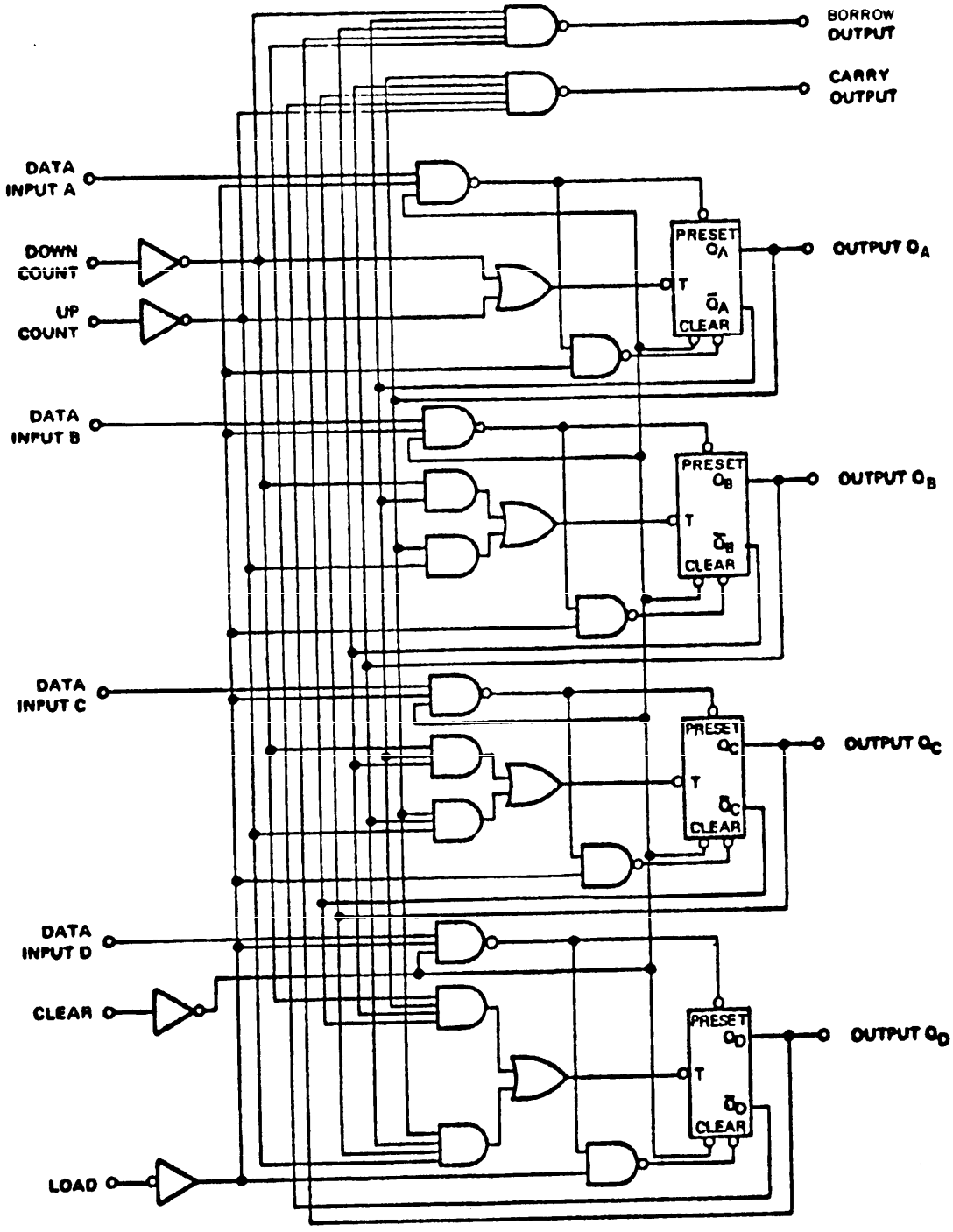
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DEVICE TYPE 08

FIGURE 2. Logic diagram - Continued.

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DEVICE TYPE 09

FIGURE 2. Logic diagram - Continued.

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DEVICE TYPE 01

Count	Output			
	D	C	B	A
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	H	L	L	L
7	H	L	L	H
8	H	L	H	L
9	H	L	H	H
10	H	H	L	L
11	H	H	L	H

DEVICE TYPE 02

Count	Output			
	D	C	B	A
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	L	H	H	L
7	L	H	H	H
8	H	L	L	L
9	H	L	L	H
10	H	L	H	L
11	H	L	H	H
12	H	H	L	L
13	H	H	L	H
14	H	H	H	L
15	H	H	H	H

NOTES FOR TYPE 01:

1. Output A connected to input BC.
2. To reset all outputs to logical L, both $R_{O(1)}$ and $R_{O(2)}$ inputs must be at logical H.
3. Either (or both) reset inputs $R_{O(1)}$ and $R_{O(2)}$ must be at logical L to count.
4. When used as a divide-by-twelve counter, output A must be externally connected to input BC. The input count pulses are applied to input A. Simultaneous divisions of 2, 6, and 12 are performed at the A, C, and D outputs as shown in the truth table above.
5. When used as a divide-by-six counter, the input count pulses are applied to input BC. Simultaneously, frequency divisions of 3 and 6 are available at the C and D outputs. Independent use of flip-flop A is available if the reset function coincides with reset of the divide-by-six counter.

NOTES FOR TYPE 02:

1. Output A connected to input B.
2. To reset all outputs to logical L, both $R_{O(1)}$ and $R_{O(2)}$ inputs must be at logical H.
3. Either (or both) reset inputs $R_{O(1)}$ and $R_{O(2)}$ must be at logical L to count.
4. When used as a 4-bit ripple-through counter, output A must be externally connected to input B. The input count pulses are applied to input A. Simultaneous divisions of 2, 4, 8, and 16 are performed at the A, B, C, and D outputs as shown in the truth table above.
5. When used as a 3-bit ripple-through counter, the input count pulses are applied to input B. Simultaneous frequency divisions of 2, 4, and 8 are available at the B, C, and D outputs. Independent use of flip-flop A is available if the reset function coincides with reset of the 3-bit ripple-through counter.

FIGURE 3. Truth tables.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:51Z
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SYNCHRONOUS TRUTH TABLE, DEVICE TYPES 03 AND 05

Inputs at time t_n									Outputs at time t_{n+1}				
Clock	Enable P	Enable T	Load	A	B	C	D	Clear	Q_A	Q_B	Q_C	Q_D	Carry output
CP	L	X	H	X	X	X	X	H	NC	NC	NC	NC	NC
CP	X	L	H	X	X	X	X	H	NC	NC	NC	NC	L
CP	H	H	H	X	X	X	X	H	Previous count plus 1 (note 1)				H if count = 9 L if count < 9
CP	X	H	L	X	X	X	X	H	A	B	C	D	H if count = 9 L if count < 9
CP	X	L	L	X	X	X	X	H	A	B	C	D	L

ASYNCHRONOUS TRUTH TABLE, DEVICE TYPE 03

Inputs at time t_n									Outputs at time t_{n+1}				
Clock	Enable P	Enable T	Load	A	B	C	D	Clear	Q_A	Q_B	Q_C	Q_D	Carry output
X	X	X	X	X	X	X	X	L	L	L	L	L	L

NOTES:
1. See UP count sequence table.
2. L = V_{IL} for inputs, V_{OL} for outputs.
3. H = V_{IH} for inputs, V_{OH} for outputs.
4. X = V_{IH} or V_{IL}.
5. CP = clock pulse.
6. NC = no change.

UP Count Sequence Table

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)
L	L	L	L
H	L	L	L
L	H	L	L
H	H	L	L
L	L	H	L
H	L	H	L
L	H	H	L
H	H	H	L
L	L	L	H
H	L	L	H

FIGURE 3. Truth tables - Continued.

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SYNCHRONOUS TRUTH TABLE, DEVICE TYPES 04 AND 06

Inputs at time t_n									Outputs at time t_{n+1}				
Clock	Enable P	Enable T	Load	A	B	C	D	Clear	Q_A	Q_B	Q_C	Q_D	Carry output
CP	L	X	H	X	X	X	X	H	NC	NC	NC	NC	NC
CP	X	L	H	X	X	X	X	H	NC	NC	NC	NC	L
CP	H	H	H	X	X	X	X	H	Previous count plus 1 (note 1)				H if count = 15 L if count < 15
CP	X	H	L	X	X	X	X	H	A	B	C	D	H if count = 15 L if count < 15
CP	X	L	L	X	X	X	X	H	A	B	C	D	L

ASYNCHRONOUS TRUTH TABLE, DEVICE TYPE 06

Inputs at time t_n									Outputs at time t_{n+1}				
Clock	Enable P	Enable T	Load	A	B	C	D	Clear	Q_A	Q_B	Q_C	Q_D	Carry output
X	X	X	X	X	X	X	X	L	L	L	L	L	L

NOTES:
1. See UP count sequence table.
2. L = V_{IL} for inputs, V_{OL} for outputs.
3. H = V_{IH} for inputs, V_{OH} for outputs.
4. X = V_{IH} or V_{IL}.
5. CP = clock pulse.
6. NC = no change.

UP Count Sequence Table

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)
L	L	L	L
H	L	L	L
L	H	L	L
H	H	L	L
L	L	H	L
H	L	H	L
L	H	H	L
H	H	H	L
L	L	L	H
H	L	L	H
L	H	L	H
H	H	L	H
L	L	H	H
H	L	H	H
L	H	H	H
H	H	H	H

FIGURE 3. Truth tables - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:51Z
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DEVICE TYPE 07

BCD count sequence (see note 1)				
	Output			
Count	D	C	B	A
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	L	H	H	L
7	L	H	H	H
8	H	L	L	L
9	H	L	L	H

Reset count (see note 2)							
Reset inputs				Output			
$R_{0(1)}$	$R_{0(2)}$	$R_{9(1)}$	$R_{9(2)}$	D	C	B	A
H	H	L	X	L	L	L	L
H	H	X	L	L	L	L	L
X	X	H	H	H	L	L	H
X	L	X	L	COUNT			
L	X	L	X	COUNT			
L	X	X	L	COUNT			
X	L	L	X	COUNT			

NC - No internal connection.

NOTES:

1. Output A connected to input BD for BCD count.
2. X indicates that either a logical H or a logical L may be present.
3. When used as a binary coded decimal decade counter, the BD input must be externally connected to the A output. The A input receives the incoming count, and a count sequence is obtained in accordance with the BCD count sequence truth table shown above. In addition to a conventional zero reset, inputs are provided to reset a BCD count for nine's complement decimal applications.
4. If a symmetrical divide-by-ten count is desired for frequency synthesizers or other applications requiring division of a binary count by a power of ten, the D output must be externally connected to the A input. The input count is then applied at the BD input and a divide-by-ten square wave is obtained at output A.
5. For operation as a divide-by-two counter and a divide-by-five counter, no external interconnections are required. Flip-flop A is used as a binary element for the divide-by-two function. The BD input is used to obtain binary divide-by-five operation at the B, C, and D outputs. In this mode, the two counters operate independently; however, all four flip-flops are reset simultaneously.

FIGURE 3. Truth tables - Continued.

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DEVICE TYPE 08 TRUTH TABLE

Inputs at time t_n								Outputs at time T_n+1							
Count up	Count down	Load	A	B	C	D	Clear	Q_A	Q_B	Q_C	Q_D	Carry		Borrow	
H	H	H	X	X	X	X	L	NC	NC	NC	NC	H		H	
H	H	H	X	X	X	X	H	L	L	L	L	H		H	
H	H	L	X	X	X	X	L	A	B	C	D	Count up L if count = 9 H if count > 9	Count down NC	Count up NC	Count down L if count = 0 H if count ≠ 0
P	H	H	X	X	X	X	L	Previous count plus 1 (note 1)				L if count = 9 H if count > 9	NA	NC	NA
H	P	H	X	X	X	X	L	Previous count minus 1 (note 2)				NA	NC	NA	L is count = 0 H if count ≠ 0

NOTES:
1. See up count sequence table.
2. See down count sequence table.
3. L = V_{IL} for inputs, V_{OL} for outputs.
4. H = V_{IH} for inputs, V_{OH} for outputs.
5. X = V_{IH} or V_{IL}.
6. NC = no change.
7. NA = not applicable.
8. P = Positive going pulse.

FIGURE 3. Truth tables - Continued.

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DEVICE TYPE 08

UP COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)	Carry
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	L

DOWN COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)	Borrow
H	L	L	H	H
L	L	L	H	H
H	H	H	L	H
L	H	H	L	H
H	L	H	L	H
L	L	H	L	H
H	H	L	L	H
L	H	L	L	H
H	L	L	L	H
L	L	L	L	L

FIGURE 3. Truth tables - Continued.

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DEVICE TYPE 09 TRUTH TABLE

Inputs at time t_n								Outputs at time T_{n+1}							
Count up	Count down	Load	A	B	C	D	Clear	Q_A	Q_B	Q_C	Q_D	Carry		Borrow	
H	H	H	X	X	X	X	L	NC	NC	NC	NC	H		H	
H	H	H	X	X	X	X	H	L	L	L	L	H		H	
H	H	L	X	X	X	X	L	A	B	C	D	Count up L if count = 15 H if count > 15	Count down NC	Count up NC	Count down L if count = 0 H if count ≠ 0
P	H	H	X	X	X	X	L	Previous count plus 1 (note 1)				L if count = 15 H if count > 15	NA	NC	NA
H	P	H	X	X	X	X	L	Previous count minus 1 (note 2)				NA	NC	NA	L is count = 0 H if count ≠ 0

NOTES:
1. See up count sequence table.
2. See down count sequence table.
3. L = V_{IL} for inputs, V_{OL} for outputs.
4. H = V_{IH} for inputs, V_{OH} for outputs.
5. X = V_{IH} or V_{IL}.
6. NC = no change.
7. NA = not applicable.
8. P = Positive going pulse.

FIGURE 3. Truth tables - Continued.

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DEVICE TYPE 09

UP COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)	Carry
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	H
L	H	L	H	H
H	H	L	H	H
L	L	H	H	H
H	L	H	H	H
L	H	H	H	H
H	H	H	H	L

DOWN COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)	Borrow
H	H	H	H	H
L	H	H	H	H
H	L	H	H	H
L	L	H	H	H
H	H	L	H	H
L	H	L	H	H
H	L	L	H	H
L	L	L	H	H
H	H	H	L	H
L	H	H	L	H
H	L	H	L	H
L	L	H	L	H
H	H	L	L	H
L	H	L	L	H
H	L	L	L	H
L	L	L	L	L

FIGURE 3. Truth tables - Continued.

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DEVICE TYPES 03 AND 05: SYNCHRONOUS DECADE COUNTERS

typical clear, preset, count, and inhibit sequences

Illustrated below is the following sequence:
1. Clear outputs to zero.
2. Preset to BCD seven.
3. Count to eight, nine, zero, one, two, and three.
4. Inhibit.

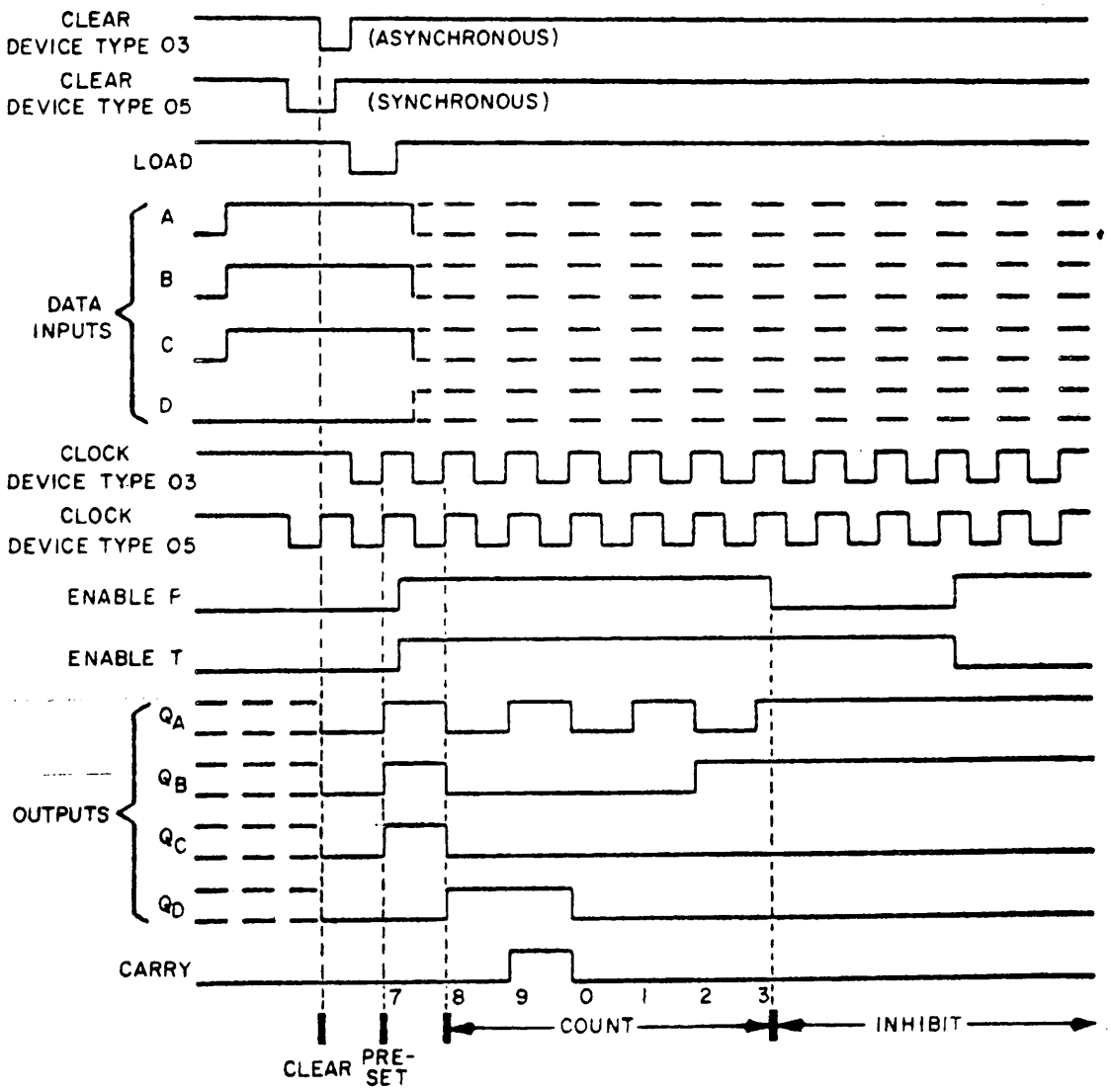


FIGURE 4. Timing diagrams.

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DEVICE TYPES 04 AND 06: SYNCHRONOUS BINARY COUNTERS

typical clear, preset, count, and inhibit sequences.

Illustrated below is the following sequence:

1. Clear outputs to zero.
2. Preset to binary twelve.
3. Count to thirteen, fourteen, fifteen, zero, one, and two.
4. Inhibit.

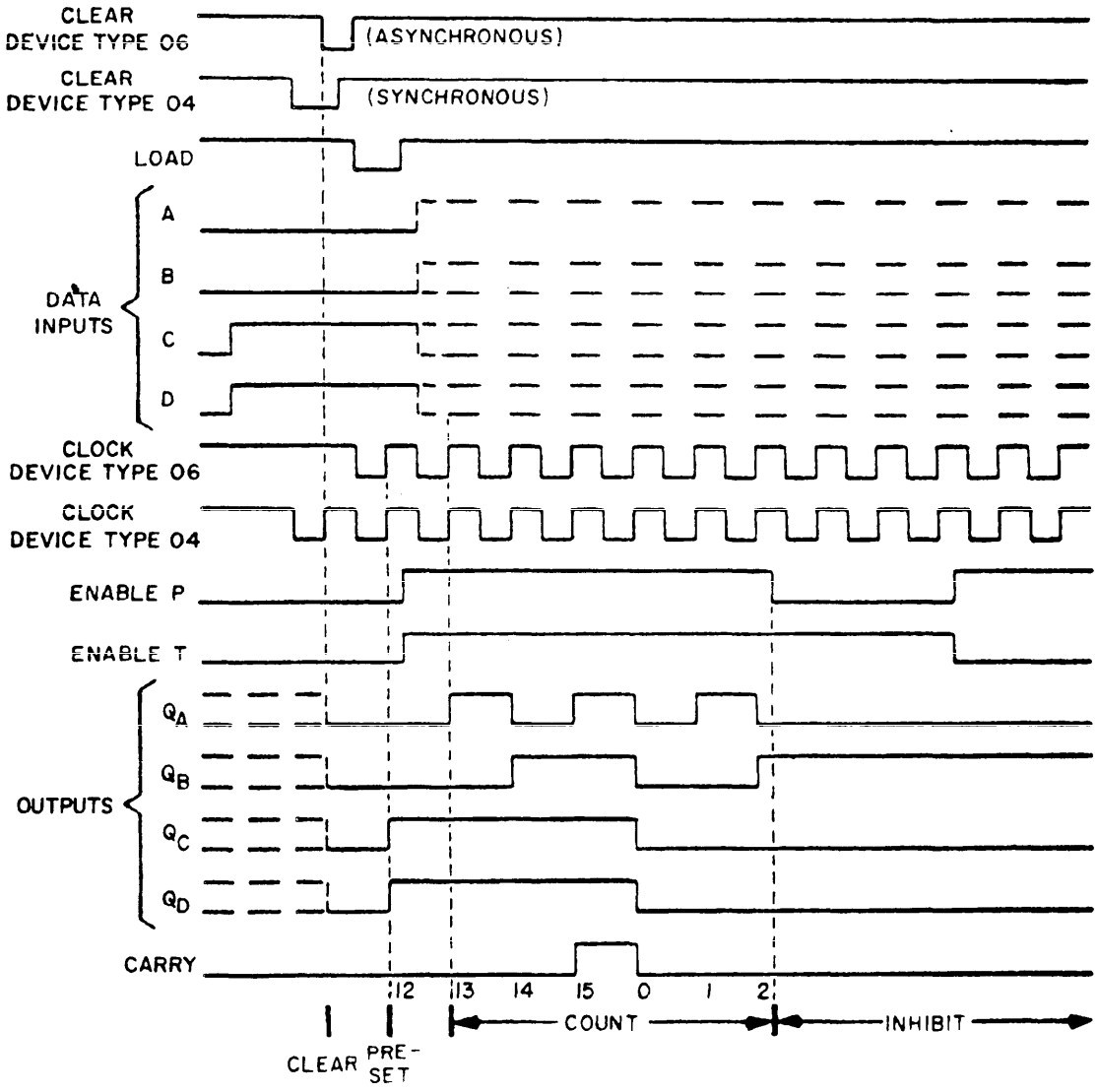
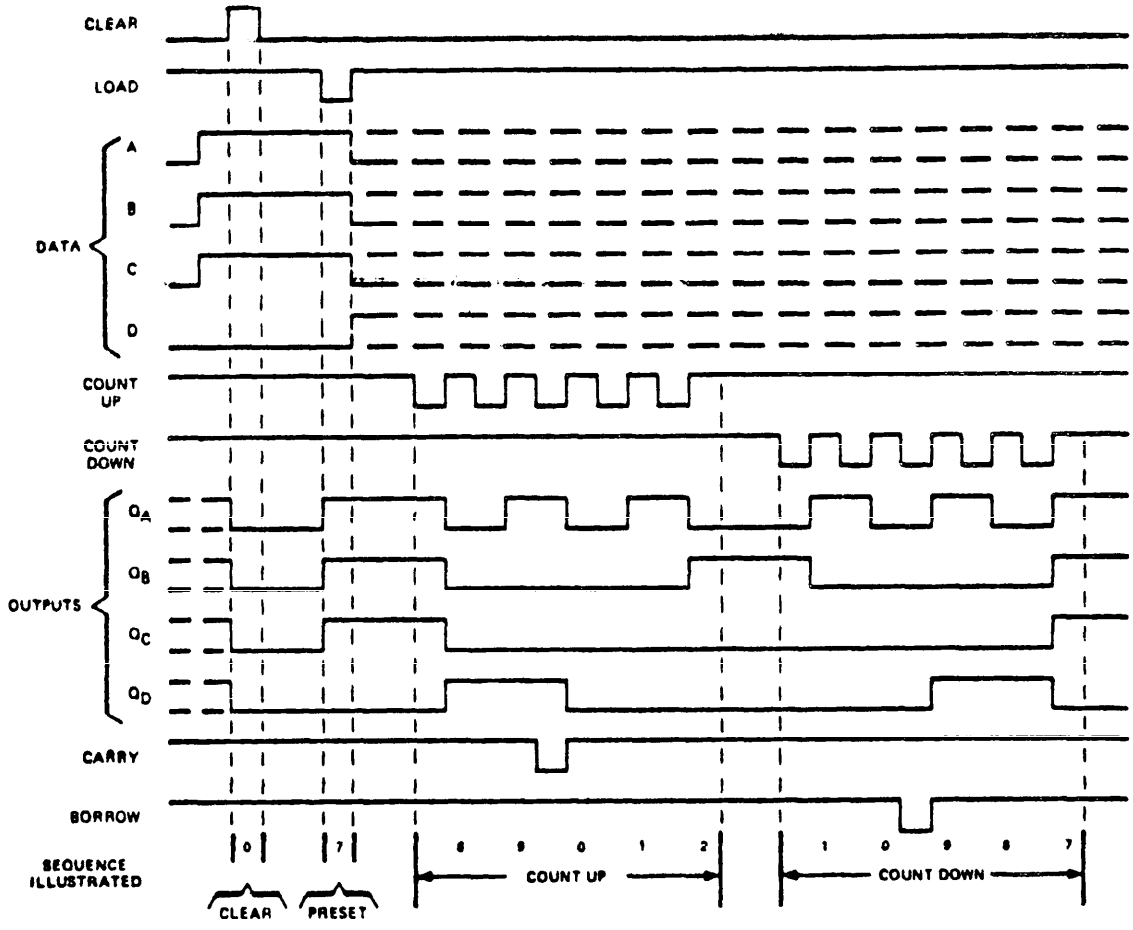


FIGURE 4. Timing diagrams - Continued.

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typical clear, load, and count sequences

Illustrated below is the following sequence:
1. Clear outputs to zero.
2. Load (preset) to BCD seven.
3. Count up to eight, nine, carry, zero, one, and two.
4. Count down to one, zero, borrow, nine, eight, and seven.



DEVICE TYPE 08 SYNCHRONOUS DECADE COUNTER

NOTES:
A. Clear overrides load, data, and count inputs.
B. When counting up, count down input must be high; when counting down, count-up input must be high.

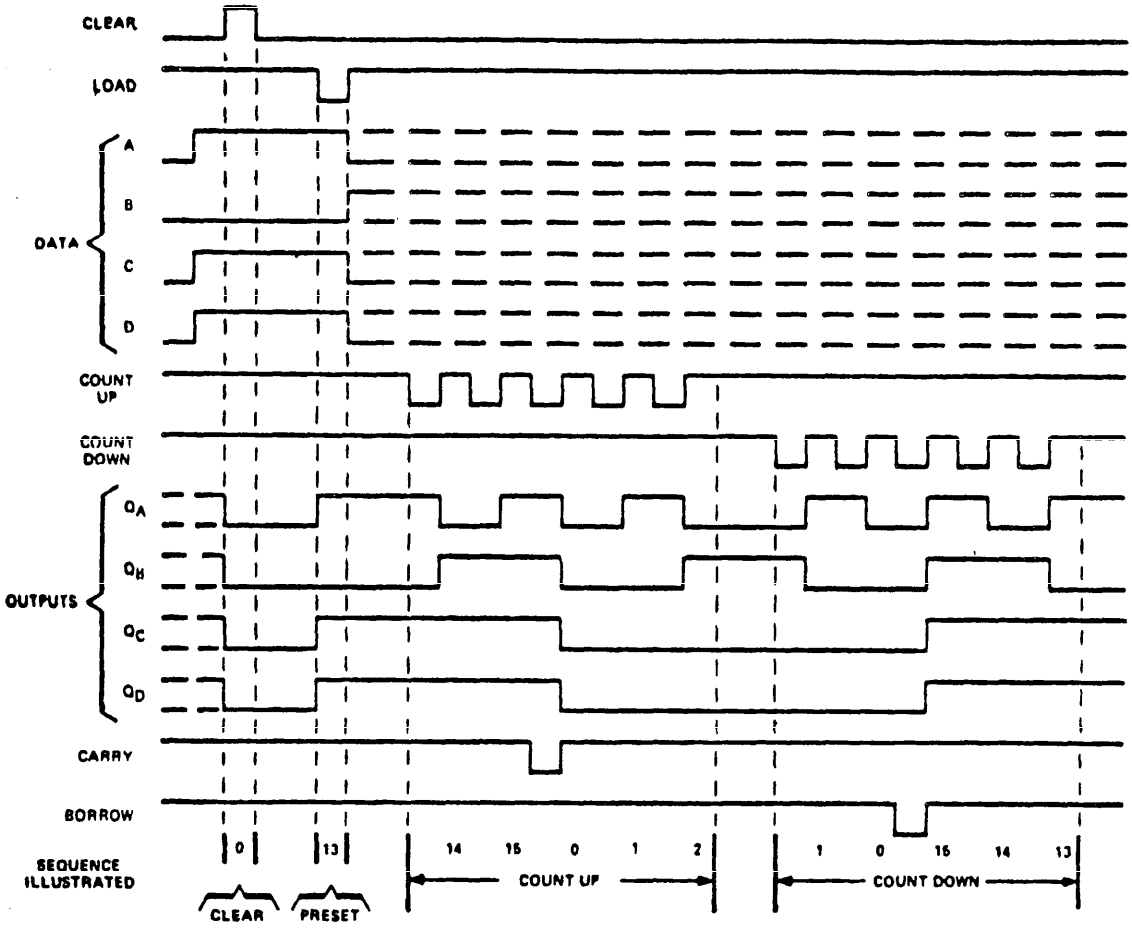
FIGURE 4. Timing diagrams - Continued.

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8/ Unit deselected less than 55 ns prior to selection.

typical clear, load, and count sequences

Illustrated below is the following sequence:
1. Clear outputs to zero.
2. Load (preset) to binary thirteen.
3. Count up to fourteen, fifteen, carry, zero, one, and two.
4. Count down to one, zero, borrow, fifteen, fouteen, and thirteen.



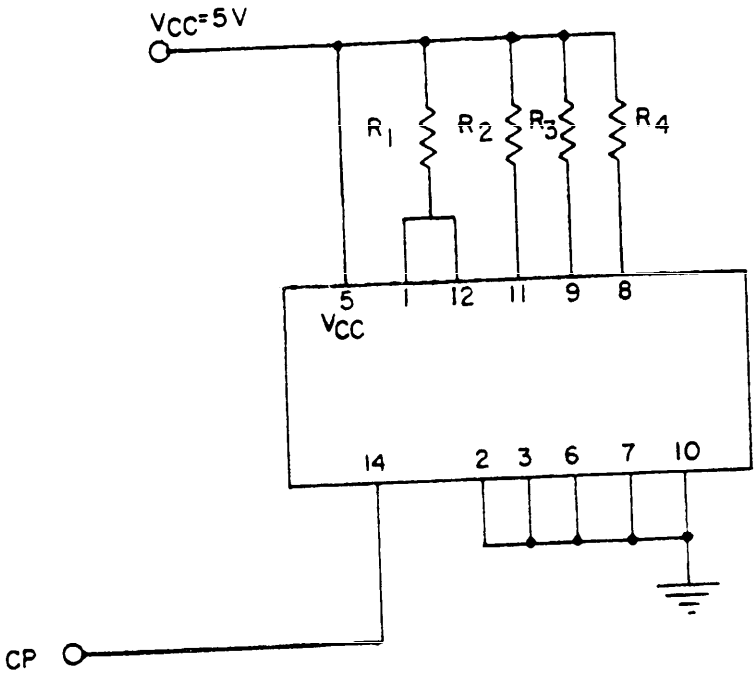
DEVICE TYPE 09: SYNCHRONOUS BINARY COUNTER

NOTES:
A. Clear overrides load, data, and count inputs.
B. When counting up, count-down input must be high; when counting down, count-up input must be high.

FIGURE 4. Timing diagrams - Continued.

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Device types 01, 02, and 07

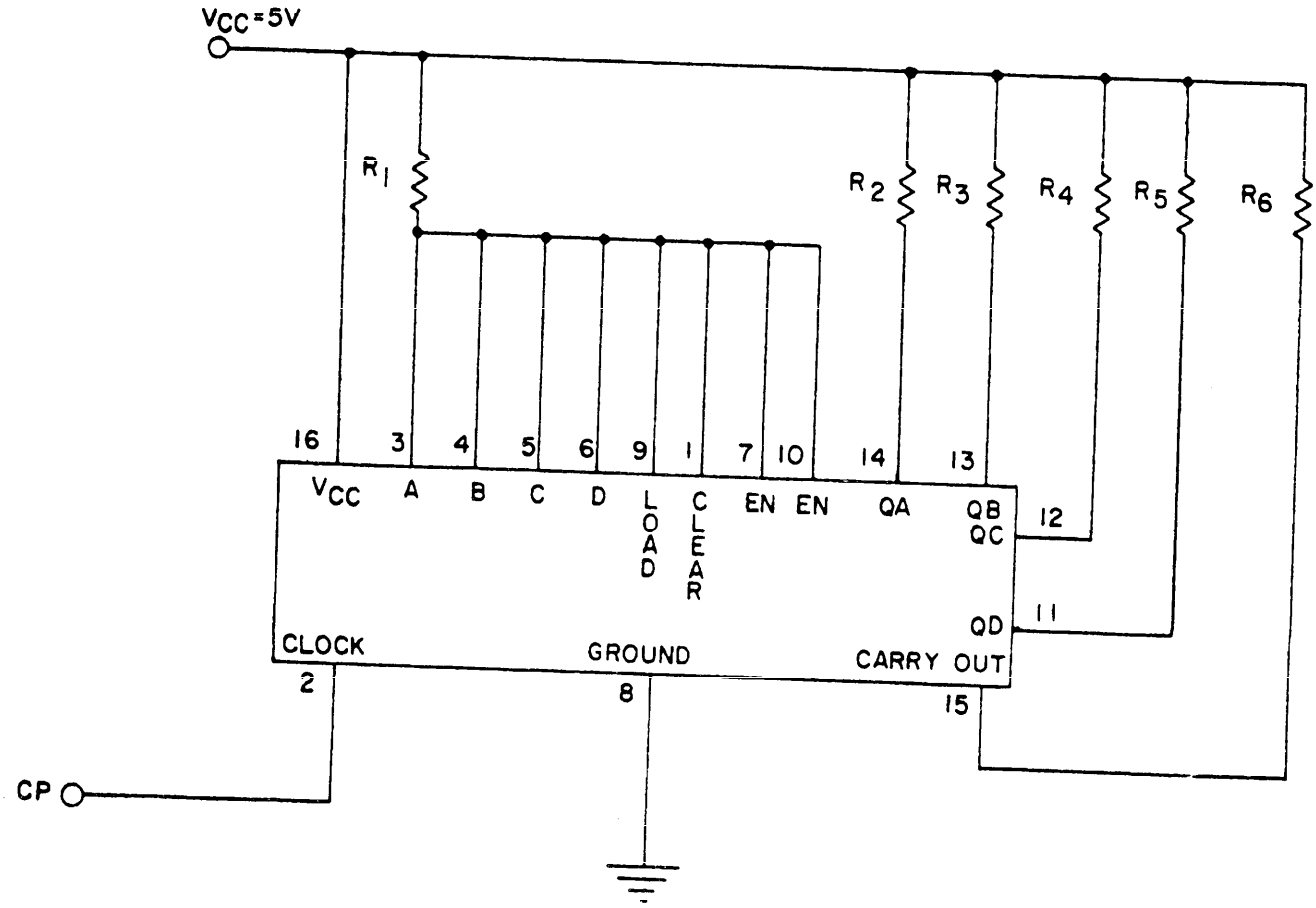


CP: 100 kHz square wave; 0-3 V
R_1 thru R_4 = 220 Ω ± 5%

FIGURE 5. Burn-in and life test circuits.

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Device types 03, 04, 05, and 06

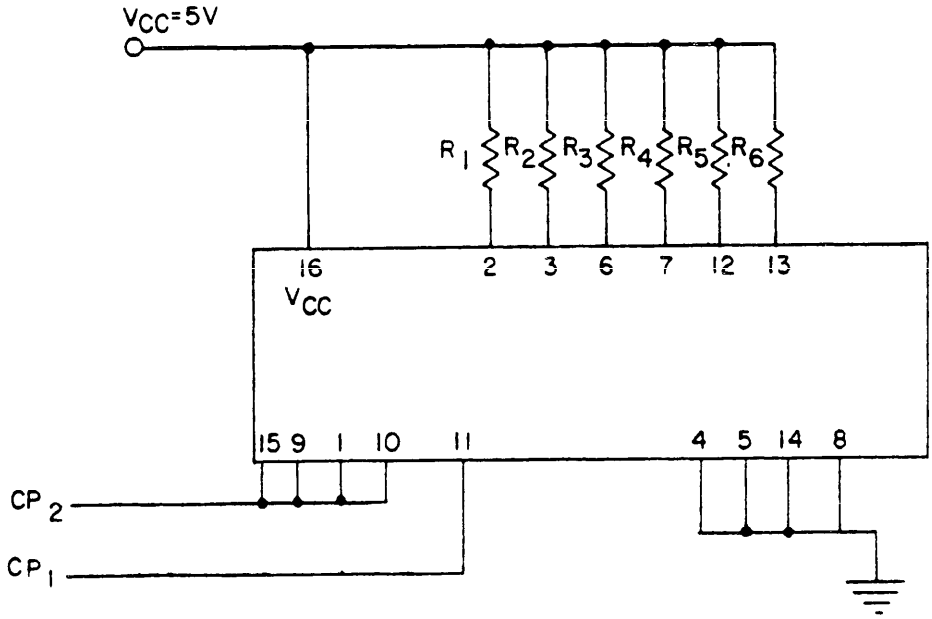


CP: 100 kHz square wave; 0-3 V
R_1 thru R_6 = 270 Ω ±5%

FIGURE 5. Burn-in and life test circuits - Continued.

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Device types 08 and 09



R_1 thru R_6 = 270 Ω ±5%

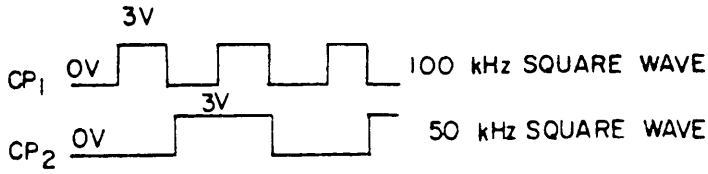
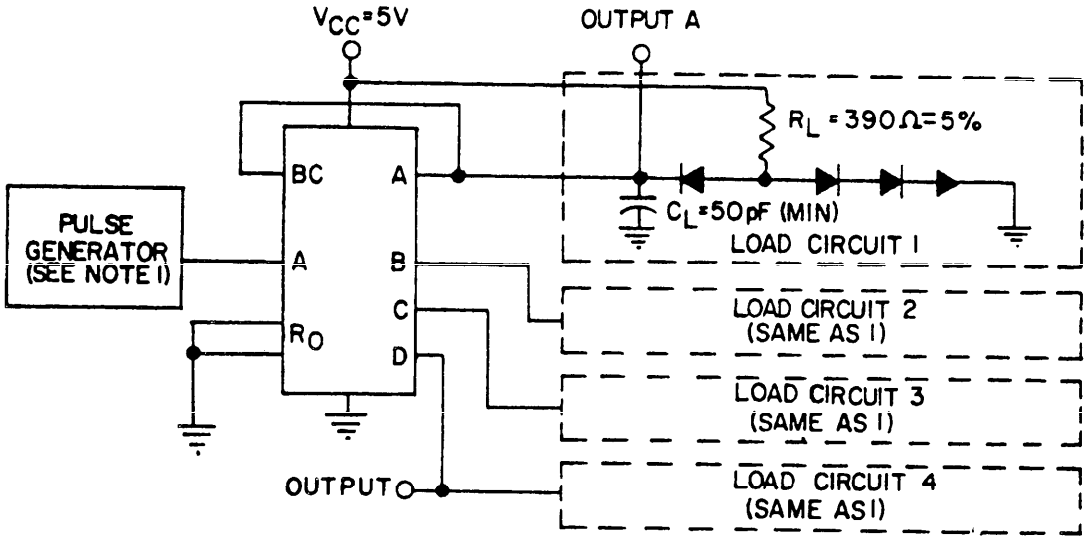
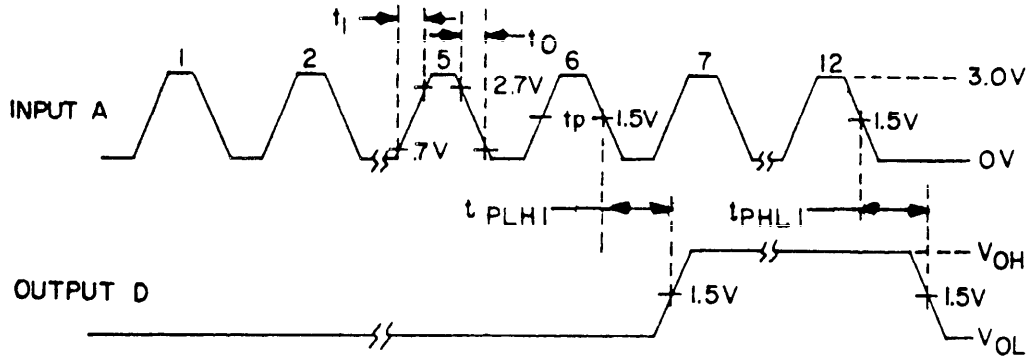


FIGURE 5. Burn-in and life test circuits - Continued.

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TEST CIRCUIT

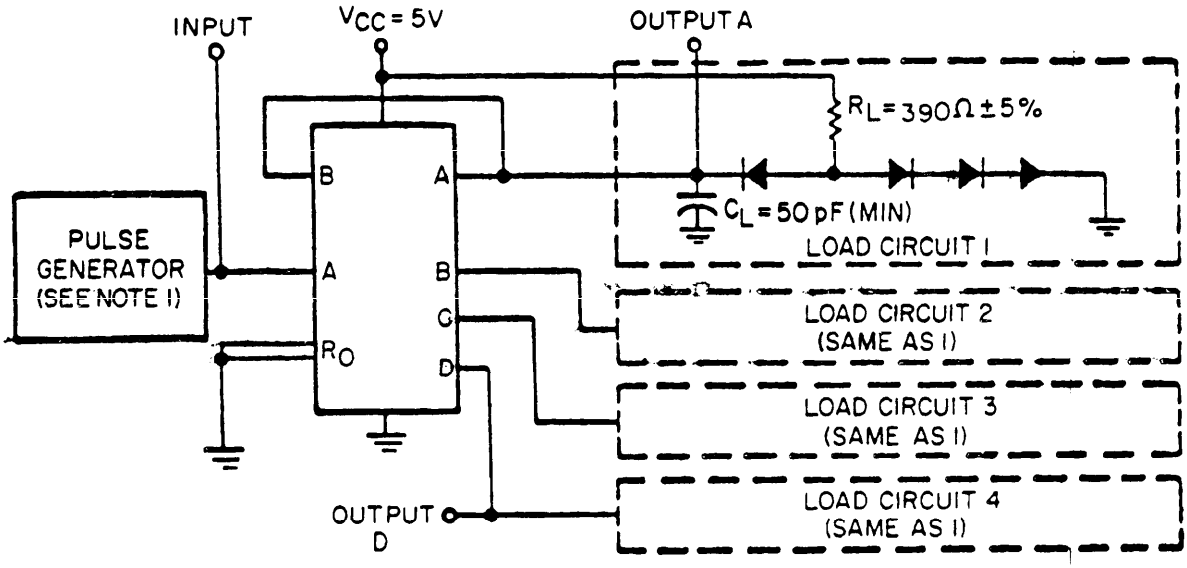


VOLTAGE WAVEFORMS

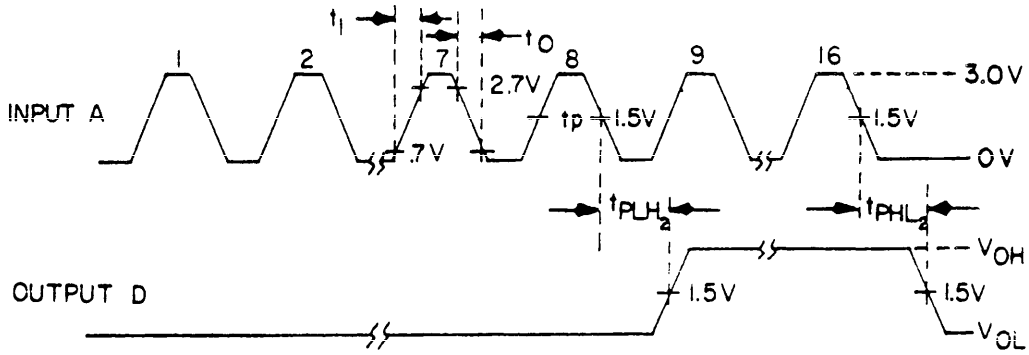
NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 15$ ns, t_p = .5 µs, PRR = 1 MHz, $Z_{out} \approx 50\Omega$.
2. All diodes are 1N3064, or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground terminal.
5. F_{MAX} pulse generator has the following characteristics: V_{gen} = 3.0 V ±0.2 V, $t_0 = t_1 \leq 10$ ns, t_p = 50 ns, PRR = 10 MHz.

FIGURE 6. Switching time test circuits and waveforms for device type 01.

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TEST CIRCUIT

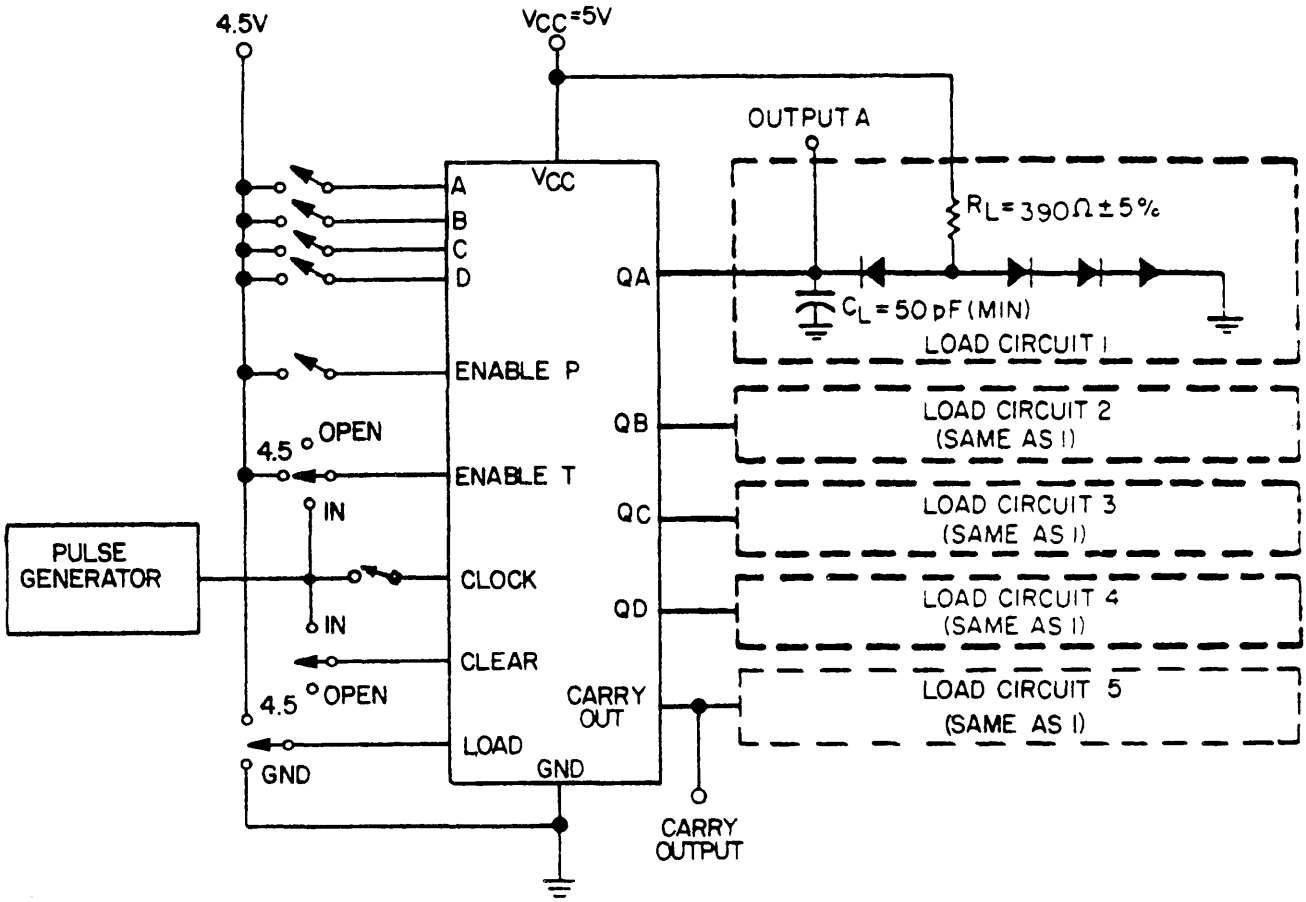


VOLTAGE WAVEFORMS

NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, $t_0 = t_1$ < 15 ns, t_p = .5 µs, PRR = 1 MHz, $Z_{out} \approx 50\Omega$.
2. All diodes are 1N3064 or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground terminal.
5. F_{MAX} pulse generator has the following characteristics: V_{gen} = 3.0 V ±0.2 V, $t_0 = t_1 \leq 10$ ns, t_p = 50 ns, PRR = 10 MHz.

FIGURE 7. Switching time test circuit and waveforms for device type 02.

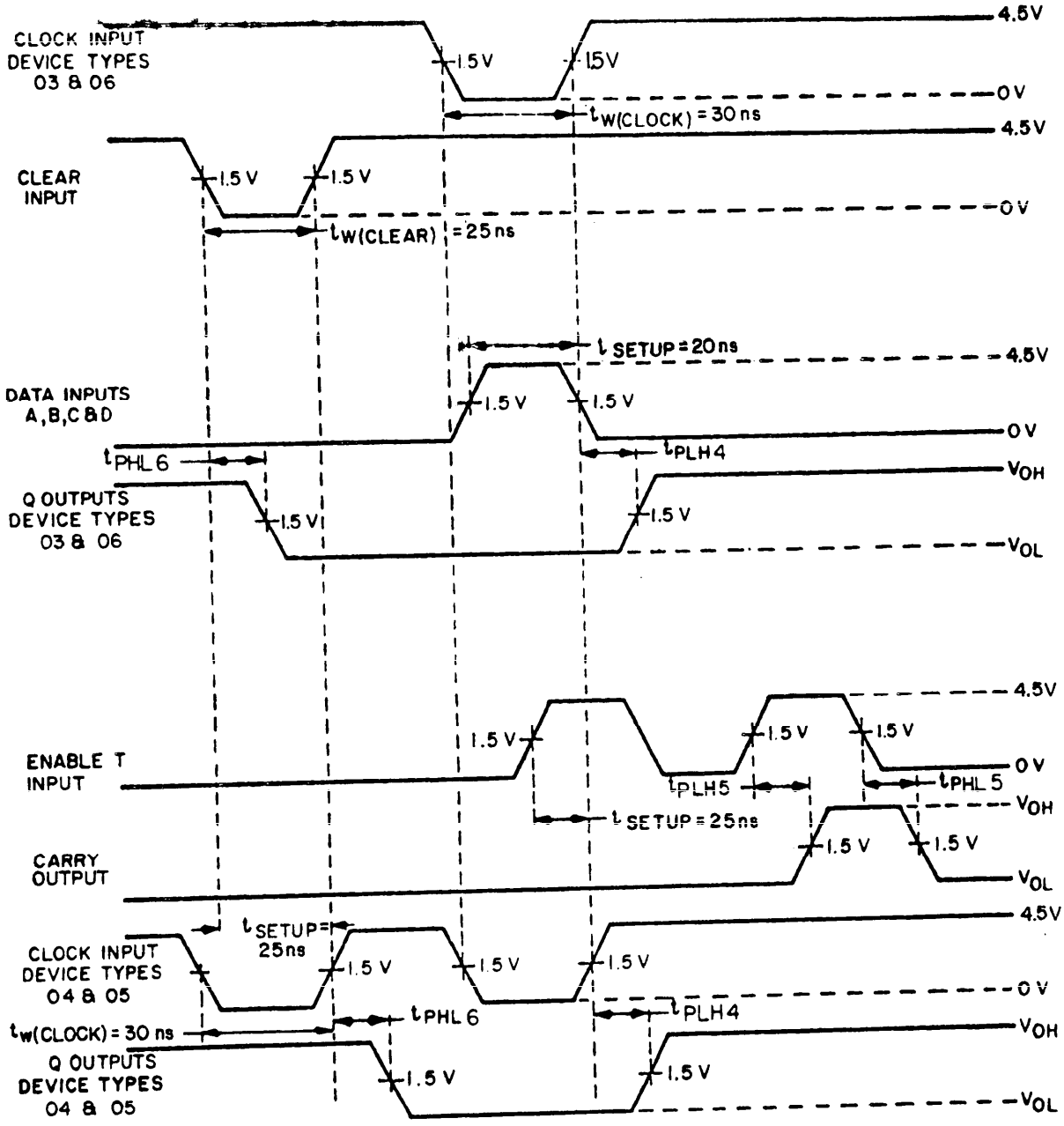
Source: https://assist.dla.mil -- Downloaded: 202610-21T12:51Z
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NOTES:
1. The pulse generator has the following characteristics: $t_r < 10$ ns, $t_f < 10$ ns, PRR < 1 MHz, duty cycle $< 50\%$, $Z_{out} \approx 50\Omega$.
2. All diodes are 1N3064, or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground terminal.
5. Load circuits on a given output are only required where the specific test given in table III indicates "OUT" on that output. Load circuits may otherwise be omitted.

FIGURE 8. Switching time test circuits and waveform for device types 03, 04, 05, and 06.

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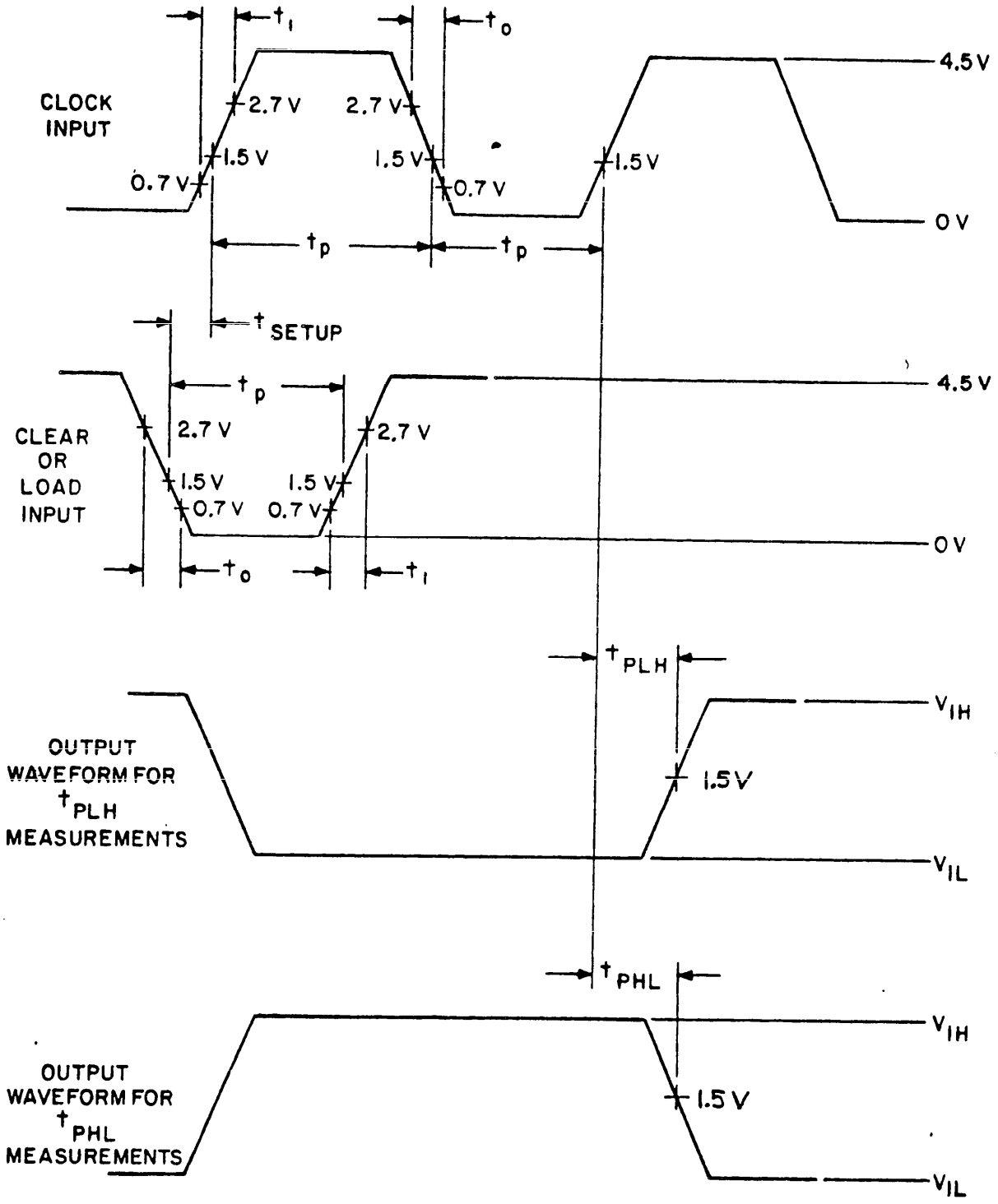


VOLTAGE WAVEFORMS

NOTES:
1. The input pulses are supplied by a generator having the following characteristics: $t_r < 10$ ns, $t_f < 10$ ns, PRR < 1 MHz, $Z_{out} \approx 50\ \Omega$.
2. Enable T duty cycle $\leq 50\%$, PRR = 1 MHz.

FIGURE 8A. Switching time test circuits and waveforms for device types 03, 04, 05, and 06 - Continued.

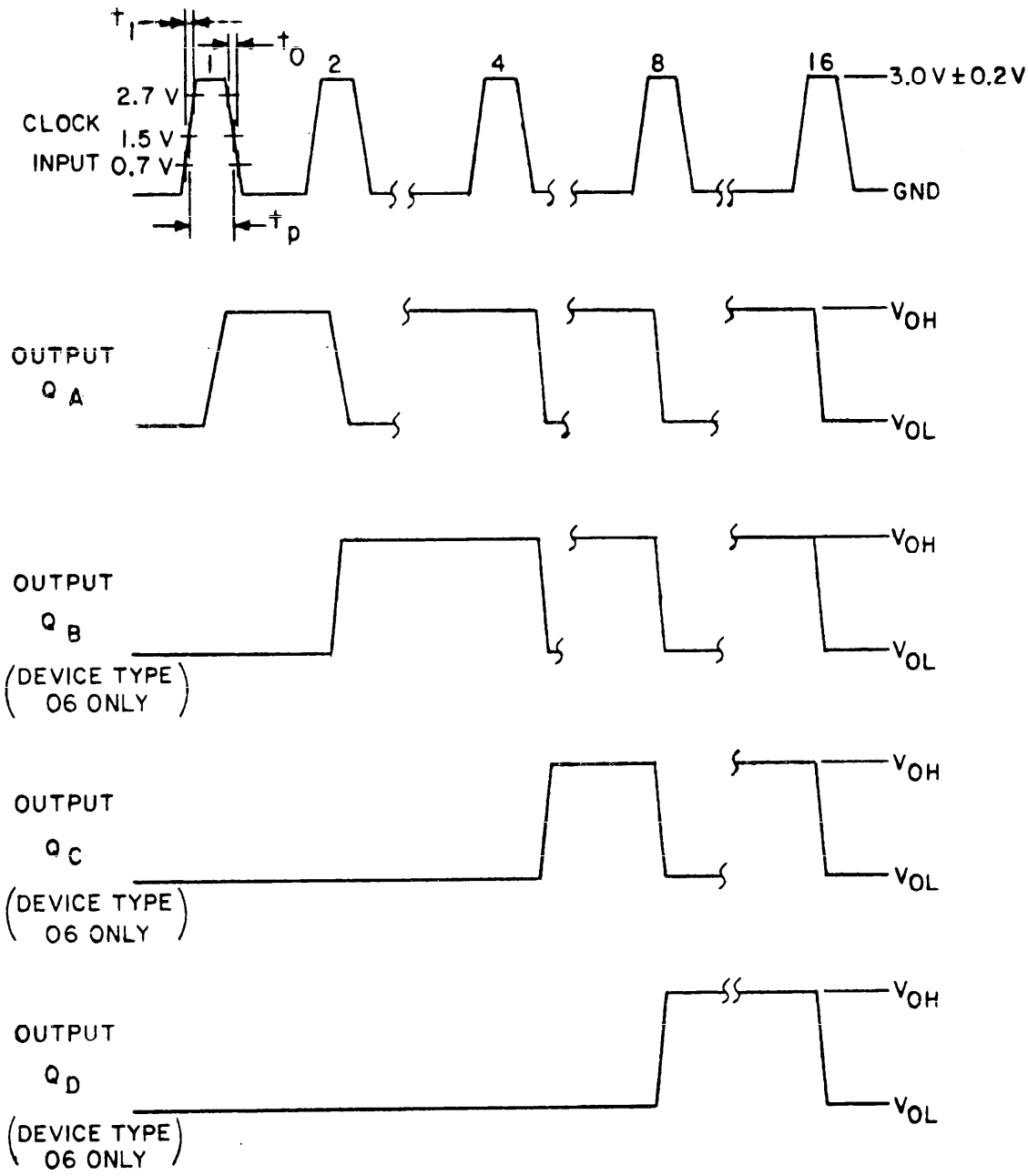
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:51Z
Check the source to verify that this is the current version before use.



NOTES:
1. The input pulses are supplied by generators having the following characteristics: $t_0 = t_1 \leq 10$ ns, PRR $\leq$ 1.0 MHz, $t_p \leq 500$ ns, $Z_{out} \approx 50\Omega$.
2. The t_{SETUP} for load pulse = 30 ns; t_{SETUP} for clear pulse = 25 ns.

FIGURE 8B. Switching time test circuits and waveforms for device types 03, 04, 05 and 06.

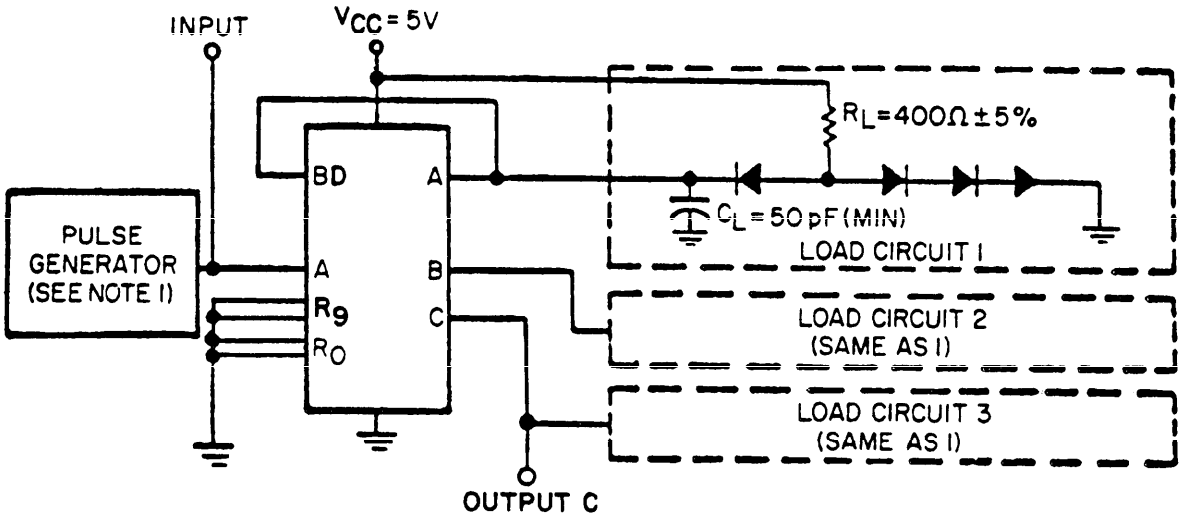
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:51Z
Check the source to verify that this is the current version before use.



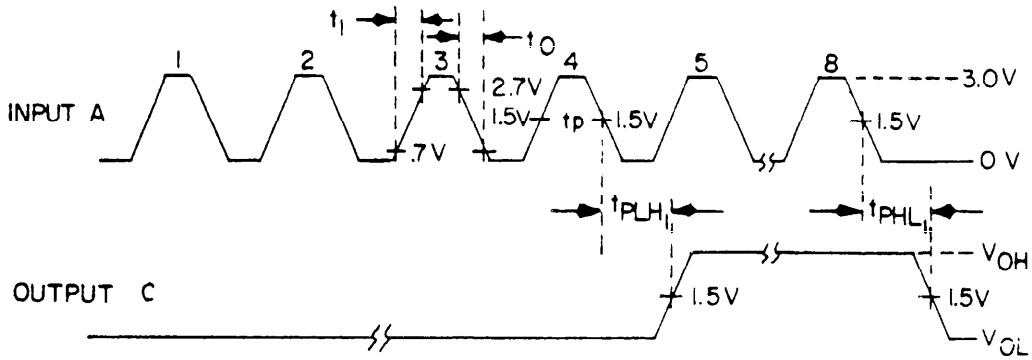
NOTES:
1. The pulse generator has the following characteristics: $t_0 = t_1 = 5 \pm 1$ nsec, $t_p = 20$ ns, $V_{gen} = 4.5$ V, $Z_{out} \approx 50\Omega$, PRR = 20 MHz.
2. The output frequency shall be:
   a. Output Q_A = one-half the input frequency.
   b. Output Q_B = one-quarter the input frequency.
   c. Output Q_C = one-eighth the input frequency.
   d. Output Q_D = one-sixteenth the input frequency.

FIGURE 8C. F_{max} waveforms for device types 03, 04, 05 and 06.

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TEST CIRCUIT

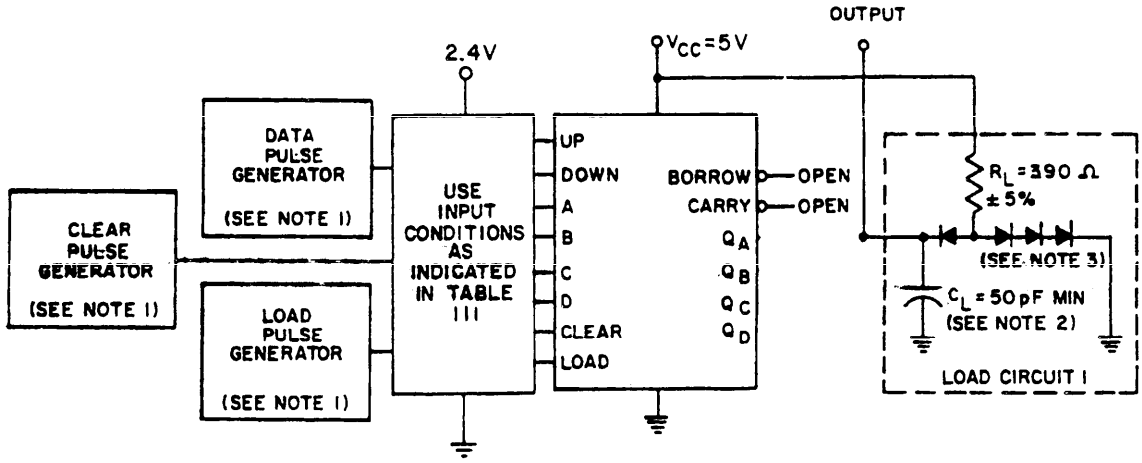


VOLTAGE WAVEFORMS

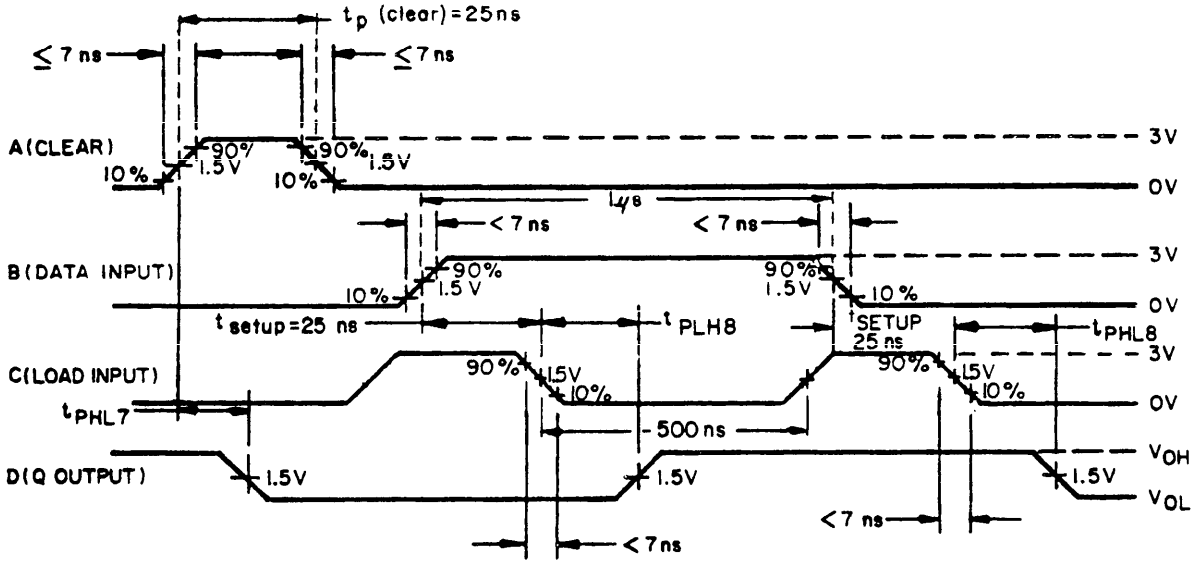
NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 15$ ns, t_p = .5 µs, PRR = 1 MHz, $Z_{out} \approx 50\Omega$.
2. All diodes are 1N3064 or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground terminal.
5. F_{MAX} pulse generator has the following characteristics: V_{gen} = 3.0 V ±0.2 V, $t_0 = t_1 \leq 10$ ns, t_p = 50 ns, PRR = 10 MHz.

FIGURE 9. Switching time test circuit and waveforms for device type 07.

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TEST CIRCUIT

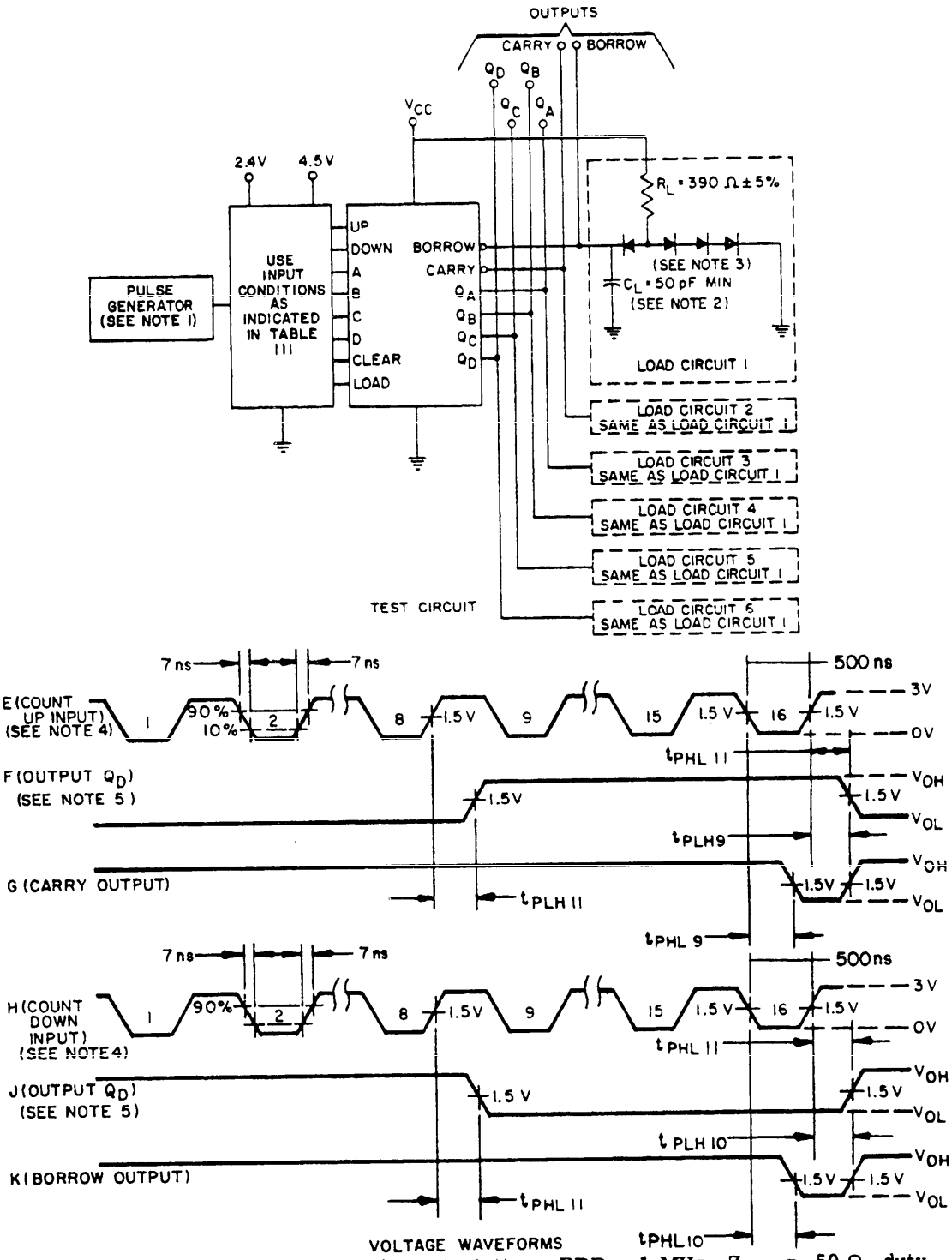


VOLTAGE WAVEFORMS

NOTES:
1. The pulse generators have the following characteristics: Z_{out} ≈50 Ω; for the data pulse generator, PRR = 500 kHz, duty cycle = 50%; for the load pulse generator, PRR = 1 MHz, duty cycle = 50%.
2. C_L includes probe and jig capacitance.
3. All diodes are 1N3064, or equivalent.

FIGURE 10. Switching time test circuits and waveforms for device types 08 and 09.

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NOTES:

1. The pulse generator has the following characteristics: PRR = 1 MHz, $Z_{out} \approx 50\ \Omega$, duty cycle = 50%.
2. C_L includes probe and jig capacitance.
3. All diodes are 1N3064, or equivalent.
4. Count-up and count-down pulse shown are for device type 09 binary counters. Count cycle for the device type 08 decade counter is 1 through 10.
5. Waveforms for outputs Q_A, Q_B, and Q_C are omitted to simplify the drawing.
6. Load circuits on a given output are only required where the specific test given in table III indicates "OUT" on that output. Load circuits may otherwise be omitted.

FIGURE 10A. Switching time test circuits and waveforms for device types 08 and 09.

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TABLE III. Group A inspection for device type 01.
Terminal conditions 13/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C Test No.	1 / 1 Input BC	2 / 2 NC	3 / 3 NC	4 / 4 NC	5 / 5 V_{CC}	6 / 6 $R_0(1)$	7 / 7 $R_0(2)$	8 / 8 Output D	9 / 9 Output C	10 / 10 GND	11 / 11 Output B	12 / 12 Output A	13 / 13 NC	14 / 14 Input A	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
1 $T_C = 25°C$	V_{OL}	3007	1	2.0 V				4.5 V	2.0 V	2.0 V	16 mA	16 mA	GND				GND	Output D		0.4	V
			2	2.0 V													GND	Output C			
			3	2.0 V										16 mA			GND	Output B			
			4	GND											16 mA		2.0 V	Output A			
	V_{OH}	3006	5	1/ 6/					1/	1/	-400 µA	-400µA					GND	Output D	2.4		V
			6	1/ 7/													GND	Output C			
			7	1/ 8/										-400µA			GND	Output B			
			8	GND											-400 µA		1/ 8/	Output A			
	I_{OS}	3011	9	2/ 6/				5.5 V	2/	2/	GND	GND					GND	Output D	-20	-57	mA
			10	2/ 7/													GND	Output C			
			11	2/ 8/										GND			GND	Output B			
			12	GND											GND		2/ 8/	Output A			
	I_{IH1}	3010	13						2.4 V	GND								$R_0(1)$		40	µA
	I_{IH1}		14						GND	2.4 V								$R_0(2)$		40	
	I_{IH2}		15						5.5 V	GND								$R_0(1)$		100	
	I_{IH2}		16						GND	5.5 V								$R_0(2)$		100	
	I_{IH3}		17														2.4 V	Input A		80	
	I_{IH4}		18														5.5 V	Input A		200	
	I_{IH5}		19	2.4 V														Input BC		160	
	I_{IH6}		20	5.5 V														Input BC		400	
	I_{IL1}	3009	21						0.4 V	4.5 V								$R_0(1)$	-14	-14/	mA
	I_{IL1}		22						4.5 V	0.4 V								$R_0(2)$			
	I_{IL2}		23						2/	2/							0.4 V	Input A			
	I_{IL3}		24	0.4 V					2/	2/								Input BC			
	I_{CC}	3005	25	GND					2/	2/							GND	V_{CC}		44	mA
	V_{IC}		26					4.5 V	-12 mA									$R_0(1)$		-1.5	V
			27							-12 mA								$R_0(2)$			
			28														-12 mA	Input A			
			29	-12 mA														Input BC			
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions 13/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / Input BC	2 / 2 / NC	3 / 3 / NC	4 / 4 / NC	5 / 5 / V_{CC}	6 / 6 / $R_0(1)$	7 / 7 / $R_0(2)$	8 / 8 / Output D	9 / 9 / Output C	10 / 10 / GND	11 / 11 / Output B	12 / 12 / Output A	13 / 13 / NC	14 / 14 / Input A	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
7 T_C = 25°C	Functional tests 4/	3014	30	B 3/				4.5 V	A 3/	A 3/	L	L	GND	L	L		B 3/				
			31	A																	
			32	B																	
			33	B					B												
			34	A										H							
			35	B																	
			36	B					A	B				L							
			37	B										L							
			38	A						B				L							
			39	B					A	B				H							
			40	A										H							
			41	B								H		L							
			42	A								H									
			43	B							H	L									
			44	A										H							
			45	B																	
			46	A										H							
			47	B								H		L					See 5/		
			48	B						A	L	L									
			49	B					B					H							
			50	A																	
			51	B										H							
			52	A								H									
			53	B										L							
			54	A								H									
			55	B							H	L									
			56	A																	
			57	B										H							
			58	A								H									
			59	B										L							
			60	A								H									
			61	B							L	L									
			62														A				
			63												H		B				
			64												H		A				
			65												L		B				
8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																				

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C	1 / 1	2 / 2	3 / 3	4 / 4	5 / 5	6 / 6	7 / 7	8 / 8	9 / 9	10 / 10	11 / 11	12 / 12	13 / 13	14 / 14	Meas. terminal	Test limits Min	Max	Unit
			Test No.	Terminal conditions 13/ Input BC	NC	NC	NC	V_{CC}	$R_0(1)$	$R_0(2)$	Output D	Output C	GND	Output B	Output A	NC	Input				
9 T_C=25°C	F_{MAX} 9/	Fig 6	66	10/ 10/				5.0 V	GND	GND	12/ OUT 11/ OUT 11/	12/ 11/ 11/	GND	12/ 11/ 11/	OUT 11/ 10/ 11/ 10/ 11/		IN	Output A	10		MHz
	t_{PLH1}		67															Output D	20	100	ns
	t_{PHL1}		68															Output D	20	100	ns
10 T_C=125°C	F_{MAX} 9/	Fig 6	69	10/ 10/							12/ OUT 11/ OUT 11/	12/ 11/ 11/		12/ 11/ 11/	OUT 11/ 10/ 11/ 10/ 11/		IN	Output A	10		MHz
	t_{PLH1}		70															Output D	20	115	ns
	t_{PHL1}		71															Output D	20	115	ns
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

1/ Momentarily apply 2.4 V, then ground prior to taking measurements to set the device in desired state. Maintain ground for measurement.
2/ Apply 4.5 V pulse then ground prior to taking measurements to set the device in the desired state. Maintain ground for measurement.
3/ A $\geq$ 2.0 V, B $\leq$ 0.8 V. Input voltages shown are the maximum for V_{IL} and the minimum for V_{IH}.
4/ Only a summary of attributes data is required.
5/ Output voltages shall be either:
(a) H = 2.4 volts minimum and L = 0.4 volt maximum when using a high speed checker double comparator, or
(b) H $\geq$ 1.5 volts and L $\leq$ 1.5 volts when using a high speed checker single comparator.
6/ Input must be applied 3 times after R_0 pulses.
7/ Input must be applied 2 times after R_0 pulses.
8/ Input must be applied once after R_0 pulses.
9/ F_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.
10/ Connect terminals together during test.
11/ See test figure 6 for terminal load.
12/ Omit specified loads for this test.
13/ Terminal conditions (pins not designated may be H $\geq$ 2.0 V, or L $\leq$ 0.8 V, or open).
14/ The limits shown shall be as follows:

Test	Min/Max limits (ma) for circuit: A	B	C	D	E
I_{IL1}	-0.4/-1.3	-0.7/-1.6	-0.3/-1.3	-0.7/-1.6	-0.7/-1.6
I_{IL2}	-1.4/-3.2	-0.4/-1.3	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2
I_{IL3}	-1.4/-4.8	-0.4/-1.3	-0.7/-4.8	-1.4/-6.4	-1.4/-6.4

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TABLE III. Group A inspection for device type 02.
Terminal conditions 13/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / Input B	2 / 2 / $R_0(1)$	3 / 3 / $R_0(2)$	4 / 4 / NC	5 / 5 / V_{CC}	6 / 6 / NC	7 / 7 / NC	8 / 8 / Output C	9 / 9 / Output B	10 / 10 / GND	11 / 11 / Output D	12 / 12 / Output A	13 / 13 / NC	14 / 14 / Input A	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	V_{OL}	3007	1	GND	2.0 V	2.0 V		4.5 V				16 mA	GND		16 mA		2.0 V	Output A		0.4	V
			2	2.0 V							16 mA						GND	Output B			
			3	2.0 V													GND	Output C			
			4	2.0 V										16 mA			GND	Output D			
	V_{OH}	3006	5	GND	1/	1/						-400µA			-400 µA		1/ 6/	Output A	2.4		V
			6	1/ 6/							-400 µA						GND	Output B			
			7	1/ 7/													GND	Output C			
			8	1/ 8/										-400µA			GND	Output D			
	I_{OS}	3011	9	GND	2/	2/		5.5 V				GND			GND		2/ 6/	Output A	-20	-57	mA
			10	2/ 6/							GND						GND	Output B			
			11	2/ 7/													GND	Output C			
			12	2/ 8/										GND			GND	Output D			
	I_{IH1}	3010	13		2.4 V	GND												$R_0(1)$		40	µA
	I_{IH1}		14		GND	2.4 V												$R_0(2)$		40	
	I_{IH2}		15		5.5 V	GND												$R_0(1)$		100	
	I_{IH2}		16		GND	5.5 V												$R_0(2)$		100	
	I_{IH7}		17		5.5 V												2.4 V	Input A		80	
	I_{IH7}		18	2.4 V														Input B		80	
	I_{IH8}		19														5.5 V	Input A		200	
	I_{IH8}		20	5.5 V														Input B		200	
	I_{IL1}	3009	21		0.4 V	4.5 V												$R_0(1)$	14/	14/	mA
	I_{IL1}		22		4.5 V	0.4 V												$R_0(2)$			
	I_{IL2}		23		2/	2/											0.4 V	Input A			
	I_{IL3}		24	0.4 V	2/	2/												Input B			
	I_{CC}	3005	25	GND	2/	2/											GND	V_{CC}		44	
	V_{IC}		26					4.5 V									-12 mA	Input A		-1.5	V
			27	-12 mA														Input B			
			28		-12 mA													$R_0(1)$			
			29			-12 mA												$R_0(2)$			
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions 13/

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C	1 / 1	2 / 2	3 / 3	4 / 4	5 / 5	6 / 6	7 / 7	8 / 8	9 / 9	10 / 10	11 / 11	12 / 12	13 / 13	14 / 14	Meas. terminal	Test limits		
			Test No.	Input B	$R_0(1)$	$R_0(2)$	NC	V_{CC}	NC	NC	Output C	Output B	GND	Output D	Output A	NC	Input A		Min	Max	Unit
7 $T_C = 25°C$	Functional tests 4/	3014	30	B 5/	A 5/	A 5/		4.5 V			L	L	GND	L	L		B 5/				
			31	A																	
			32	B		B															
			33	B								H									
			34	A																	
			35	B		A						L									
			36	B	B																
			37	B		B						H									
			38	A																	
			39	A		B						H									
			40	B	B	B					H	H									
			41	A								L		H							
			42	B							H	L									
			43	A								H									
			44	B								H									
			45	A							L	L		H							
			46	B								L									
			47	A								H									
			48	B							H	L									
			49	A								H									
			50	B								L									
			51	A								H									
			52	B								H									
			53	A							L	L		L						See 3/	
			54	B								L									
			55	A								H									
			56	B								H									
			57	A							H	L									
			58	B								L									
			59	A								H									
			60	B								H									
			61	A							L	L		H							
			62	B							L	L		H							
			63	A								H									
			64	B	A																
			65	B							H	L									
			66	A								L									
			67	B								H									
			68	A								H									
			69	B							L	L		L							
			70	A		A															
			71	A		A															
			72	B	B	B									H		A				
			73												H		B				
			74												L		A				
			75														B				
			76																		
8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																				

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case A, B, D / Case C / Test No.	1 / 1 / Input B	2 / 2 / $R_0(1)$	3 / 3 / $R_0(2)$	4 / 4 / NC	5 / 5 / V_{CC}	6 / 6 / NC	7 / 7 / NC	8 / 8 / Output C	9 / 9 / Output B	10 / 10 / GND	11 / 11 / Output D	12 / 12 / Output A	13 / 13 / NC	14 / 14 / Input A	Meas. terminal	Test limits Min	Max	Unit
			Terminal conditions 13/																		
9 $T_C = 25°C$	F_{MAX} 9/	Fig 7	77		GND	GND		5.0 V			12/	12/	GND		OUT 11/		IN	Output A	10		MHz
	t_{PLH2}		78	10/							11/	11/		OUT 11/	10/ 11/		IN	Output D	20	135	ns
	t_{PHL2}		79	10/							11/	11/		OUT 11/	10/ 11/		IN	Output D	20	135	ns
10 $T_C = 125°C$	F_{MAX} 9/	Fig 7	80								12/	12/			OUT 11/		IN	Output A	10		MHz
	t_{PLH2}		81	10/							11/	11/		OUT 11/	10/ 11/		IN	Output D	20	135	ns
	t_{PHL2}		82	10/							11/	11/		OUT 11/	10/ 11/		IN	Output D	20	135	ns
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																				

1/ Momentarily apply 2.4 V, then ground prior to taking measurements to set the device in desired state. Maintain ground for measurement.
2/ Apply 4.5 V pulse, then ground prior to taking measurements to set the device in desired state. Maintain ground for measurement.
3/ Output voltages shall be either:
(a) H = 2.4 volts minimum and L = 0.4 volt maximum when using a high speed checker double comparator, or
(b) H $\geq$ 1.5 volts and L $\leq$ 1.5 volts when using a high speed checker single comparator.
4/ Only a summary of attributes data is required.
5/ A $\geq$ 2.0 V, B $\leq$ 0.8 V. Input voltages shown are the maximum for V_{IL} and the minimum for V_{IH}.
6/ Input pulse must be applied 1 time after R_0 pulses.
7/ Input pulse must be applied 2 times after R_0 pulses.
8/ Input pulse must be applied 4 times after R_0 pulses.
9/ F_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.
10/ Connect terminals together during test.
11/ See test figure 7 for terminal load.
12/ Omit specified loads for this test.
13/ Terminal conditions (pins not designated may be H $\geq$ 2.0 V, or L $\leq$ 0.8 V, or open).
14/ The limits shown are as follows:

Test	Min/Max limits (mA) for circuit: A	B	C	D	E
I_{IL1}	-0.4/-1.3	-0.7/-1.6	-0.4/-1.3	-0.7/-1.6	-0.7/-1.6
I_{IL2}	-0.7/-3.2	-0.4/-1.3	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2
I_{IL3}	-0.7/-3.2	-0.4/-1.3	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2

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TABLE III. Group A inspection for device type 03.
Terminal conditions 2/

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clear	2 Clock	3 DATA INPUTS A	4 DATA INPUTS B	5 DATA INPUTS C	6 DATA INPUTS D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
1 T_C = 25°C	V_{OH}	3006	1	4.5 V	1/			2.0 V	2.0 V		GND	GND		-800 µA	-800 µA	-800 µA	-800 µA	-800µA	4.5 V	Q_D	2.4		V
			2																	Q_C			
			3				2.0 V													Q_B			
			4			2.0 V														Q_A			
			5			2.0 V	0.8 V	0.8 V	2.0 V				2.0 V							Carry output			
	V_{OL}	3007	6						0.8 V	0.8 V			0.8 V	16 mA						Q_D		0.4	
			7					0.8 V							16 mA					Q_C			
			8				0.8 V									16 mA				Q_B			
			9			0.8 V											16 mA			Q_A			
			10															16 mA		Carry output			
	V_{IC}		11			-12 mA														Input A		-1.5	
			12				-12 mA													Input B			
			13					-12 mA												Input C			
			14						-12 mA											Input D			
			15							-12 mA										Enable P			
			16									-12 mA								Load			
			17										-12 mA							Enable T			
			18	-12 mA																Clear			
			19		-12 mA															Clock			
	I_{IH9}	3010	20		2.4 V														5.5 V	Clock		80	µA
	I_{IH9}		21	1/		GND	GND	GND	GND	GND		GND	2.4 V							Enable T		80	
	I_{IH10}		22		5.5 V															Clock		200	
	I_{IH10}		23	1/		GND	GND	GND	GND	GND		GND	5.5 V							Enable T		200	
	I_{IH11}		24	2.4 V																Clear		40	
			25			2.4 V						5.5 V								A			
			26				2.4 V													B			
			27					2.4 V												C			
			28						2.4 V											D			
			29							2.4 V			GND							Enable P			
			30									2.4 V								Load			
	I_{IH12}		31	5.5 V																Clear		100	
			32			5.5 V						5.5 V								A			
			33				5.5 V													B			
			34					5.5 V												C			
			35						5.5 V											D			
			36							5.5 V			GND							Enable P			
			37									5.5 V								Load			
	I_{IL5}	3009	38		0.4 V															Clock	-1.0	-2.3	mA
	I_{IL6}		39			0.4 V						GND								Input A	-0.4	-1.3	
			40				0.4 V													Input B			
			41					0.4 V												Input C			
			42						0.4 V											Input D			
	I_{IL7}		43							0.4 V			4.5 V							Enable P	-0.7	-1.6	
			44									0.4 V								Load	-0.5		
			45	0.4 V																Clear	-0.7		
	I_{IL8}		46	4.5 V	1/	4.5 V			4.5 V	4.5 V		GND	0.4 V							Enable T	-1.0	-3.2	

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TABLE III. Group A inspection for device type 03 - Continued

Terminal conditions 2/

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clear	2 Clock	3 DATA INPUTS A	4 DATA INPUTS B	5 DATA INPUTS C	6 DATA INPUTS D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
1 T_C = 25°C	I_{OS}	3011	47	4.5 V	1/			4.5 V	4.5 V		GND	GND		GND	GND				5.5 V	Q_D	-20	-57	mA
			48				4.5 V									GND				Q_C			
			49														GND			Q_B			
			50			4.5 V														Q_A			
			51			4.5 V	GND	GND	4.5 V				4.5 V					GND		Carry output			
	I_{CCH}	3005	52	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		5.5 V	5.5 V							V_{CC}		85	
	I_{CCH}		53	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		GND	5.5 V									85	
	I_{CCL}		54	GND	5.5 V	GND	GND	GND	GND	GND		GND	GND									91	
	I_{CCL}		55	GND	GND	GND	GND	GND	GND	GND		GND	GND									91	
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = 25°C	Functional tests 4/	3014	56	B 5/	A 5/	A 5/	A 5/	A 5/	A 5/	A 5/	GND	A 5/	A 5/	L	L	L	L	L	4.5 V			See 3/	
			57	A	A	B	B	B	B				B				L						
			58		B	A	A	A	A				A				L						
			59		A	A	A	A	A				A				H						
			60		A	B	B	B	B				B				H						
			61		B	A	A	A	A				A				H						
			62		A	A	A	A	A				A			H	L						
			63		A	B	B	B	B				B				L						
			64		B	A	A	A	A				A				L						
			65		A	A	A	A	A				A				H						
			66		A	B	B	B	B				B				H						
			67		B	A	A	A	A				A				H						
			68		A	A	A	A	A				A		H	L	L						
			69		A	B	B	B	B				B				L						
			70		B	A	A	A	A				A				L						
			71		A	A	A	A	A				A				H						
			72		A	B	B	B	B				B				H						
			73		B	A	A	A	A				A				H						
			74		A	A	A	A	A				A			H	L						
			75		A	B	B	B	B				B				L						
			76		B	A	A	A	A				A				L						
			77		A	A	A	A	A				A				H						
			78		A	B	B	B	B				B				H						
			79		B	A	A	A	A				A				H						
			80		A	A	A	A	A				A	H	L	L	L						
			81		A	B	B	B	B				B				L						
			82		B	A	A	A	A				A				L						
			83		A	A	A	A	A				A				H	H					
			84		A	B	B	B	B				B				H	L					
			85		B	A	A	A	A				A				H	H					
			86		A									L			L	L					
			87	B	A					B													
			88	A	A					B													
			89	A	B					B													
			90	A	A					B													
			91	A	A	B	B	B	B	A			B										

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions 2/

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clear	2 Clock	3 DATA INPUTS A	4 DATA INPUTS B	5 DATA INPUTS C	6 DATA INPUTS D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
7 T_C = 25°C	Functional tests 4/	3014	92	A 5/	A 5/	A 5/	A 5/	A 5/	A 5/	A 5/	GND	A 5/	B 5/	L	L	L	L	L	4.5 V			See 3/	
			93	B	B	A	A	A	A			A											
			94	A	A	A	A	A	A			A											
			95	A	A	B	B	B	B			B											
			96	B	B	B	B	B									H						
			97	A	B	B	B	B									H						
			98	A	A	A	A	A															
			99	B											H	H							
			100	A								A											
			101	A								A											
			102	B								A											
			103	A								A											
			104	A		B	B	B	A			B											
			105	B								B											
			106	A								B		H	L	L							
			107	A								A											
			108	B								A											
			109	A								A											
			110	B	A	A	A	A				B	A	L			L						
			111	A	A									L			L						
			112	A	B									L			L						
			113	A	A									H	H	H	H	X					
			114	B	A							A		L	L	L	L	L					
			115	A	A		B	B	B	B		B	B				L						
			116		B				B	B			B				L						
			117		A				B	B			B				H						
			118		A	B			A	A			A				H						
			119		B				A	A				H			H						
			120		A				A	A				H			L						
			121		A		A		B	B				H									
			122		B		A		B					H		H							
			123		A		A		B					L		H							
			124		A		B	A	A					L		H							
			125		B		B							L		H							
			126		A		B					B		H	H	L							
			127		A		A					B	B			L							
			128		B								B			L							
			129		A								B			H							
			130		A	A	B	B		A			A			H							
			131		B		B	B		A			A			H							
			132		A		B	B		A			A		L	L	H	H					
			133		A		A	A	B	B			B		L	L		L					
			134		B			A	B	B					L	L		L					
			135		A			A	B	B				L	H	H		H					
			136		A	B		B	A	A				L	H	H							
			137		B	B			A					L	H	H							
			138		A	B			A					H	L		L						
			139		A	A			B				A	H			L						
			140		B	A								H			L						
			141		A	A								L			H						
			142		A	B	B	A						L			H						

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	Terminal conditions 2/ 1 Clear	2 Clock	3 A	4 DATA B	5 INPUTS C	6 D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
7 T_C = 25°C	Functional tests 4/	3014	143 144 145 146 147 148 149 150	A 5/	B 5/ A A B A A B A	B 5/ B A A A B B B	B 5/ A A A	A 5/	B 5/	A 5/	GND	B 5/	A 5/	L	L H	H L	H L L L H H H L	L	4.5 V			See 3/	
8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																						
9 T_C = 25°C	F_{MAX} 6/	Fig 8 & 8C	151	4.5 V	IN	GND	GND	GND	4.5 V	4.5 V	4.5 V	GND	4.5 V	4.5 V				OUT		5.0 V	Q_A	20		MHz
	t_{PLH3}	Fig 8 & 8B	152			4.5 V	GND		4.5 V				IN					OUT			Carry	3	45	ns
	t_{PHL3}		153				GND		GND								OUT				Carry		45	
	t_{PHL4}		154				4.5 V	GND										OUT	OUT		Q_A		30	
	t_{PHL4}		155				4.5 V		4.5 V						OUT	OUT					Q_B			
	t_{PHL4}		156				GND	GND					4.5 V								Q_C			
	t_{PHL4}		157	8/					4.5 V				IN								Q_D			
	t_{PLH4}		158	4.5 V			GND	GND				4.5 V		OUT							Q_A		28	
	t_{PLH4}		159			4.5 V			GND						OUT	OUT					Q_B			
	t_{PLH4}		160				GND	GND									OUT				Q_C			
	t_{PLH4}		161				4.5 V	4.5 V													Q_D			
	t_{PLH4}	Fig 8	162	8/		IN	4.5 V	4.5 V	4.5 V			GND					OUT				Q_A			
	t_{PLH4}	and 8A	163			4.5 V	IN	4.5 V													Q_B			
	t_{PLH4}		164				4.5 V	IN	IN												Q_C			
	t_{PLH4}		165				4.5 V	4.5 V	4.5 V												Q_D			
	t_{PLH5}		166	4.5 V	7/		GND	GND	IN				IN	OUT	OUT	OUT		OUT			Carry		18	
	t_{PHL5}		167	4.5 V	7/		GND	GND	4.5 V				IN				OUT				Carry		21	
	t_{PHL6}		168	IN	IN		4.5 V	4.5 V					4.5 V								Q_A		39	
	t_{PHL6}		169																		Q_B			
	t_{PHL6}		170												OUT	OUT					Q_C			
	t_{PHL6}		171											OUT							Q_D			

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	Terminal conditions 2/ 1 Clear	2 Clock	3 A	4 DATA INPUTS B	5 DATA INPUTS C	6 D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
10 T_C = 125°C	F_{MAX} 6/	Fig 8 & 8C	172	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V				OUT		5.0 V	Q_A	20		MHz
	t_{PLH3}	Fig 8 & 8B	173			GND	GND	GND	4.5 V			IN						OUT		Carry	3	58	ns
	t_{PHL3}		174			4.5 V			4.5 V									OUT		Carry		58	
	t_{PHL4}		175						GND								OUT			Q_A		38	
	t_{PHL4}		176				4.5 V									OUT				Q_B			
	t_{PHL4}		177				4.5 V	4.5 V							OUT					Q_C			
	t_{PHL4}		178				GND	GND	4.5 V					OUT						Q_D			
	t_{PLH4}		179	8/								4.5 V					OUT			Q_A		36	
	t_{PLH4}		180	4.5 V		4.5 V	GND	GND	GND			IN				OUT				Q_B			
	t_{PLH4}		181				4.5 V	GND							OUT					Q_C			
	t_{PLH4}		182				4.5 V	4.5 V	GND					OUT						Q_D			
	t_{PLH4}	Fig 8 and 8A	183	8/		IN	4.5 V	4.5 V	4.5 V			GND					OUT			Q_A			
	t_{PLH4}		184			4.5 V	IN	4.5 V	4.5 V							OUT				Q_B			
	t_{PLH4}		185				4.5 V	IN	4.5 V						OUT					Q_C			
	t_{PLH4}		186				4.5 V	4.5 V	IN					OUT						Q_D			
	t_{PLH5}		187	4.5 V	7/		GND	GND	4.5 V				IN					OUT		Carry		27	
	t_{PHL5}		188	4.5 V	7/		GND	GND					IN					OUT		Carry		27	
	t_{PHL6}		189	IN	IN		4.5 V	4.5 V					4.5 V				OUT			Q_A		47	
	t_{PHL6}		190													OUT				Q_B			
	t_{PHL6}		191												OUT					Q_C			
	t_{PHL6}		192											OUT						Q_D			
11	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																						

1/ Normal clock pulse: ($V_{IL} \leq 0.8$ V, $V_{IH} \geq 2.0$ V).
2/ Terminal conditions (pins not designated may be H $\geq$ 2.0 V, or L $\leq$ 0.8 V, or open).
3/ Output voltages shall be either:
(a) H = 2.4 volts minimum and L = 0.4 volt maximum when using a high speed checker double comparator, or
(b) H $\geq$ 1.5 volts and L $\leq$ 1.5 volts when using a high speed checker single comparator.
(c) X = Don't care.
4/ Only a summary of attributes data is required.
5/ A $\geq$ 2.0 V, B $\leq$ 0.8 V. Input voltages shown are the maximum for V_{IL} and the minimum for V_{IH}.
6/ F_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.
7/ Apply 1 clock pulse prior to input pulses.
8/ Apply momentary GND, then 4.5 volts prior to input pulses, maintain 4.5 volts during test.

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TABLE III. Group A inspection for device type 04.
Terminal conditions 2/

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clear	2 Clock	3 DATA INPUTS A	4 DATA INPUTS B	5 DATA INPUTS C	6 DATA INPUTS D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
1 $T_C = 25°C$	V_{OH}	3006	1	4.5 V	1/			2.0 V	2.0 V		GND	GND		-800 µA	-800 µA	-800µA	-800 µA	-800µA	4.5 V	Q_D Q_C Q_B Q_A Carry output	2.4		V
			2 3 4			2.0 V	2.0 V																
			5			2.0 V	2.0 V	2.0 V	2.0 V				2.0 V										
	V_{OL}	3007	6 7 8 9 10			0.8 V	0.8 V	0.8 V	0.8 V	0.8 V			0.8 V	16 mA	16 mA	16 mA	16 mA	16 mA		Q_D Q_C Q_B Q_A Carry output		0.4	
	V_{IC}		11 12 13 14 15 16 17 18 19	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA		-12 mA	-12 mA							Input A Input B Input C Input D Enable P Load Enable T Clear Clock		-1.5	
	I_{IH9} I_{IH9} I_{IH10} I_{IH10}	3010	20 21 22 23	1/ 1/	2.4 V 5.5 V	GND GND	GND GND	GND GND	GND GND	GND GND		GND GND	2.4 V 5.5 V					5.5 V	Clock Enable T Clock Enable T		80 80 200 200	µA
	I_{IH11}		24 25 26 27 28 29 30	2.4 V		2.4 V	2.4 V	2.4 V	2.4 V	2.4 V		5.5 V 2.4 V	GND							Clear A B C D Enable P Load		40	
	I_{IH12}		31 32 33 34 35 36 37	5.5 V		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		5.5 V 5.5 V	GND							Clear A B C D Enable P Load		100	
	I_{IL5} I_{IL6} I_{IL7} I_{IL7} I_{IL7} I_{IL8}	3009	38 39 40 41 42 43 44 45 46	0.4 V 4.5 V	0.4 V 1/	0.4 V 4.5 V	0.4 V 4.5 V	0.4 V 4.5 V	0.4 V 4.5 V	0.4 V 4.5 V		GND 0.4 V GND	4.5 V 0.4 V							Clock Input A Input B Input C Input D Enable P Load Clear Enable T	-1.0 -0.4 -0.7 -0.5 -0.7 -1.0	-2.3 -1.3 -1.6 -1.6 -1.6 -3.2	mA

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	Terminal conditions 2/ 1 Clear	2 Clock	3 DATA INPUTS A	4 DATA INPUTS B	5 DATA INPUTS C	6 DATA INPUTS D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 $T_C = 25°C$	I_{OS}	3011	47	4.5 V	1/			4.5 V	4.5 V		GND	GND	GND	GND					5.5 V	Q_D	-20	-57	mA
			48												GND					Q_C			
			49				4.5 V													Q_B			
			50			4.5 V											GND			Q_A			
			51			4.5 V	4.5 V	4.5 V	4.5 V				4.5 V					GND		Carry output			
	I_{CCH}	3005	52	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		5.5 V	5.5 V							V_{CC}		85	
	I_{CCH}		53	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		GND	5.5 V									85	
	I_{CCL}		54	GND	5.5 V	GND	GND	GND	GND	GND		GND	GND									91	
	I_{CCL}		55	GND	GND	GND	GND	GND	GND	GND		GND	GND									91	
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 $T_C = 25$ C	Functional tests 4/	3014	56	B 5/	B 5/	B 5/	B 5/	B 5/	B 5/	A 5/	GND	B 5/	A 5/	X	X	X	X	X	4.5 V			See 3/	
			56A	B	A	B	B	B	B					L	L	L	L	L					
			57	A	B	B	B	B	B														
			58	A	A	B	B	B	B				B										
			59	B	A	A	A	A	A			A	A				H						
			60	B	B	A	A	A	A				A				H						
			61	B	A	A	A	A	A				B				H						
			62	A	A	B	B	B	B				B				L						
			63		B	A	A	A	A				A				L						
			64		A	A	A	A	A				A				H						
			65		A	B	B	B	B				B				H						
			66		B	A	A	A	A				A			H	L						
			67		A	A	A	A	A				A				L						
			68		A	B	B	B	B				B				L						
			69		B	A	A	A	A				A				H						
			70		A	A	A	A	A				A				H						
			71		A	B	B	B	B				B				H						
			72		B	A	A	A	A				A				H						
			73		A	A	A	A	A				A		H		L						
			74		A	B	B	B	B				B				L						
			75		B	A	A	A	A				A				L						
			76		A	A	A	A	A				A				H						
			77		A	B	B	B	B				B				H						
			78		B	A	A	A	A				A				H						
			79		A	A	A	A	A				A			H	L						
			80		A	B	B	B	B				B				L						
			81		B	A	A	A	A				A				L						
			82		A	A	A	A	A				A				H						
			83		A	B	B	B	B				B				H						
			84		B	A	A	A	A				A				H						
			85		A	A	A	A	A				A	H	L	L	L						
			86		A	B	B	B	B				B				L						
			87		B	A	A	A	A				A				L						
			88		A	A	A	A	A				A				H						
			89		A	B	B	B	B				B				H						
			90		B	A	A	A	A				A				H						
			91		A	A	A	A	A				A			H	L						

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions 2/

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clear	2 Clock	3 DATA INPUTS A	4 DATA INPUTS B	5 DATA INPUTS C	6 DATA INPUTS D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
7 T_C = 25°C	Functional tests 4/	3014	92	A 5/	A 5/	B 5/	B 5/	B 5/	B 5/	A 5/	GND	A 5/	B 5/	H	L	H	L	L	4.5 V				
			93		B	A	A	A	A	A			A				L						
			94		A	A	A	A	A				A				H						
			95		A	B	B	B	B				B				H						
			96		B	A	A	A	A				A				H						
			97		A	A	A	A	A				A		H	L	L						
			98		A	B	B	B	B				B				L						
			99		B	A	A	A	A				A				L						
			100		A	A	A	A	A				A				H						
			101		A	B	B	B	B				B				H						
			102		B	A	A	A	A				A				H						
			103		A	A	A	A	A				A			H	L						
			104		A	B	B	B	B				B				L						
			105		B	A	A	A	A				A				L						
			106		A	A	A	A	A				A				H	H					
			107		A	B	B	B	B				B				H	L					
			108		B	A	A	A	A				A				H	H					
			109		A								A	L	L	L	L	L					
			110		B								B										
			111		A																		
			112		B																		
			113		A		B	B	B			A											
			114		B							B					H						
			115		A																		
			116		A		A	A													See 3/		
			117		B										H	H							
			118		A							A											
			119		A							A											
			120		B							B											
			121		A		B	B	A														
			122		B		B	B							L	L							
			123		A		B	B						H	L	L							
			124		A		A	A					A	H	L	L							
			125		B		A	A					A	H	L	L							
			126		A	A	A	A						H	H	H		H					
			127		A	B																	
			128		B	A																	
			129		B	A	B	B	B														
			130		A	A	B	B	B					L	L	L		L					
			131		B	A	B	B	B														
			132		A	A	A	A	A														
			133		B	A			A														
			134		A	A			B														
			135		B	B	A	B															
			136		B	A																	
			137		A																		
			138		B											H							
			139		A											H							
			140		B	B	B	A								H							
			141		B	B	B	A								H							

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
						DATA INPUTS															Min	Max	Unit
				Clear	Clock	A	B	C	D	Enable P	GND	Load	Enable T	Output Q_D	Output Q_C	Output Q_B	Output Q_A	Carry output	V_{CC}				
7 T_C = 25°C	Functional tests 4/	3014	142	A 5/	A 5/	B 5/	B 5/	A 5/	B 5/	A 5/	GND	B 5/	A 5/	L	H	L	L	L	4.5 V			See 3/	
			143		A	A											L						
			144		B	A											L						
			145		A	A											H						
			146		A	B	A										H						
			147		B		A										H						
			148		A		A									H	L						
			149		A		B	B	A							H							
			150		B		B							L		H							
			151		A		B							H	L	L							
			152		A		A									L							
			153		B											L							
			154		A											H							
			155		A	A																	
			156		B	A																	
			157		A	A											H						
			158		A	B	B	A									H						
			159		B	B											H						
			160		A	B										L	L						
			161		A	A											L						
			162		B												L						
			163		A												H						
			164		A							A					H						
			165		B							A					H						
			166		A							A				H	L						
8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																						
9 T_C = 25°C	F_{MAX} 6/	Fig 8 & 8C	167	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V				OUT		5.0 V	Q_A	20		MHz
	t_{PLH3}	Fig 8 & 8B	168			GND	4.5 V	4.5 V	4.5 V			IN						OUT		Carry	3	45	ns
	t_{PHL3}		169			4.5 V	4.5 V	4.5 V	4.5 V									OUT		Carry		45	
	t_{PHL4}		170				GND	GND	GND								OUT			Q_A		30	
	t_{PHL4}		171				4.5 V	GND								OUT				Q_B			
	t_{PHL4}		172				4.5 V	4.5 V							OUT					Q_C			
	t_{PHL4}		173				4.5 V	4.5 V	4.5 V					OUT						Q_D			
	t_{PLH4}		174	IN								4.5 V					OUT			Q_A		28	
	t_{PLH4}		175	4.5 V		4.5 V	GND	GND	GND			IN				OUT				Q_B			
	t_{PLH4}		176	4.5 V			4.5 V	GND							OUT					Q_C			
	t_{PLH4}		177	4.5 V				4.5 V						OUT						Q_D			
	t_{PLH4}	Fig 8	178	8/		IN			4.5 V			GND					OUT			Q_A			
	t_{PLH4}	and 8A	179			4.5 V	IN		4.5 V							OUT				Q_B			
	t_{PLH4}		180				4.5 V	IN	4.5 V						OUT					Q_C			
	t_{PLH4}		181				4.5 V	4.5 V	IN					OUT						Q_D			
	t_{PLH5}		182	4.5 V	7/				4.5 V				IN					OUT		Carry		18	
	t_{PHL5}		183	4.5 V	7/								IN					OUT		Carry		21	
	t_{PHL6}		184	IN	IN								4.5 V				OUT			Q_A		30	
	t_{PHL6}		185													OUT				Q_B			
	t_{PHL6}		186												OUT					Q_C			
	t_{PHL6}		187											OUT						Q_D			

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TABLE III. Group A inspection for device type 04 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	Terminal conditions 2/ 1 Clear	2 Clock	3 DATA INPUTS A	4 DATA INPUTS B	5 DATA INPUTS C	6 DATA INPUTS D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
10 T_C = 125°C	F_{MAX} 6/	Fig 8 & 8C	188	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V				OUT		5.0 V	Q_A	20		MHz
	t_{PLH3}	Fig 8 & 8B	189			GND	4.5 V	4.5 V	4.5 V			IN						OUT		Carry	3	58	ns
	t_{PHL3}		190			4.5 V	4.5 V	4.5 V	4.5 V									OUT		Carry		58	
	t_{PHL4}		191				GND	GND	GND								OUT			Q_A		38	
	t_{PHL4}		192				4.5 V	GND							OUT					Q_B			
	t_{PHL4}		193					4.5 V	4.5 V					OUT						Q_C			
	t_{PHL4}		194					4.5 V												Q_D			
	t_{PLH4}		195	IN					4.5 V			4.5 V			OUT					Q_A		36	
	t_{PLH4}		196	4.5 V		4.5 V	GND	GND	GND			IN					OUT			Q_B			
	t_{PLH4}		197				4.5 V	GND												Q_C			
	t_{PLH4}		198				4.5 V	4.5 V												Q_D			
	t_{PLH4}	Fig 8 and 8A	199	8/		IN	4.5 V	4.5 V	4.5 V			GND					OUT			Q_A			
	t_{PLH4}		200			4.5 V	IN	4.5 V	4.5 V						OUT					Q_B			
	t_{PLH4}		201				4.5 V	IN	IN											Q_C			
	t_{PLH4}		202					4.5 V	4.5 V											Q_D			
	t_{PLH5}		203	4.5 V	7/				IN				IN	OUT		OUT				Carry		27	
	t_{PHL5}		204	4.5 V	7/				4.5 V				IN					OUT		Carry		27	
	t_{PHL6}		205	IN	IN								4.5 V				OUT	OUT		Q_A		38	
	t_{PHL6}		206												OUT					Q_B			
	t_{PHL6}		207													OUT				Q_C			
	t_{PHL6}		208											OUT						Q_D			
11	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																						

1/ Normal clock pulse: ($V_{IL} \leq 0.8$ V, $V_{IH} \geq 2.0$ V).
2/ Terminal conditions (pins not designated may be H $\geq$2.0 V, or L $\leq$0.8 V, or open).
3/ Output voltages shall be either:
(a) H = 2.4 volts minimum and L = 0.4 volt maximum when using a high speed checker double comparator, or
(b) H $\geq$1.5 volts and L $\leq$1.5 volts when using a high speed checker single comparator.
(c) X = Don't care
4/ Only a summary of attributes data is required.
5/ A $\geq$2.0 V, B $\leq$0.8 V. Input voltages shown are the maximum for V_{IL} and the minimum for V_{IH}.
6/ F_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.
7/ Apply 1 clock pulse prior to input pulses.
8/ Momentarily ground, then apply 4.5 V.

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TABLE III. Group A inspection for device type 05.
Terminal conditions 2/

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clear	2 Clock	3 DATA INPUTS A	4 DATA INPUTS B	5 DATA INPUTS C	6 DATA INPUTS D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 VCC	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
1 T_C = 25°C	V_{OH}	3006	1	4.5 V	1/			2.0 V	2.0 V		GND	GND		-800 µA	-800 µA				4.5 V	Q_D	2.4		V
			2				2.0 V									-800 µA				Q_C			
			3														-800 µA			Q_B			
			4			2.0 V														Q_A			
			5			2.0 V	0.8 V	0.8 V	2.0 V				2.0 V					-800µA		Carry output			
	V_{OL}	3007	6					0.8 V	0.8 V	0.8 V			0.8 V	16 mA						Q_D		0.4	
			7												16 mA					Q_C			
			8				0.8 V									16 mA				Q_B			
			9			0.8 V											16 mA			Q_A			
			10															16 mA		Carry output			
	V_{IC}		11			-12 mA														Input A		-1.5	
			12				-12 mA													Input B			
			13					-12 mA												Input C			
			14						-12 mA											Input D			
			15							-12 mA										Enable P			
			16									-12 mA								Load			
			17										-12 mA							Enable T			
			18	-12 mA																Clear			
			19		-12 mA															Clock			
	I_{IH9}	3010	20		2.4 V														5.5 V	Clock		80	µA
	I_{IH9}		21	1/		GND	GND	GND	GND	GND		GND	2.4 V							Enable T		80	
	I_{IH10}		22		5.5 V															Clock		200	
	I_{IH10}		23	1/		GND	GND	GND	GND	GND		GND	5.5 V							Enable T		200	
	I_{IH11}		24	2.4 V																Clear		40	
			25			2.4 V						5.5 V								A			
			26				2.4 V													B			
			27					2.4 V												C			
			28						2.4 V											D			
			29							2.4 V			GND							Enable P			
			30									2.4 V								Load			
	I_{IH12}		31	5.5 V																Clear		100	
			32			5.5 V						5.5 V								A			
			33				5.5 V													B			
			34					5.5 V												C			
			35						5.5 V											D			
			36							5.5 V			GND							Enable P			
			37									5.5 V								Load			
	I_{IL5}	3009	38		0.4 V															Clock	-1.0	-2.3	mA
	I_{IL6}		39			0.4 V						GND								Input A	-0.4	-1.3	
			40				0.4 V													Input B			
			41					0.4 V												Input C			
			42						0.4 V											Input D			
	I_{IL7}		43							0.4 V			4.5 V							Enable P	-0.7	-1.6	
	I_{IL7}		44									0.4 V								Load	-0.7	-1.6	
	I_{IL7}		45	0.4 V																Clear	-0.7	-1.6	
	I_{IL8}		46	4.5 V	1/	4.5 V			4.5 V	4.5 V		GND	0.4 V							Enable T	-1.0	-3.2	

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions 2/

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clear	2 Clock	3 DATA INPUTS A	4 DATA INPUTS B	5 DATA INPUTS C	6 DATA INPUTS D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
1 T_C = 25°C	I_{OS}	3011	47	4.5 V	D 1/			4.5 V	4.5 V		GND	GND	GND	GND	GND				5.5 V	Q_D	-20	-57	mA
			48																	Q_C			
			49				4.5 V									GND				Q_B			
			50			4.5 V											GND			Q_A			
			51			4.5 V			4.5 V									GND		Carry output			
	I_{CCH}	3005	52	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		5.5 V	5.5 V							V_{CC}		85	
	I_{CCH}		53	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		GND	5.5 V									85	
	I_{CCL}		54	GND	5.5 V	GND	GND	GND	GND	GND		GND	GND									91	
	I_{CCL}		55	GND	GND	GND	GND	GND	GND	GND		GND	GND									91	
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = 25 C	Func-tional tests 4/	3014	56	B 5/	B 5/	B 5/	B 5/	B 5/	B 5/	A 5/	GND	B 5/	A 5/	X	X	X	X	X	4.5 V			See 3/	
			56A	B	A									L	L	L	L	L					
			57	A	B																		
			58	A	A																		
			59	B	A	A	A	A	A			A											
			60	B	B	A	A	A	A														
			61	B	A	A	A	A	A														
			62	A	A	B	B	B	B				B										
			63		B	A	A	A	A				A										
			64		A	A	A	A	A				A				H						
			65		A	B	B	B	B				B				H						
			66		B	A	A	A	A				A				H						
			67		A	A	A	A	A				A			H	L						
			68		A	B	B	B	B				B				L						
			69		B	A	A	A	A				A				L						
			70		A	A	A	A	A				A				H						
			71		A	B	B	B	B				B				H						
			72		B	A	A	A	A				A				H						
			73		A	A	A	A	A				A		H	L	L						
			74		A	B	B	B	B				B				L						
			75		B	A	A	A	A				A				L						
			76		A	A	A	A	A				A				H						
			77		A	B	B	B	B				B				H						
			78		B	A	A	A	A				A				H						
			79		A	A	A	A	A				A			H	L						
			80		A	B	B	B	B				B				L						
			81		B	A	A	A	A				A				L						
			82		A	A	A	A	A				A				H						
			83		A	B	B	B	B				B				H						
			84		B	A	A	A	A				A				H						
			85		A	A	A	A	A				A	H	L	L	L						
			86		A	B	B	B	B				B				L						
			86A		B	B	B	B	B				B				L						
			87		B 5/	A	A	A	A				A				L						
			88		A	A	A	A	A				A				H						
			89		A	B	B	B	B				B				H						
			90		B	A	A	A	A				A				H						
			91		A	A	A	A	A				A	L			L						

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions 2/

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clear	2 Clock	3 DATA INPUTS A	4 DATA INPUTS B	5 DATA INPUTS C	6 DATA INPUTS D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
7 T_C=25°C	Functional tests 4/	3014	92	A 5/	A 5/	A 5/	A 5/	A 5/	A 5/	B 5/	GND	A 5/	A 5/	L	L	L	L	L	4.5 V			See 3/	
			93		B					B			A										
			94		A					A			B										
			95		B												H						
			96		A		B	B	B			B			H	H							
			97		B		B	B															
			98		A		B	B															
			99		A		A	A															
			100		B										L								
			101		A										H								
			102		A							A											
			103		B																		
			104		A		B	B	A			B											
			105		B		B	B				B		H									
			106		A							B			L	L							
			107		A							A											
			108		B																		
			109		A																		
			110	B	B		A	A					A					H					
			111	B	A									L			L	L					
			112	A	A		B	B				B		L			L						
			113		B									L			L						
			114		A									H			H						
			115		B																		
			116		A																		
			117	B	B																		
			118	B	A									L			L						
			119	A	A	B		B	B				B										
			120		B		A		B							H							
			121		A		A		B							H							
			122		A		B		A							H							
			123		B				A							H							
			124		A				A					H		L							
			125		A				B				A	H									
			126		B				B					H									
			127		A				B					L									
			128		A		A		A					L									
			129		B		A		A					L		H							
			130		A		A		A					H	H	H							
			131		A	A	B		B					H	H								
			132		B	A	B							H	H								
			133		A	A	B		B					L	L		H						
			134		A	B	B		A						L		H						
			135		B	B	B								L		H						
			136		A	A	A								H	L	L						
			137		A	A	A										L						
			138		B	A	A										H						
			139		A	A	A										H						
			140		A	B	A										H						
			141		B	B	A										H						

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TABLE III. Group A inspection for device type 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	Terminal conditions 2/ 1 Clear	2 Clock	3 DATA INPUTS A	4 DATA INPUTS B	5 DATA INPUTS C	6 DATA INPUTS D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
7 T_C = 25°C	Functional tests 4/	3014	142	A 5/	A 5/	B 5/	A 5/	A 5/	B 5/	A 5/	GND	B 5/	A 5/	L	H	H	L	L	4.5 V			See 3/	
			143		A		B	B	A					H		H							
			144		B											H							
			145		A										L	L							
			146		A	A																	
			147		B																		
			148		A												H						
			149		A		A	A				A					H						
			150		B		A	A				A					H						
			151		A		A	A				A		L			L						
8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																						
9 T_C = 25°C	F_{MAX} 6/	Fig 8 & 8C	152	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V				OUT		5.0 V	Q_A	20	45	MHz
	t_{PLH3}	Fig 8 & 8B	153			GND	GND	GND	4.5 V			IN						OUT		Carry	3	45	ns
	t_{PHL3}		154			4.5 V			4.5 V									OUT		Carry		45	
	t_{PHL4}		155						GND								OUT			Q_A		30	
	t_{PHL4}		156				4.5 V									OUT				Q_B			
	t_{PHL4}		157				4.5 V	4.5 V							OUT					Q_C			
	t_{PHL4}		158				GND	GND	4.5 V					OUT						Q_D			
	t_{PLH4}		159	IN								4.5 V					OUT			Q_A		28	
	t_{PLH4}		160	4.5 V		4.5 V	GND	GND	GND			IN				OUT				Q_B			
	t_{PLH4}		161				4.5 V	GND							OUT					Q_C			
	t_{PLH4}		162				4.5 V	4.5 V						OUT						Q_D			
	t_{PLH4}	Fig 8 and 8A	163	8/		IN	4.5 V	4.5 V	4.5 V			GND					OUT			Q_A			
	t_{PLH4}		164			4.5 V	IN	4.5 V	4.5 V							OUT				Q_B			
	t_{PLH4}		165				4.5 V	IN	4.5 V						OUT					Q_C			
	t_{PLH4}		166				4.5 V	4.5 V	IN					OUT						Q_D			
	t_{PLH5}		167	4.5 V	7/		GND	GND	4.5 V				IN					OUT		Carry		18	
	t_{PHL5}		168	4.5 V	7/		GND	GND					IN					OUT		Carry		21	
	t_{PHL6}		169	IN	IN		4.5 V	4.5 V					4.5 V				OUT			Q_A		30	
	t_{PHL6}		170													OUT				Q_B			
	t_{PHL6}		171												OUT					Q_C			
	t_{PHL6}		172											OUT						Q_D			

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TABLE III. Group A inspection for device type 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	Terminal conditions 2/ 1 Clear	2 Clock	3 A	4 B	5 C	6 D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
						DATA INPUTS																	
10 T_C = 125°C	F_{MAX} 6/	Fig 8 & 8C	173	4.5 V	IN	GND	GND	GND	4.5 V	4.5 V	GND	4.5 V	4.5 V				OUT	OUT	5.0 V	Q_A	20		MHz
	t_{PLH3}	Fig 8 & 8B	174			4.5 V			4.5 V			IN						OUT		Carry	3	58	ns
	t_{PHL3}		175						GND								OUT	OUT		Carry		58	
	t_{PHL4}		176				4.5 V									OUT				Q_A		38	
	t_{PHL4}		177				4.5 V	4.5 V							OUT					Q_B			
	t_{PHL4}		178				GND	GND	4.5 V			4.5 V		OUT			OUT			Q_C		36	
	t_{PHL4}		179			4.5 V						IN				OUT				Q_D			
	t_{PLH4}		180	IN			GND	GND	GND						OUT					Q_A			
	t_{PLH4}		181	4.5 V			4.5 V	GND									OUT			Q_B			
	t_{PLH4}		182				4.5 V	4.5 V						OUT	OUT					Q_C			
	t_{PLH4}		183																	Q_D			
	t_{PLH4}	Fig 8 and 8A	184	8/		IN	4.5 V	4.5 V	4.5 V			GND					OUT			Q_A			
	t_{PLH4}		185			4.5 V	IN	4.5 V	4.5 V						OUT	OUT				Q_B			
	t_{PLH4}		186				4.5 V	IN	4.5 V					OUT						Q_C			
	t_{PLH4}		187				4.5 V	4.5 V	IN				IN					OUT		Q_D			
	t_{PLH5}		188	4.5 V	7/		GND	GND	4.5 V				IN			OUT				Carry		27	
	t_{PHL5}		189	4.5 V	7/		GND	GND					4.5 V					OUT		Carry		27	
	t_{PHL6}		190	IN	IN		4.5 V	4.5 V									OUT			Q_A		38	
	t_{PHL6}		191												OUT	OUT				Q_B			
	t_{PHL6}		192											OUT						Q_C			
	t_{PHL6}		193																	Q_D			
11	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																						

1/ Normal clock pulse: ($V_{IL} \leq 0.8$ V, $V_{IH} \geq 2.0$ V).
2/ Terminal conditions (pins not designated may be H $\geq$2.0 V, or L $\leq$ 0.8 V, or open).
3/ Output voltages shall be either:
(a) H = 2.4 volts minimum and L = 0.4 volt maximum when using a high speed checker double comparator, or
(b) H $\geq$1.5 volts and L $\leq$1.5 volts when using a high speed checker single comparator.
4/ Only a summary of attributes data is required.
5/ A $\geq$2.0 V, B $\leq$0.8 V. Input voltages shown are the maximum for V_{IL} and the minimum for V_{IH}.
6/ F_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.
7/ Apply 1 clock pulse prior to input pulses.
8/ Momentarily ground, then apply 4.5 V.

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TABLE III. Group A inspection for device type 06.

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
						DATA INPUTS															Min	Max	Unit
				Clear	Clock	A	B	C	D	Enable P	GND	Load	Enable T	Output Q_D	Output Q_C	Output Q_B	Output Q_A	Carry output	V_{CC}				
1 $T_C = 25°C$	V_{OH}	3006	1	4.5 V	1/			2.0 V	2.0 V		GND	GND		-800 µA	-800 µA	-800 µA	-800 µA	-800µA	4.5 V	Q_D	2.4		V
			2		1/		2.0 V													Q_C			
			3		1/	2.0 V														Q_B			
			4		1/	2.0 V	2.0 V	2.0 V	2.0 V				2.0 V							Q_A			
			5		1/															Carry output			
	V_{OL}	3007	6		1/				0.8 V	0.8 V			0.8 V	16 mA	16 mA	16 mA	16 mA	16 mA		Q_D		0.4	
			7		1/			0.8 V												Q_C			
			8		1/		0.8 V													Q_B			
			9		1/	0.8 V														Q_A			
			10																	Carry output			
	V_{IC}		11			-12 mA														Input A		-1.5	
			12				-12 mA													Input B			
			13					-12 mA												Input C			
			14						-12 mA											Input D			
			15							-12 mA										Enable P			
			16									-12 mA								Load			
			17										-12 mA							Enable T			
			18	-12 mA																Clear			
			19		-12 mA															Clock			
	I_{IH9}	3010	20		2.4 V															Clock		80	µA
	I_{IH9}		21	1/		GND	GND	GND	GND	GND		GND	2.4 V						5.5 V	Enable T		80	
	I_{IH10}		22		5.5 V															Clock		200	
	I_{IH10}		23	1/		GND	GND	GND	GND	GND		GND	5.5 V							Enable T		200	
	I_{IH11}		24	2.4 V																Clear		40	
			25			2.4 V														A			
			26				2.4 V													B			
			27					2.4 V												C			
			28						2.4 V											D			
			29							2.4 V										Enable P			
			30									5.5 V	GND							Load			
	I_{IH12}		31	5.5 V																Clear		100	
			32			5.5 V														A			
			33				5.5 V													B			
			34					5.5 V												C			
			35						5.5 V											D			
			36							5.5 V										Enable P			
			37									5.5 V	GND							Load			
	I_{IL5}	3009	38		0.4 V															Clock	-1.0	-2.3	mA
	I_{IL6}		39			0.4 V														Input A	-0.4	-1.3	
			40				0.4 V													Input B			
			41					0.4 V												Input C			
			42						0.4 V											Input D			
	I_{IL7}		43							0.4 V										Enable P	-0.7	-1.6	
	I_{IL7}		44									0.4 V	4.5 V							Load	-0.5		
	I_{IL7}		45	0.4 V																Clear	-0.7		
	I_{IL8}		46	4.5 V	1/	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V		GND	0.4 V							Enable T	-1.0	-3.2	

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.

Terminal conditions 2/

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clear	2 Clock	3 DATA INPUTS A	4 DATA INPUTS B	5 DATA INPUTS C	6 DATA INPUTS D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	I_{OS}	3011	47	4.5 V	1/			4.5 V	4.5 V		GND	GND		GND					5.5 V	Q_D	-20	-57	mA
			48												GND					Q_C			
			49				4.5 V													Q_B			
			50			4.5 V											GND			Q_A			
			51			4.5 V	4.5 V	4.5 V	4.5 V				4.5 V					GND		Carry output			
	I_{CCH}	3005	52	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		5.5 V	5.5 V							V_{CC}		85	
	I_{CCH}		53	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		GND	5.5 V									85	
	I_{CCL}		54	GND	5.5 V	GND	GND	GND	GND	GND		GND	GND									91	
	I_{CCL}		55	GND	GND	GND	GND	GND	GND	GND		GND	GND									91	
2	Same tests, terminal conditions and limits as subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = 25°C	Functional tests 4/	3014	56	B 5/	A 5/	A 5/	A 5/	A 5/	A 5/	A 5/	GND	A 5/	A 5/	L	L	L	L	L	4.5 V			See 3/	
			57	A	A	B	B	B	B				B				L						
			58		B	A	A	A	A				A				L						
			59		A	A	A	A	A				A				H						
			60		A	B	B	B	B				B				H						
			61		B	A	A	A	A				A			H							
			62		A	A	A	A	A				A			H	L						
			63		A	B	B	B	B				B				L						
			64		B	A	A	A	A				A				L						
			65		A	A	A	A	A				A				H						
			66		A	B	B	B	B				B				H						
			67		B	A	A	A	A				A				H						
			68		A	A	A	A	A				A		H	L	L						
			69		A	B	B	B	B				B				L						
			70		B	A	A	A	A				A				L						
			71		A	A	A	A	A				A				H						
			72		A	B	B	B	B				B				H						
			73		B	A	A	A	A				A			H							
			74		A	A	A	A	A				A			H	L						
			75		A	B	B	B	B				B				L						
			76		B	A	A	A	A				A				L						
			77		A	A	A	A	A				A				H						
			78		A	B	B	B	B				B				H						
			79		B	A	A	A	A				A				H						
			80		A	A	A	A	A				A	H	L	L	L						
			81		A	B	B	B	B				B				L						
			82		B	A	A	A	A				A				L						
			83		A	A	A	A	A				A				H						
			84		A	B	B	B	B				B				H						
			85		B	A	A	A	A				A				H						
			86		A	A	A	A	A				A			H	L						
			87		A	B	B	B	B				B				L						
			88		B	A	A	A	A				A				L						
			89		A	A	A	A	A				A				H						
			90		A	B	B	B	B				B				H						
			91		B	A	A	A	A				A				H						

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.

Terminal conditions 2/

Subgroup	Symbol	MIL-STD-883 method	Case E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.			DATA INPUTS															Min	Max	Unit
				Clear	Clock	A	B	C	D	Enable P	GND	Load	Enable T	Output Q_D	Output Q_C	Output Q_B	Output Q_A	Carry output	V_{CC}				
7 T_C = 25°C	Functional tests 4/	3014	92	A 5/	A 5/	A 5/	A 5/	A 5/	A 5/	A 5/	GND	A 5/	A 5/	H	H	L	L	L	4.5 V			See 3/	
			93		A	B	B	B	B				B				L						
			94		B	A	A	A	A				A				L						
			95		A	A	A	A	A				A				H						
			96		A	B	B	B	B				B				H						
			97		B	A	A	A	A				A			H	L						
			98		A	A	A	A	A				A				L						
			99		A	B	B	B	B				B				L						
			100		B	A	A	A	A				A				H	H					
			101		A	A	A	A	A				A				H	L					
			102		A	B	B	B	B				B				H	H					
			103		B	A	A	A	A				A				H	L					
			104		A									L	L	L	L	L					
			105	B	A																		
			106	A	A					B													
			107		B					B			B										
			108		A					A													
			109		B		B	B	B														
			110		A		B	B	B														
			111		B		B	B				B											
			112		A		B	B									H						
			113		A		A	A															
			114		B										H	H							
			115		A																		
			116		A							A											
			117		B							A											
			118		A		B	B	A			A											
			119		B		B	B				B		H	L	L	L						
			120		A		B	B					A	L	H	L	L	H					
			121	B	A		A	A						L	H	L	L	H					
			122	A	A									L	L	L	L	L					
			123		B									L	L	L	L	L					
			124		A									H	H	H	H	H					
			125		B									H	H	H	H	H					
			126	B	A							A		L	L	L	L	L					
			127	A	A	B						B		L	L	L		L					
			128		B									L	L	L							
			129		A									H	H	H							
			130		A			B	B					H	H	H							
			131		B									H	H								
			132		A									L	L								
			133		A	A																	
			134		B	A																	
			135		A	A											H						
			136		A	B	B	A									H						
			137		B	B											H						
			138		A	B											L						
			139		A	A											L						
			140		B	A											L						
			141		A	A											H						

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	Terminal conditions 2/ 1 Clear	2 Clock	3 DATA INPUTS A	4 DATA INPUTS B	5 DATA INPUTS C	6 DATA INPUTS D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
7 $T_C = 25°C$	Functional tests 4/	3014	142	A 5/	A 5/	B 5/	A 5/	A 5/	B 5/	A 5/	GND	B 5/	A 5/	L	H	L	H	L	4.5 V			See 3/	
			143		B	A	A	A	B							L	H						
			144		A	A	A	A	B							H	L						
			145		A	B	B		A							H							
			146		B	B										H							
			147		A	B								H	L	L							
			148		A	A										L							
			149		B	A	A									L							
			150		A	A	A									H	H						
			151		A	B	B	A								H	H						
			152		B	B										H	H						
			153		B	A									H	L	L						
			154		A	A											L						
			155		A	B	A										L						
			156		A	A											H						
8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																						
9 $T_C = 25$ C	F_{MAX} 6/	Fig 8 & 8C Fig 8 & 8B	157	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V	OUT	OUT	OUT	OUT		5.0 V	Q_A Q_B Q_C Q_D	20		MHz
			158																				
			159																				
			160																				
	t_{PLH3}		161			GND	4.5 V	4.5 V	4.5 V			IN					OUT		Carry	3	45	ns
	t_{PHL3}		162			4.5 V	4.5 V	4.5 V	4.5 V									OUT		Carry		45	
	t_{PHL4}		163				GND	GND	GND								OUT			Q_A		30	
	t_{PHL4}		164				4.5 V	GND								OUT				Q_B			
	t_{PHL4}		165					4.5 V							OUT					Q_C			
	t_{PHL4}		166					4.5 V	4.5 V					OUT						Q_D			
	t_{PLH4}		167	8/								4.5 V					OUT			Q_A		28	
	t_{PLH4}		168	4.5 V		4.5 V	GND	GND	GND			IN				OUT				Q_B			
	t_{PLH4}		169				4.5 V	GND							OUT					Q_C			
	t_{PLH4}		170				4.5 V	4.5 V						OUT						Q_D			
	t_{PLH4}	Fig 8 and 8A	171	8/		IN	4.5 V	4.5 V	4.5 V			GND				OUT	OUT			Q_A			
	t_{PLH4}		172			4.5 V	IN	4.5 V	4.5 V											Q_B			
	t_{PLH4}		173				4.5 V	IN	4.5 V						OUT					Q_C			
	t_{PLH4}		174					4.5 V	IN					OUT						Q_D			
	t_{PLH5}		175	4.5 V	7/				4.5 V				IN					OUT		Carry		18	
	t_{PHL5}		176	4.5 V	7/								IN					OUT		Carry		21	
	t_{PHL6}		177	IN	IN								4.5 V				OUT			Q_A		39	
	t_{PHL6}		178													OUT				Q_B			
	t_{PHL6}		179												OUT					Q_C			
	t_{PHL6}		180											OUT						Q_D			

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TABLE III. Group A inspection for device type 06 -Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test no.	Terminal conditions 2/ 1 Clear	2 Clock	3 Data inputs A	4 Data inputs B	5 Data inputs C	6 Data inputs D	7 Enable P	8 GND	9 Load	10 Enable T	11 Output Q_D	12 Output Q_C	13 Output Q_B	14 Output Q_A	15 Carry output	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
10 . $T_C = 125°C$	6/ F_{MAX}	Fig 8 & 8C Fig 8 & 8B	181 182 183 184	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V	OUT	OUT	OUT	OUT		5.0 V	Q_A Q_B Q_C Q_D	20		MHz
	t_{PLH3} t_{PHL3} t_{PHL4} t_{PHL4} t_{PHL4} t_{PLH4} t_{PLH4} t_{PLH4} t_{PLH4}		185 186 187 188 189 190 191 192 193 194	8/ 4.5 V 4.5 V 4.5 V		GND 4.5 V 4.5 V 4.5 V	4.5 V 4.5 V GND 4.5 V GND 4.5 V GND 4.5 V	4.5 V 4.5 V GND GND 4.5 V 4.5 V GND GND GND GND	4.5 V 4.5 V GND GND 4.5 V 4.5 V GND GND GND GND			IN 4.5 V IN IN IN		OUT OUT OUT	OUT OUT OUT	OUT OUT OUT	OUT OUT OUT	OUT OUT		Carry Carry Q_A Q_B Q_C Q_D Q_A Q_B Q_C Q_D	3	58 58 38 36	ns
	t_{PLH4} t_{PLH4} t_{PLH4} t_{PLH4} t_{PLH5} t_{PHL5} t_{PHL6} t_{PHL6} t_{PHL6} t_{PHL6}	Fig 8 and 8A	195 196 197 198 199 200 201 202 203 204	8/ 4.5 V 4.5 V IN	7/ 7/ IN	IN 4.5 V	4.5 V IN 4.5 V	4.5 V 4.5 V IN 4.5 V	4.5 V 4.5 V 4.5 V IN 4.5 V			GND	IN IN 4.5 V	OUT OUT	OUT OUT	OUT OUT	OUT OUT	OUT OUT		Q_A Q_B Q_C Q_D Carry Carry Q_A Q_B Q_C Q_D		27 27 47	

1/ Normal clock pulse: ($V_{IL} \leq 0.8$ V, $V_{IH} \geq 2.0$ V).
2/ Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.8 V, or open).
3/ Output voltages shall be either:
(a) H = 2.4 volts minimum and L = 0.4 volt maximum when using a high speed checker double comparator, or
(b) H ≥ 1.5 volts and L ≤ 1.5 volts when using a high speed checker single comparator.
4/ Only a summary of attributes data is required.
5/ A ≥ 2.0 V, B ≤ 0.8 V. Input voltages shown are the maximum for V_{IL} and the minimum for V_{IH}.
6/ F_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be as follows: $F_{MAX/2}$ at Q_A, $F_{MAX/4}$ at Q_B, $F_{MAX/8}$ at Q_C, $F_{MAX/16}$ at Q_D.
7/ Apply 1 clock pulse prior to input pulses.
8/ Apply momentary GND, then 4.5 volts prior to input pulses, maintain 4.5 volts during test.

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TABLE III. Group A inspection for device type 07.

Terminal conditions 12/

Subgroup	Symbol	MIL-STD-883 method	Case C, D Test No.	1 Input BD	2 Reset $R_0(1)$	3 Reset $R_0(2)$	4 NC	5 V_{CC}	6 Reset $R_g(1)$	7 Reset $R_g(2)$	8 Output C	9 Output B	10 GND	11 Output D	12 Output A	13 NC	14 Input A	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	V_{OL}	3007	1	2.0 V	2.0 V	2.0 V		4.5 V	GND	GND	16 mA	16 mA	GND	16 mA	16 mA		GND	Output C		0.4	Vdc
			2	2.0 V					GND	GND							GND	Output B			
			3	2.0 V					4.5 V	0.8 V							GND	Output D			
			4	GND					0.8 V	4.5 V							2.0 V	Output A			
			5	2.0 V	4.5 V	0.8 V			2.0 V	2.0 V	16 mA						GND	Output C			
			6	2.0 V	0.8 V	4.5 V						16 mA					GND	Output B			
	V_{OH}	3006	7	2.0 V	4.5 V	0.8 V								-0.4mA			GND	Output D	2.4		V
			8	GND	0.8 V	4.5 V									-0.4 mA		2.0 V	Output A			
			9	2.0 V		0.8 V								-0.4mA			GND	Output D			
			10	GND		0.8 V									-0.4 mA		2.0 V	Output A			
			11	1/ 4/	1/	1/			0.8 V	0.8 V	-0.4 mA						GND	Output C			
			12	1/ 3/	1/	1/			0.8 V	0.8 V		-0.4mA					GND	Output B			
	I_{IL1}	3009	13		0.4 V	4.5 V		5.5 V										Reset $R_0(1)$	13/	13/	mA
			14		4.5 V	0.4 V												Reset $R_0(2)$			
			15						0.4 V	4.5 V								Reset $R_g(1)$			
			16						4.5 V	0.4 V								Reset $R_g(2)$			
	I_{IL2}		17		GND	GND			2/	2/							0.4 V	Input A			
	I_{IL3}		18	0.4 V	GND	GND			2/	2/								Input BD			
	V_{IC}		19		-12 mA			4.5 V										Reset $R_0(1)$		-1.5	V
			20			-12 mA												Reset $R_0(2)$			
			21						-12 mA									Reset $R_g(1)$			
			22							-12 mA								Reset $R_g(2)$			
			23		GND	GND			GND	GND							-12 mA	Input A			
			24	-12 mA	GND	GND			GND	GND								BD Input			

See footnotes at end of device type 07.

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TABLE III. Group A inspection for device type 07 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case C, D Test No.	1 Input BD	2 Reset $R_0(1)$	3 Reset $R_0(2)$	4 NC	5 V_{CC}	6 Reset $R_9(1)$	7 Reset $R_9(2)$	8 Output C	9 Output B	10 GND	11 Output D	12 Output A	13 NC	14 Input A	Meas. terminal	Test limits Min	Max	Unit
Terminal conditions 12/																					
1 T_C=25°C	I_{IH1}	3010	25		2.4 V	GND		5.5 V					GND					Reset $R_0(1)$		40	µA
			26		GND	2.4 V												Reset $R_0(2)$			
			27						2.4 V	GND								Reset $R_9(1)$			
			28						GND	2.4 V								Reset $R_9(2)$		80	
	I_{IH3}		29		GND	GND			GND	GND							2.4 V	Input A		160	
	I_{IH5}		30	2.4 V	GND	GND			GND	GND								BD Input			
	I_{IH2}		31		5.5 V	GND												Reset $R_0(1)$		100	
			32		GND	5.5 V												Reset $R_0(2)$			
			33						5.5 V	GND								Reset $R_9(1)$			
			34						GND	5.5 V								Reset $R_9(2)$		200	
	I_{IH4}		35		GND	GND			GND	GND							5.5 V	Input A		400	
	I_{IH6}		36	5.5 V	GND	GND			GND	GND								BD Input			
	I_{OS}	3011	37		GND	GND			4.5 V	4.5 V				GND	GND			Output D	-20	-57	mA
			38		GND	GND			4.5 V	4.5 V								Output A			
			39	2/ 3/	2/	2/			GND	GND		GND						Output B			
			40	2/ 4/	2/	2/			GND	GND	GND							Output C			
	I_{CC}		41	GND	2/	2/			GND	GND							GND	V_{CC}		44	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C=25°C	Functional tests 5/	3014	42	A 6/	A 6/	A 6/		4.5 V	A 6/	A 6/	L	L	GND	H	H		B 6/	See 7/		See 7/	
			43	B					A	A				H	H						
			44	B					B	B				L	L						
			45	A	B	B			A	B				H	H						
			46	B	B	B			B	B				L	L						
			47	B	A	A			A	A				L	H						
			48	A	A	A			B	B				H	L						
			49	B	A	B			A	B				H	L						
			50	A	B	A			B	A				L	H						
			51	B					A	B				H	H						
			52	A					B	A				L	L						
			53	B					A	B				H	H						
			54	A					B	A				L	L						
			55	B					A	B				H	H						
			56	A					B	A				L	L						
			57	B					A	B				H	H						
			58	A					B	A				L	L						
			59	B					A	B				H	H						
			60	B		B			A	B				L	L						
			61	A					A	B				H	H						
			62	B	A				A	B				L	L						

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions 12/

Subgroup	Symbol	MIL-STD-883 method	Case C, D Test No.	1 Input BD	2 Reset $R_0(1)$	3 Reset $R_0(2)$	4 NC	5 V_{CC}	6 Reset $R_9(1)$	7 Reset $R_9(2)$	8 Output C	9 Output B	10 GND	11 Output D	12 Output A	13 NC	14 Input A	Meas. terminal	Test limits Min	Max	Unit
7 T_C = 25°C	Functional tests 5/	3014	63	A 6/	A 6/	B 6/		4.5 V	A 6/	B 6/	L	H	GND	L	L		B 6/				
			64	B							H	L									
			65	A								L									
			66	B								H									
			67	A								H									
			68	B							L	L		H							
			69	A										H							
			70	B										L							
			71	B	B				B												
			72	A	B																
			73	B	A					A		H									
			74	A								H									
			75	B							H	L									
			76	A								L									
			77	B								H									
			78	B		A			A	B	L	L									
			79	B	B				A												
			80	A								H									
			81	B								H									
			82	A								H									
			83	B							H	L									
			84	A								L									
			85	B								H									
			86	A		B				A	L	L		H	H						
			87	A		B				B				H	H						
			88	A	A	A			B					L	L				See 7/		
			89	B	A	A															
			90	A	B	B															
			91	B								H									
			92	A								H									
			93	B							H	L									
			94	A								L									
			95	B								H									
			96	A								H									
			97	B	A	A					L	L									
			98	B	B	B						L									
			99	A								L									
			100	B								H									
			101	A								H									
			102	B							H	L									
			103	A								L									
			104	B								H									
			105	A								H									
			106	A					A	A	L	L		H	H						
			107	B					A	A											
			108	B					B	B											
			109	A										L	L						
			110	A	A	A								L	L						
			111	B	A	A								L	L						
• 8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																				

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TABLE III. Group A inspection for device type 07 - Continued.

Terminal conditions 12/

Subgroup	Symbol	MIL-STD-883 method	Case C, D Test No.	1 Input BD	2 Reset $R_0(1)$	3 Reset $R_0(2)$	4 NC	5 V_{CC}	6 Reset $R_9(1)$	7 Reset $R_9(2)$	8 Output C	9 Output B	10 GND	11 Output D	12 Output A	13 NC	14 Input A	Meas. terminal	Test limits Min	Max	Unit
9 $T_C = 25°C$	F_{MAX} 8/	Fig 9	112		GND	GND		5.0 V	GND	GND	11/	11/	GND	11/	OUT 10/		IN	Output A	10		MHz
	t_{PLH1}		113	9/							OUT 10/	10/		11/	9/ 10/		IN	Output C	20	100	ns
	t_{PHL1}		114	9/							OUT 10/	10/		11/	9/ 10/		IN	Output C	20	100	ns
10 $T_C = 125°C$	F_{MAX} 8/		115								11/	11/		11/	OUT 10/		IN	Output A	10		MHz
	t_{PLH1}		116	9/							OUT 10/	10/		11/	9/ 10/		IN	Output C	20	115	ns
	t_{PHL1}		117	9/							OUT 10/	10/		11/	9/ 10/		IN	Output C	20	115	ns
11	Same test terminal conditions and limits as subgroup 10, except T_C = -55°C.																				

1/ Momentarily apply 2.0 V, then ground prior to taking measurements to set the device in desired state. Maintain ground for measurement.
2/ Apply 4.5 V pulse then ground prior to taking measurements to set the device in the desired state. Maintain ground for measurement.
3/ Apply one pulse after reset (R_0) pulses.
4/ Apply two pulses after reset (R_0) pulses.
5/ Only a summary of attributes data is required.
6/ A $\geq$ 2.0 V, B $\leq$ 0.8 V. Input voltages shown are the maximum for V_{IL} and the minimum for V_{IH}.
7/ Output voltages shall be either:
(a) H = 2.4 volts minimum and L = 0.4 volt maximum when using a high speed checker double comparator, or
(b) H $\geq$ 1.5 volts and L $\leq$ 1.5 volts when using a high speed checker single comparator.
8/ F_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.
9/ Connect terminals together during test.
10/ See test figure 9 for terminal load.
11/ Omit specified loads for this test.
12/ Terminal conditions (pins not designated may be H $\geq$ 2.0 V, or L $\leq$ 0.8 V, or open).
13/ The limits shown shall be as follows:

Test	Min/Max limits (mA) for circuit A	B	C
I_{IL1}	-0.4/-1.3	-0.4/-1.3	-0.7/-1.6
I_{IL2}	-1.4/-3.2	-0.7/-3.2	-0.7/-3.2
I_{IL3}	-1.4/-4.8	-0.7/-4.8	-1.4/-6.4

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TABLE III. Group A inspection for device type 08.
Terminal conditions 17/

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Data B Input	2 Output Q_B	3 Output Q_A	4 Count Down Input	5 Count Up Input	6 Output Q_C	7 Output Q_D	8 GND	9 Data D Input	10 Data C Input	11 Load	12 Carry Output	13 Borrow Output	14 Clear Input	15 Data A Input	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 $T_C = 25°C$	I_{IL2}	3010	1								GND			GND			GND	0.4 V	5.5 V	Data A Input	-0.7	-1.6	mAdc
	I_{IL6}		2	0.4 V																Data B Input			
			3										0.4 V							Data C Input			
			4									0.4 V								Data D Input			
	I_{IL7}		5											0.4 V						Load	18/	18/	
	I_{IL8}		6														0.4 V			Clear Input			
	I_{IL5}		7					0.4 V												Count Up Input			
			8				0.4 V													Count Down Input			
	I_{IH11}		9											4.5 V			4.5 V	2.4 V		Data A Input		40	µAdc
			10	2.4 V																Data B Input			
			11										2.4 V							Data C Input			
			12									2.4 V								Data D Input			
			13											2.4 V						Load			
			14														2.4 V			Clear Input			
			15					2.4 V												Count Up Input			
			16				2.4 V													Count Down Input			
	I_{IH12}		17											4.5 V			4.5 V	5.5 V		Data A Input		100	µA
			18	5.5 V																Data B Input			
			19										5.5 V							Data C Input			
			20									5.5 V								Data D Input			
			21											5.5 V						Load			
			22														5.5 V			Clear Input			
			23					5.5 V												Count Up Input			
			24				5.5 V													Count Down Input			

See footnotes at end of device type 08.

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TABLE III. Group A inspection for device type 08 - Continued.
Terminal conditions 17/

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Data B Input	2 Output Q_B	3 Output Q_A	4 Count Down Input	5 Count Up Input	6 Output Q_C	7 Output Q_D	8 GND	9 Data D Input	10 Data C Input	11 Load	12 Carry Output	13 Borrow Output	14 Clear Input	15 Data A Input	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	V_{OH}	3006	25	2.0 V	-400µA	-400µA					GND			0.8 V			0.8 V	2.0 V	4.5 V	Output Q_A	2.4		Vdc
			26						-400 µA				2.0 V							Output Q_B			
			27							-400 µA										Output Q_C			
			28									2.0 V			-400 µA					Output Q_D			
			29					2.0 V												Carry Output			
			30				2.0 V									-400µA				Borrow Output			
	V_{OL}	3007	31	0.8 V	16 mA	16 mA								0.8 V			0.8 V	0.8 V		Output Q_A		0.4	
			32						16 mA				0.8 V							Output Q_B			
			33							16 mA		0.8 V								Output Q_C			
			34									2.4 V	2.4 V	GND	16 mA			2.4 V		Output Q_D			
			35	2.4 V				0.8 V												Carry Output			
			36				0.8 V									16 mA	2.0 V			Borrow Output			
			37				0.8 V							2.0 V		16 mA	2.0 V			Borrow Output			
	I_{OS}	3011	38	2.4 V	GND	GND								GND			GND	2.4 V	5.5 V	Output Q_A	-20	-65	mAdc
			39						GND				2.4 V							Output Q_B			
			40							GND		2.4 V			GND					Output Q_C			
			41					2.4 V												Output Q_D			
			42													GND				Carry Output			
			43				2.4 V													Borrow Output			
	I_{CC}	3010	44											GND						V_{CC}		89	
	V_{IC}		45	-12 mA														-12 mA	4.5 V	Data A Input		-1.5	V
			46										-12 mA							Data B Input			
			47									-12 mA		-12 mA			-12 mA			Data C Input			
			48																	Data D Input			
			49																	Load			
			50																	Clear Input			
			51					-12 mA												Count Up Input			
			52				-12 mA													Count Down Input			
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C, and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C, and V_{IC} tests are omitted.																						

See foonotes at end of device type 08.

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TABLE III. Group A inspection for device type 08 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Data B Input	2 Output Q_B	3 Output Q_A	4 Count Down Input	5 Count Up Input	6 Output Q_C	7 Output Q_D	8 GND	9 Data D Input	10 Data C Input	11 Load	12 Carry Output	13 Borrow Output	14 Clear Input	15 Data A Input	16 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
7 $T_C = 25°C$	Functional tests 2/	3014	53	A 3/	L	L	A 3/	A 3/	L	L	GND	A 3/	A 3/	A 3/	H	H	A 3/	A 3/	4.5 V			See 1/	
			54			L		A									B						
			55			L		B															
			56			H		A															
			57			H		B															
			58		H	L		A															
			59			L		B															
			60			H		A															
			61			H		B															
			62		L	L		A	H														
			63			L		B															
			64			H		A															
			65			H		B															
			66		H	L		A															
			67			L		B															
			68			H		A															
			69			H		B															
			70		L	L		A	L	H													
			71			L		B															
			72			H		A															
			73			H		B							L								
			74			L		A		L					H								
			75			L	B			L						L							
			76			H	A			H						H							
			77			H	B																
			78			L	A																
			79			L	B																
			80		H	H	A		H	L													
			81			H	B																
			82			L	A																
			83			L	B																
			84		L	H	A																
			85			H	B																
			86			L	A																
			87			L	B																
			88		H	H	A		L														
			89			H	B																
			90			L	A																
			91			L	B																
			92		L	H	A																
			93			H	B																
			94			L	A																
			95			L	B			H						L							
			96			H	A			H						H							
			97		H	H			H	H				B			A					
			98		L	L			L	L				B			B					
			99		H	H			H	H				B			A					
			100		H	H			H	H				A			B					
			101		L	L			L	L				A			A					
			102		L	L			L	L				A			B					

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TABLE III. Group A inspection for device type 08 - Continued.
Terminal conditions 17/

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Data B Input	2 Output Q_B	3 Output Q_A	4 Count Down Input	5 Count Up Input	6 Output Q_C	7 Output Q_D	8 GND	9 Data D Input	10 Data C Input	11 Load	12 Carry Output	13 Borrow Output	14 Clear Input	15 Data A Input	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
7 T_C = 25°C	Functional tests 2/	3014	103	A 3/	H	H	A 3/	A 3/	H	H	GND	A 3/	A 3/	B 3/	H	H	B 3/	A 3/	4.5 V				
			104	A	H	H			H	H		A	A	A		L		A				
			105	B	H	H			H	H		B	B	B		H		B				
			106		H	L			L	L													
			107		L	L	B																
			108				A	B															
			109					A															
			110											A									
			111				B										A						
			112																				
			113				A	B								L							
			114													H							
			115				A																
			116				A	B								L							
																H							
9 T_C = 25° C	F_{MAX} 14/	Fig 10A	117			OUT	2.4 V	4/						2.4 V			GND		5.0 V	Output Q_A	20		MHz
	F_{MAX} 14/	Fig 10A	118				4/	2.4 V						2.4 V			GND			Output Q_A	20		MHz
	t_{PHL7}	Fig 10	119				GND	GND						5/			5/	2.4 V		Output Q_A	3	44	ns
	t_{PLH8}	Fig 10	120				GND	GND						6/			GND	6/		Output Q_A		44	
	t_{PHL8}	Fig 10	121				GND	GND						7/			GND	7/		Output Q_A		44	
	t_{PHL9}	Fig 10A	122	GND			2.4 V	8/				2.4 V	GND	8/	OUT		8/ GND	GND		Carry Output		28	
	t_{PLH9}		123					9/				2.4 V		9/	OUT		9/			Carry Output		30	
	t_{PHL10}		124				10/	2.4 V				GND		2.4 V		OUT	10/			Borrow Output		28	
	t_{PLH10}		125				11/	2.4 V				GND		2.4 V		OUT	11/			Borrow Output		28	
	t_{PLH11}		126	GND		OUT	2.4 V	15/				GND		15/			GND	GND		Output Q_A		42	
			127	GND	OUT													2.4 V		Output Q_B			
			128	2.4 V					OUT									2.4 V		Output Q_C			
			129	2.4 V						OUT			2.4 V				2.4 V		Output Q_D			
	t_{PHL11}		130	GND			16/	2.4 V		OUT		2.4 V	GND	16/			GND			Output Q_D		51	
			131	GND			16/	2.4 V	OUT			GND	2.4 V				GND			Output Q_C			
			132	2.4 V	OUT		16/	2.4 V				GND	GND				GND			Output Q_B			
			133	GND		OUT	16/	2.4 V				GND	GND					2.4 V		Output Q_A			

See footnotes at end of device type 08.

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TABLE III. Group A inspection for device type 08--Continued.

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test no.	Terminal conditions 17/ 1 Data B Input	2 Output Q_B	3 Output Q_A	4 Count Down Input	5 Count Up Input	6 Output Q_C	7 Output Q_D	8 GND	9 Data D Input	10 Data C Input	11 Load	12 Carry Output	13 Borrow Output	14 Clear Input	15 Data A Input	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
10	F_{MAX} 14/	Fig 10A	134			OUT	2.4 V	4/			GND			2.4 V			GND		5.0 V	Output Q_A	20		MHz
T_C = 125 °C	F_{MAX} 14/	Fig 10A	135				4/	2.4 V						2.4 V			GND			Output Q_A	20		MHz
	t_{PHL7}	Fig 10	136				GND	GND						5/			5/	2.4 V		Output Q_A	3	57	ns
	t_{PLH8}	Fig 10	137				GND	GND						6/			GND	6/		Output Q_A		66	
	t_{PHL8}	Fig 10	138				GND	GND						7/			GND	7/		Output Q_A		62	
	t_{PHL9}	Fig 10A	139	GND			2.4 V	8/				2.4 V	GND	8/	OUT		8/ GND	GND		Carry Output		39	
	t_{PLH9}		140				2.4 V	9/				2.4 V		9/	OUT		9/			Carry Output		45	
	t_{PHL10}		141				10/	2.4 V				GND		2.4 V		OUT	10/			Borrow Output		39	
	t_{PLH10}		142				11/	2.4 V				GND		2.4 V		OUT	11/			Borrow Output		42	
	t_{PLH11}		143	GND		OUT	2.4 V	15/				GND	GND	15/			GND	GND		Output Q_A		63	
			144	GND	OUT		2.4 V	15/				GND	GND	15/				2.4 V		Output Q_B		63	
			145	2.4 V			2.4 V	15/	OUT			GND	GND	15/				2.4 V		Output Q_C		63	
			146	2.4 V			2.4 V	15/		OUT		GND	2.4 V	15/				2.4 V		Output Q_D		63	
	t_{PHL11}		147	GND			16/	2.4 V		OUT		2.4 V	GND	16/				GND		Output Q_D		71	
			148	GND			16/	2.4 V	OUT			GND	2.4 V	16/				GND		Output Q_C		71	
			149	2.4 V	OUT		16/	2.4 V				GND	GND	16/				GND		Output Q_B		71	
			150	GND		OUT	16/	2.4 V				GND	GND	16/				2.4 V		Output Q_A		71	
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55 °C.																						

1/ Output voltages shall be either:
(a) H = 2.4 volts minimum and L = 0.4 volt maximum when using a high speed checker double comparator, or
(b) H $\geq$1.5 volts and L <1.5 volts when using a high speed checker single comparator.
2/ Only a summary of attributes data is required.
3/ A $\geq$2.0 V, B $\leq$0.8 V. Input voltages shown are the maximum for V_{IL} and the minimum for V_{IH}.
4/ See figure 10A. Apply waveform E for tests 117 and 134, and waveform H for tests 118 and 135.
5/ See figure 10, waveforms A, B, C, and D. Load A_{IN}, then apply waveform A and observe the drop in Q_A output.
6/ See figure 10, waveforms B, C, and D. Load A_{IN}, and observe t_{PLH} of waveform D at output Q_A.
7/ Repeat steps as in note 6. Observe the pulse delay of the drop in Q_A when the load pulse is applied.
8/ See figures 10 and 10A, waveforms A, B, C, E, F, and G. First use clear pulse, then load pulse, then one count up pulse. Also see figure 10A, note 4.
9/ See note 5, add one more count up pulse for t_{PLH9} as shown on figure 10A, waveforms F and G.
10/ See figures 10 and 10A. First clear pulse, then one count down pulse.
11/ See note 7. Clear pulse first, figure 10 waveforms A, B, and C, add one more count down pulse for t_{PLH10} as shown on figure 10A, waveforms J and K.
12/ See figures 10 and 10A. First a clear pulse, then one count up pulse for Q_A. On next three tests enter count up pulses to appropriate count to check Q_B, Q_C, and Q_D.
13/ See figure 10A waveforms H and J. On next three tests enter count down pulse, waveform H and observe t_{PHL11}, waveform J on outputs Q_C, Q_B, and Q_A.
14/ F_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.
15/ Load pulse before count up input, then 4.5 V.
16/ Load pulse before count down input, then 4.5 V.
17/ Terminal conditions (pins not designated may be H $\geq$2.0 V, or L $\leq$ 0.8 V, or open).
18/ The limits shall be: -0.5 mA minimum, -1.3 mA maximum for circuit C,and -0.7 mA minimum, -1.6 mA maximum for all other circuits.

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TABLE III. Group A inspection for device type 09.
Terminal conditions 17/

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Data B Input	2 Output Q_B	3 Output Q_A	4 Count Down Input	5 Count Up Input	6 Output Q_C	7 Output Q_D	8 GND	9 Data D Input	10 Data C Input	11 Load	12 Carry Output	13 Borrow Output	14 Clear Input	15 Data A Input	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	I_{IL2}	3010	1								GND			GND			GND	0.4 V	5.5 V	Data A Input	-0.7	-1.6	mAdc
	I_{IL6}		2	0.4 V																Data B Input			
			3										0.4 V							Data C Input			
			4									0.4 V								Data D Input			
	I_{IL7}		5											0.4 V						Load	18/	18/	
	I_{IL8}		6														0.4 V			Clear Input			
	I_{IL5}		7					0.4 V												Count Up Input			
	I_{IL5}		8				0.4 V													Count Down Input			
	I_{IH11}		9											4.5 V			4.5 V	2.4 V		Data A Input		40	µAdc
			10	2.4 V																Data B Input			
			11										2.4 V							Data C Input			
			12									2.4 V								Data D Input			
			13											2.4 V						Load			
			14														2.4 V			Clear Input			
			15					2.4 V												Count Up Input			
			16				2.4 V													Count Down Input			
	I_{IH12}		17											4.5 V			4.5 V	5.5 V		Data A Input		100	µA
			18	5.5 V																Data B Input			
			19										5.5 V							Data C Input			
			20									5.5 V								Data D Input			
			21											5.5 V						Load			
			22														5.5 V			Clear Input			
			23					5.5 V												Count Up Input			
			24				5.5 V													Count Down Input			

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TABLE III. Group A inspection for device type 09 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Data B Input	2 Output Q_B	3 Output Q_A	4 Count Down Input	5 Count Up Input	6 Output Q_C	7 Output Q_D	8 GND	9 Data D Input	10 Data C Input	11 Load	12 Carry Output	13 Borrow Output	14 Clear Input	15 Data A Input	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	V_{OH}	3006	25 26 27 28 29 30	2.0 V	-400 µA	-400 µA	2.0 V	2.0 V	-400 µA	-400 µA	GND	2.0 V	2.0 V	0.8 V	-400 µA	-400µA	0.8 V	2.0 V	4.5 V	Output Q_A Output Q_B Output Q_C Output Q_D Carry Output Borrow Output	2.4		Vdc
	V_{OL}	3007	31 32 33 34 35 36	0.8 V 2.4 V	16 mA	16 mA	0.8 V	0.8 V	16 mA	16 mA		0.8 V 2.4 V	0.8 V 2.4 V	0.8 V GND	16 mA	16 mA	0.8 V 2.0 V	0.8 V 2.4 V		Output Q_A Output Q_B Output Q_C Output Q_D Carry Output Borrow Output		0.4	
	I_{OS}	3011	37 38 39 40 41 42	2.4 V	GND	GND	2.4 V	2.4 V	GND	GND		2.4 V	2.4 V	GND	GND	GND	0.8 V	2.4 V	5.5 V	Output Q_A Output Q_B Output Q_C Output Q_D Carry Output Borrow Output	-20	-65	mAdc
														GND						V_{CC}		89	
	I_{CC}	3010	43															-12 mA	4.5 V	Data A Input Data B Input Data C Input Data D Input Load Clear Input Count Up Input Count Down Input		-1.5	V
	V_{IC}		44 45 46 47 48 49 50 51	-12 mA			-12 mA	-12 mA				-12 mA	-12 mA	-12 mA			-12 mA						
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

Terminal conditions 17/

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TABLE III. Group A inspection for device type 09 - Continued.
Terminal conditions 17/

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Data B Input	2 Output Q_B	3 Output Q_A	4 Count Down Input	5 Count Up Input	6 Output Q_C	7 Output Q_D	8 GND	9 Data D Input	10 Data C Input	11 Load	12 Carry Output	13 Borrow Output	14 Clear Input	15 Data A Input	16 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
7 T_C = 25°C	Functional tests 2/	3014	52	B 3/	L	L	A 3/	A 3/	L	L	GND	B 3/	B 3/	B 3/	H	H	B 3/	B 3/	4.5 V			See 1/	
			53			L								A				B					
			54			H								B				A					
			55			H								A				A					
			56	A	H	L								B				B					
			57			L								A				B					
			58			H								B				A					
			59			H								A				A					
			60	B	L	L			H				A	B				B					
			61			L								A				B					
			62			H								B				A					
			63			H								A				A					
			64	A	H	L								B				B					
			65			L								A				B					
			66			H								B				A					
			67			H				H				A				A					
			68	B	L	L			L			A	B	B				B					
			69			L								A				B					
			70			H								B				A					
			71			H								A				A					
			72	A	H	L								B				B					
			73			L								A				A					
			74			H								B				B					
			75			H								A				A					
			76	B	L	L			H				A	B				B					
			77			L								A				B					
			78			H								B				A					
			79			H								A				A					
			80	A	H	L								B				B					
			81		H	L								A				A					
			82		H	H								B				B					
			83		L	L			L	L				B			A	A					
			84											A				B					
			85																				
			86				B									L							
			87		H	H	A		H	H						H							
			88			H	B																
			89			L	A																
			90			L	B																
			91		L	H	A																
			92			H	B																
			93			L	A																
			94			L	B																
			95		H	H	A		L														
			96			H	B																
			97			L	A																
			98			L	B																
			99		L	H	A																
			100		L	H	B																
			101		L	L	A																

See footnotes at end of device type 09.

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TABLE III. Group A inspection for device type 09 - Continued.
Terminal conditions 17/

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Data B Input	2 Output Q_B	3 Output Q_A	4 Count Down Input	5 Count Up Input	6 Output Q_C	7 Output Q_D	8 GND	9 Data D Input	10 Data C Input	11 Load	12 Carry Output	13 Borrow Output	14 Clear Input	15 Data A Input	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
7 T_C = 125°C	Functional tests 2/	3014	102	A 3/	L	L	B 3/	A 3/	L	H	GND	A 3/	A 3/	A 3/	H	H	B 3/	A 3/	4.5 V			See 1/	
			103		H	H	A		H	L													
			104			H	B																
			105			L	A																
			106		↓	L	B																
			107		L	H	A																
			108			H	B																
			109			L	A																
			110		↓	L	B		↓														
			111		H	H	A		L														
			112			H	B																
			113			L	A																
			114		↓	L	B																
			115		L	H	A																
			116			H	B																
			117			L	A	↓															
			118			L		B															
			119			H		A															
			120		↓	H		B															
			121		H	L		A															
			122			L		B															
			123			H		A															
			124		↓	H		B	↓														
			125		L	L		A	H														
			126			L		B															
			127			H		A															
			128		↓	H		B															
			129		H	L		A															
			130		H	L		B															
			131		H	H		A															
			132		H	H		B	↓	↓													
			133		L	L		A	L	H													
			134			L		B															
			135			H		A															
			136		↓	H		B															
			137		H	L		A															
			138			L		B															
			139			H		A															
			140		↓	H		B	↓														
			141		L	L		A	H														
			142			L		B															
			143			H		A															
			144		↓	H		B															
			145		H	L		A															
			146			L		B															
			147			H		A															
			148		↓	H		B	↓	↓					L								
			149		L	L		A	L	L					H								
			150		L	L		A	L	L					H								
			151		L	L	↓	B	L	L	↓	↓	↓	↓	H	↓	A A	↓	↓	↓			

See footnotes at end of device type 09.

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TABLE III. Group A inspection for device type 09 -Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test no.	1 Data B Input	2 Output Q_B	3 Output Q_A	4 Count Down Input	5 Count Up Input	6 Output Q_C	7 Output Q_D	8 GND	9 Data D Input	10 Data C Input	11 Load	12 Carry Output	13 Borrow Output	14 Clear Input	15 Data A Input	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
7 $T_C = 125°C$	Functional tests 2/	3014	152	A 3/	L	L	A 3/	A 3/	L	L	GND	A 3/	A 3/	A 3/	H	H	A 3/	A 3/	4.5 V				
			153				B									L							
			154				A									H							
			155														B						
			156		H	H			H	H				B							See 1/		
			157					B							L								
			158					A							H								
			159				B	A							H								
			160				A	A							H								
9 $T_C = 25°C$	F_{MAX} 14/	Fig 10A	161			OUT	2.4 V	4/						2.4 V			GND		5.0 V	Output Q_A	20		MHz
	F_{MAX} 14/	Fig 10A	162				4/	2.4 V						2.4 V			GND			Output Q_A	20		MHz
	t_{PHL7}	Fig 10	163				GND	GND						5/			5/	2.4 V		Output Q_A	3	44	ns
	t_{PLH8}	Fig 10	164				GND	GND						6/			GND	6/		Output Q_A		44	
	t_{PHL8}	Fig 10	165				GND	GND						7/			GND	7/		Output Q_A		44	
	t_{PHL9}	Fig 10A	166	2.4 V			2.4 V	8/				2.4 V	2.4 V	8/	OUT		8/ GND	GND		Carry Output		28	
	t_{PLH9}		167	2.4 V				9/				2.4 V	2.4 V	9/	OUT		9/			Carry Output		30	
	t_{PHL10}		168	GND			10/	2.4 V				GND	GND	2.4 V		OUT	10/			Borrow Output		28	
	t_{PLH10}		169	GND			11/	2.4 V				GND	GND	2.4 V		OUT	11/			Borrow Output		28	
	t_{PLH11}		170	GND		OUT	2.4 V	15/				GND	GND	2.4 V			GND	GND		Output Q_A		42	
			171	GND	OUT								GND					2.4 V		Output Q_B			
			172	2.4 V					OUT				GND					2.4 V		Output Q_C			
			173	2.4 V						OUT			2.4 V					2.4 V		Output Q_D			
	t_{PHL11}		174	GND			16/	2.4 V		OUT		2.4 V	GND	16/				GND		Output Q_D		51	
			175	GND					OUT			GND	2.4 V	16/				GND		Output Q_C			
			176	2.4 V	OUT							GND	GND	16/				GND		Output Q_B			
			177	GND		OUT						GND	GND	16/				2.4 V		Output Q_A			

See footnotes at end of device type 09.

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TABLE III. Group A inspection for device type 09 -Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test no.	Terminal conditions 17/ 1 Data B Input	2 Output Q_B	3 Output Q_A	4 Count Down Input	5 Count Up Input	6 Output Q_C	7 Output Q_D	8 GND	9 Data D Input	10 Data C Input	11 Load	12 Carry Output	13 Borrow Output	14 Clear Input	15 Data A Input	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
10	F_{MAX} 14/	Fig 10A	178			OUT	2.4 V	4/			GND			2.4 V			GND		5.0 V	Output Q_B	20		MHz
T_C = 125 °C	F_{MAX} 14/	Fig 10A	179				4/	2.4 V						2.4 V			GND			Output Q_A	20		MHz
	t_{PHL7}	Fig 10	180				GND	GND						5/			5/	2.4 V		Output Q_A	3	57	ns
	t_{PLH8}	Fig 10	181				GND	GND						6/			GND	6/		Output Q_A		66	
	t_{PHL8}	Fig 10	182				GND	GND						7/			GND	7/		Output Q_A		62	
	t_{PHL9}	Fig 10A	183	2.4 V			2.4 V	8/				2.4 V	2.4 V	8/	OUT		8/GND	GND		Carry Output		39	
	t_{PLH9}		184	2.4 V			2.4 V	9/				2.4 V	2.4 V	9/	OUT		9/			Carry Output		45	
	t_{PHL10}		185	GND			10/	2.4 V				GND	GND	2.4 V		OUT	10/			Borrow Output		39	
	t_{PLH10}		186	GND			11/	2.4 V				GND	GND	2.4 V		OUT	11/			Borrow Output		42	
	t_{PLH11}		187	GND		OUT	2.4 V	15/				GND	GND	2.4 V			GND	GND		Output Q_A		63	
			188	GND	OUT		2.4 V						GND	2.4 V				2.4 V		Output Q_B		63	
			189	2.4 V			2.4 V		OUT				GND	2.4 V				2.4 V		Output Q_C		63	
			190	2.4 V			2.4 V			OUT		GND	2.4 V	2.4 V				2.4 V		Output Q_D		63	
	t_{PHL11}		191	GND			16/	2.4 V		OUT		2.4 V	GND	16/				GND		Output Q_D		71	
			192	GND					OUT			GND	2.4 V	16/				GND		Output Q_C		71	
			193	2.4 V	OUT							GND	GND	16/				GND		Output Q_C		71	
			194	GND		OUT						GND	GND	16/				2.4 V		Output Q_A		71	
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55° C.																						

1/ Output voltages shall be either:
(a) H = 2.4 volts minimum and L = 0.4 volt maximum when using a high speed checker double comparator, or
(b) H ≥1.5 volts and L ≤ 1.5 volts when using a high speed checker single comparator.
2/ Only a summary of attributes data is required.
3/ A ≥2.0 V, B ≤ 0.8 V. Input voltages shown are the maximum for V_{IL} and the minimum for V_{IH}.
4/ See figure 10A. Apply waveform E for tests 162 and 179, and waveform H for tests 163 and 180.
5/ See figure 10, waveforms A, B, C, and D. Load A_{IN}, then apply waveform A and observe the drop in Q_A output.
6/ See figure 10, waveform B, C, and D. Load A_{IN}, and observe t_{PLH} of waveform D at output Q_A.
7/ Repeat steps as in note 6. Observe the pulse delay of the drop in Q_A when the load pulse is applied.
8/ See figures 10 and 10A, waveforms A, B, C, E, F, and G. First use clear pulse, then load pulse, then one count up pulse. Also see figure 10B, note 7.
9/ See note 8, add one more count up pulse for t_{PLH9} as shown on figure 10A, waveforms F and G.
10/ See figures 10 and 10A. First clear pulse, then one count down pulse.
11/ See note 10. Clear pulse first, figure 10, waveforms A, B, and C, add one more count down pulse for t_{PLH10} as shown on figure 10A, waveforms J and K.
12/ See figures 10 and 10A. First a clear pulse, then one count up pulse for Q_A. On next three tests enter count up pulses to appropriate count to check Q_B, Q_C, and Q_D.
13/ See figure 10A, waveforms H and J. On next three tests enter count down pulse, waveform H and observe t_{PHL11}, waveform J on outputs Q_C, Q_B, and Q_A.
14/ F_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.
15/ Load pulse before count up input, then 4.5 V.
16/ Load pulse before count down input, then 4.5 volts.
17/ Terminal conditions (pins not designated may be H ≥2.0 V, or L ≤ 0.8 V, or open).
18/ The limits shall be: -0.5 mA minimum, -1.3 mA maximum for circuit C, and -0.7 mA minimum, -1.6 mA maximum for all other circuits.

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6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-35810 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	5492
02	5493
03	54160
04	54163A
05	54162
06	54161
07	5490
08	54192
09	54193

6.7 Manufacturers designation. Manufacturers circuits' included in this specification are designated as shown in table IV herein.

TABLE IV. Manufacturers' designation.

Circuit Device type	A Texas Instruments, Incorporated	B National Semiconductor Corporation	C Motorola Incorporated	D Advanced Micro Devices, Incorporated	E Signetics Corporation	F Fairchild Semi-conductor
01	X	X	X		X	
02	X	X	X		X	
03	X	X	X	X	X	
04	X	X	X	X	X	
05	X	X	X	X	X	X
06	X	X	X	X	X	X
07	X	X	X		X	
08			X		X	
09			X		X	

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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